CLASSIFICATION OF QUADRATIC-GROWTH WEIGHTED SIGNORINI SOLUTIONS IN THIN DIMENSION TWO WITH NEGATIVE WEIGHT EXPONENT

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Abstract. We classify all even global weighted Signorini solutions in bulk $\mathbb{R}^3$, with thin space $\mathbb{R}^2$, negative weight exponent $\eta = 1 - 2s \in (-1, 0)$, and at most quadratic growth. No blowdown, rank, convexity, or fractional tail is assumed. A unique positive-semidefinite quadratic blowdown is automatic. The solutions are exactly pressure-free polynomials, full-contact states, regular halfspaces, strips, parabolas, and ellipses, with explicit pressures and unique parameters. The noncompact rank-one exhaustion rests on a positive-derivative slit estimate and recovery of a transverse quadratic channel from negative infinity, which establish an ordinary tail and the canonical remainder extension. A finite positive derivative-pressure moment, mesoscopic curvature, affine matching, and signed ordering then yield exact rigidity. The remaining ranks reduce to established subquadratic classification and planar Riesz equilibrium theory; quadratic uniqueness uses known negative-energy epiperimetry. An all-order finite-tail classification is obtained as a companion result.

1. Introduction and the classification theorem

Global obstacle solutions encode the contact geometries compatible with an elliptic equation and its growth at infinity. We classify every even global weighted Signorini solution of at most quadratic growth in bulk $\mathbb{R}^3$ with thin space $\mathbb{R}^2$ and weight exponent $-1 < \eta < 0$. The alternatives are pressure-free polynomials, full-contact states, regular halfspaces, strips, parabolas, and ellipses. Neither a quadratic blowdown nor its rank is prescribed. The analytic difficulty is the noncompact rank-one branch, where an ordinary fractional tail must be recovered from intrinsic local-energy information.

Fix $1/2 < s < 1$, put $\eta = 1 - 2s$, and write $x \in \mathbb{R}^2$, $Y \in \mathbb{R}$, $q = (x, Y)$, $\iota(x) = (x, 0)$, and

$$L_\eta = \operatorname{div}(|Y|^\eta \nabla), \qquad Q_{\mathsf{B}}(x, Y) = x^T \mathsf{B} x - \frac{\operatorname{tr}(\mathsf{B})}{1+\eta} Y^2.$$

Here B is a symmetric matrix. Intrinsic pressure is denoted by $\widehat{\mu}$. The source equation below is understood in the weighted-energy sense: after localization its positive source is an energy-dual measure, admissible trace constraints use quasi-continuous representatives, and complementarity gives the local zero-obstacle variational inequality. Lemma 3.1 records this interpretation precisely.

Theorem 1.1 (Complete quadratic-growth classification)**.** *Let $1/2 < s < 1$ and $\eta = 1 - 2s$. Suppose*

$$\begin{gathered} U \in C(\mathbb{R}^3) \cap H^1_{\mathrm{loc}}(\mathbb{R}^3, |Y|^\eta), \qquad U(x, Y) = U(x, -Y), \\ u := U|_{Y=0} \geq 0, \qquad L_\eta U = -2\iota_* \widehat{\mu}, \qquad \widehat{\mu} \geq 0 \text{ is locally finite Radon}, \\ \operatorname{supp} \widehat{\mu} \subset K := \{u = 0\}, \qquad |U(q)| \leq C(1 + |q|^2). \end{gathered} \tag{1.1}$$

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There is a unique symmetric positive-semidefinite matrix B *for which* $R^{-2}U(R\cdot) \to Q_{\mathsf{B}}$ *locally uniformly and strongly in local weighted* H^1*. Exactly one of the six families in Section 2 occurs, with the unique parameters and normal-sign conventions stated there. Conversely every listed parameter choice gives a solution of* (1.1)*. The quantitative quadratic remainder is given by Proposition 4.1.*

The theorem includes $\mathsf{B} = 0$ and all pressure-free degeneracies. Its assumptions contain no convexity, density, tail, canonical-extension, or contact-geometry condition. The dimension, negative weight exponent, and quadratic growth are essential restrictions of the statement. The all-order ordinary-tail classification for a fixed rank-one obstacle is proved separately in Section 26.

1.1. **Proof mechanism.** Local variational compactness and Weiss dissipation first give a homogeneous quadratic limit. A compact annular Green test removes its pressure without presupposing convexity. Even harmonic Liouville gives a positive-semidefinite quadratic. Carducci's established negative-energy epiperimetric inequality then makes this limit unique, including when it is zero, and supplies a strict subquadratic remainder.

For zero rank, only finite differences and derivatives are put in the ordinary fractional operator domain. Their equations and growth verify the subquadratic classification theorem of Caffarelli–Ros-Oton–Serra. For positive-definite rank, full uniform asymptotics force compact contact. A compact-potential representation, justified initially for measure pressure, leads to the known ellipse equilibrium. These two reductions leave rank one as the noncompact core.

In rank-one coordinates the quadratic is $Q = a[Z^2 - Y^2/(1+\eta)]$, $a > 0$. Intrinsic contact analysis either gives an invariant state or an exact zero ray in a unique recession direction. Difference quotients show that $W = -U_X$ is nonnegative before its energy is inferred. A slit spectral window then improves its growth. A bound on U_X alone leaves an uncontrolled transverse quadratic channel. Integration from negative infinity fixes that channel using the known quadratic limit, and yields $U - Q = O(R^{s+1/2+\varepsilon})$. Choosing $\varepsilon < s - 1/2$ proves the ordinary tail. A test in an eventual positivity cone removes the possible zero-trace vertical mode and identifies the canonical extension of the remainder.

A finite positive derivative-pressure moment is needed to reach the exact endpoint. Its proof uses full transverse projection, normal-scale pressure bounds, and a measure-valued radial identity. The same moment fixes the lower-order canonical field. Mesoscopic selection identifies curvature, affine matching removes the remaining discrepancy, and a separate signed projection and two-phase argument orders actual profiles. Common contact then forces equality. The spectral domain, source moment, matching, and signed ordering arguments are proved in full below.

1.2. **Relation to earlier work.** The local weighted theory of Caffarelli–Salsa–Silvestre [3, Lemmas 2.7, 2.9 and 4.1, Proposition 4.3] supplies regularity and harmonic Liouville inputs. Their homogeneous classification [3, Theorem 5.7] begins with a homogeneous system, including convexity; it does not provide the global nonhomogeneous exhaustion here. The uniqueness and power gain at quadratic infinity use Carducci's negative-energy inequality [5, Theorem 5.2]. Its exponent need not give an ordinary tail. Directional remainder estimates also have direct antecedents in Fernández-Real–Jhaveri [15, Lemma 2.9 and Remark 2.10]. The zero-rank terminal halfspace theorem is due to Caffarelli–Ros-Oton–Serra [2, Theorem 4.1]; we prove the extension-to-operator interface needed to apply it.

The positive-definite pressure ellipse belongs to established planar Riesz equilibrium theory. Carrillo–Shu [7, Theorem 3.1 and Remark 3.2] cover arbitrary positive-definite quadratic confinement, including our exponent $2-2s \in (0,1)$. Frank, Mateu, Mora, Rondi, Scardia, and Verdera [16, Theorem 1.1] give a broader equilibrium theorem and credit that planar antecedent. Our direct angular and matrix proof makes the intrinsic reduction, all parameters, and pressure constants

explicit; it is not a new ellipse law. Its interval input is Carrillo–Huang's identity [6, equation (16)].

Eberle–Ros-Oton–Weiss [10, Theorems 1 and 3] classify compact coincidence sets in their weighted polynomial-growth problem; their printed theorem has thin dimension at least three. Eberle–Yu [12, Theorem 1.2 and Remark 1.4] treat compact contacts in the unweighted at-most-quadratic thin problem. Neither compact-contact hypothesis includes the noncompact strip and parabolic contacts. The central contribution here is their intrinsic rank-one exhaustion and automatic-tail bridge, completed by the rank-zero, blowdown, and equilibrium reductions above.

The classical obstacle classifications of Eberle–Figalli–Weiss [9] and Salib–Weiss [18, Theorem 1.3] provide methodological comparisons for matching and ordering. The equations are different: the present weighted source and slit spectrum require their own estimates, and no universal sharp weighted Friedland–Hayman inequality is asserted. The compact-perturbation results of Eberle, Salib, Weiss, and Shahgholian [11, Theorems 1.1–1.2] concern the classical equation. For the all-order companion, Fall's ordinary-tail Liouville theorem [14, Theorem 1.1] and the interval torsion identities of Dyda [8, Theorem 1 and Table 1] are further inputs.

1.3. **Conventions and proof organization.** The coefficient $a > 0$ is reserved for the rank-one obstacle and $b > 0$ for parabolic curvature; neither denotes the weight exponent. We distinguish the intrinsic pressure $\widehat{\mu}$ from the ordinary fractional pressure μ. Unqualified balls are Euclidean and centered at zero. Constants may depend on the fixed order, solution, and stated exponents; no endpoint-uniform bounds are claimed.

Sections 3–6 establish the automatic quadratic limit and close the zero and positive-definite branches. Sections 7–11 establish the rank-one analytic interface. The cylindrical and noncylindrical proofs then give Proposition 24.2. The complete theorem is finished in Section 25, before the late ordinary-tail companion is deduced. In particular that companion is not an input to the full intrinsic classification.

2. The six model families

Throughout this catalogue $1/2 < s < 1$, $t = 1-s$, and $\eta = 1-2s$. We use $\widehat{f}(\xi) = \int e^{-ix\cdot\xi} f(x)\,\mathrm{d}x$, with inverse factor $(2\pi)^{-n}$, and $\mathcal{A}_s^{(n)} = (-\Delta)^s$ for the multiplier $|\xi|^{2s}$. Omit the superscript in dimension two. Put

$$e_s = 2^{1-2s}\frac{\Gamma(1-s)}{\Gamma(s)}, \qquad k_s = \frac{\Gamma(1-s)}{4^s\pi\Gamma(s)} = \frac{e_s}{2\pi}, \qquad K_s(x) = k_s|x|^{-2t},$$

$$d_s = -\frac{\Gamma(1/2-s)}{4^s\sqrt{\pi}\Gamma(s)} > 0, \qquad A_s^* = \frac{2^{1-2s}\Gamma(3/2-s)\Gamma(2-s)}{\sqrt{\pi}}.$$

The conormal factor e_s, planar Riesz coefficient k_s, and strip coefficient d_s are distinct. Euler's beta function is denoted by $B(\cdot,\cdot)$, whereas quadratic matrices use B. The ordinary tail class is

$$\mathcal{T}_s = \left\{ f : \int_{\mathbb{R}^2} |f(x)|(1+|x|^2)^{-1-s}\,\mathrm{d}x < \infty \right\}.$$

The equivalent weights $(1+|x|)^{-2-2s}$ and $(1+|x|^{2+2s})^{-1}$ are also used, as is L_s^1 in the dimension in question. For an ordinary-tail trace the even Poisson extension is

$$\mathcal{E}_s f(x,Y) = \frac{s}{\pi}\int_{\mathbb{R}^2} \frac{|Y|^{2s} f(\xi)}{(|x-\xi|^2+Y^2)^{1+s}}\,\mathrm{d}\xi.$$

It has conormal $e_s\mathcal{A}_s f$ and even source $-2e_s\iota_*(\mathcal{A}_s f)$ whenever interpreted through the local energy formulation below. Only remainders with an established absolute tail enter this integral. For

classical symmetric integrals the normalization is

$$c_{n,s} = \frac{4^s\Gamma(n/2+s)}{\pi^{n/2}|\Gamma(-s)|}, \qquad \mathcal{A}_s^{(n)} f(x) = \frac{c_{n,s}}{2}\int_{\mathbb{R}^n} \frac{2f(x) - f(x+y) - f(x-y)}{|y|^{n+2s}}\,\mathrm{d}y.$$

Otherwise the ordinary-tail operator is defined by compact-test duality.

2.1. **Pressure-free polynomials.** Choose $\mathsf{B} = \mathsf{B}^T \geq 0$, $\boldsymbol{\ell} \in \operatorname{range}\mathsf{B}$, and $c \geq \boldsymbol{\ell}^T\mathsf{B}^+\boldsymbol{\ell}/4$, with B^+ the Moore–Penrose inverse. Then

$$U(x,Y) = Q_{\mathsf{B}}(x,Y) + \boldsymbol{\ell}\cdot x + c, \qquad \widehat{\mu} = 0.$$

The literal coefficients are unique. If $c > \boldsymbol{\ell}^T\mathsf{B}^+\boldsymbol{\ell}/4$, contact is empty. In the equality case it is $-\mathsf{B}^+\boldsymbol{\ell}/2 + \ker\mathsf{B}$. This includes every nonnegative constant and the zero state. In particular the zero state has full contact, without positive pressure.

2.2. **Full contact with positive pressure.** For a unique $c_0 > 0$,

$$U(x,Y) = -c_0|Y|^{2s}, \qquad K = \mathbb{R}^2, \qquad \mathrm{d}\widehat{\mu} = 2sc_0\,\mathrm{d}x.$$

Its quadratic matrix is zero. The strict amplitude excludes the zero state already assigned to the polynomial family.

2.3. **Regular halfspaces.** For $r = (T^2+Y^2)^{1/2}$ define the explicit even extension

$$F_s(T,Y) = \frac{(r+T)^s(T-sr)}{2^s(1-s)}. \tag{2.1}$$

It is assigned its continuous value at the origin and has thin trace T_+^{1+s}. For unique $\alpha > 0$, oriented $e \in S^1$, and $h_{\mathrm{reg}} \in \mathbb{R}$, the regular family is

$$U(x,Y) = \alpha F_s(e\cdot x - h_{\mathrm{reg}}, Y), \qquad K = \{e\cdot x \leq h_{\mathrm{reg}}\},$$

$$\mathrm{d}\widehat{\mu} = \alpha r_s(h_{\mathrm{reg}} - e\cdot x)_+^{1-s}\,\mathrm{d}x, \qquad r_s = \frac{2^{1-2s}s(1+s)}{1-s}.$$

The normal points into the positivity halfspace; its sign is not a gauge. Tangential translation is not a parameter. Here $\mathsf{B} = 0$. The trace has degree $1+s > 2s$ and is not put in the ordinary Poisson integral.

2.4. **Positive-pressure strips.** Choose $\mathsf{B} = an\otimes n$, $a > 0$, a unit normal n, an affine centerline $\{n\cdot x = h\}$, and $L > 0$. The data (n,h) and $(-n,-h)$ are identified. Write $k = a/A_s^*$ and

$$p_L(z) = -d_s k\left[\int_{-L}^{L} |z-y|^{2s-1}(L^2-y^2)^{1-s}\,\mathrm{d}y - B(s,2-s)L^2\right]. \tag{2.2}$$

Set $P(x) = p_L(n\cdot x - h) - 2ah(n\cdot x) + ah^2$. Then

$$U = Q_{\mathsf{B}} + \mathcal{E}_s P, \qquad K = \{|n\cdot x - h| \leq L\},$$

$$\mathrm{d}\widehat{\mu} = e_s\frac{a}{A_s^*}[L^2 - (n\cdot x - h)^2]_+^{1-s}\,\mathrm{d}x.$$

The unique intrinsic data are B, centerline, and L. There is no translation parameter along the centerline. Pressure per unit centerline length is $e_s(a/A_s^*)B(1/2,2-s)L^{3-2s}$; total planar pressure is infinite. The centered potential has growth $2s-1$, and the affine gauge has an ordinary tail since $2s > 1$.

2.5. **Parabolic contacts.** Choose $\mathsf{B} = an \otimes n$, $a > 0$, an oriented unit vector $\tau \in \ker \mathsf{B}$, a vertex $v \in \mathbb{R}^2$, and $b > 0$. The transverse normal n is determined only up to sign. In centered coordinates (X, Z) put

$$\rho_b(X, Z) = \frac{a}{A_s^* b^t}(X - bZ^2)_+^t$$

and define

$$P_b(x) = \int_{\mathbb{R}^2} [K_s(x - y) - K_s(-y) - \nabla K_s(-y) \cdot x]\rho_b(y)\,\mathrm{d}y. \tag{2.3}$$

The bracket is one absolutely convergent combined integrand; its three terms are not separate integrals. Set $p(x) = P_b(\tau \cdot (x - v), n \cdot (x - v))$. The solution is

$$U(x, Y) = Q_{\mathsf{B}}(x - v, Y) + \mathcal{E}_s p(x, Y),$$
$$K = \{\tau \cdot (x - v) \geq b[n \cdot (x - v)]^2\}, \qquad \mathrm{d}\widehat{\mu} = \frac{e_s a}{A_s^* b^t}[\tau \cdot (x - v) - b(n \cdot (x - v))^2]_+^t\,\mathrm{d}x.$$

The matrix, oriented opening, vertex, and curvature are unique. The sign of n has no effect. The centered remainder has growth $\lambda = s + 1/2 < 2s$ and is even in Z. With $\beta = 3/2 - s$ and $D_s = B(1/2, 2 - s)/A_s^*$, its lower-order canonical amplitude is $M = aD_s b^{-\beta} > 0$. This is a free amplitude, not a finite total pressure mass.

For later use choose an initial unit vector $\tau_0 \in \ker \mathsf{B}$, write $(X, Z) = (\tau_0 \cdot x, n \cdot x)$, and put $\tau = \varepsilon\tau_0$, $v = (\varepsilon d)\tau_0 + hn$. Relative to the unshifted quadratic the same remainder is

$$P(X, Z) = P_b(\varepsilon X - d, Z - h) - 2ahZ + ah^2. \tag{2.4}$$

Thus the coordinate formula has exactly the physical vertex stated above, with no extra reflection or center parameter.

2.6. **Positive-pressure ellipses.** Choose $\mathsf{B} = \mathsf{B}^T > 0$, a center $z \in \mathbb{R}^2$, and a fractional pressure mass $m_{\mathrm{ell}} > 0$. Define

$$J_s = 2B(1/2 - t, 1/2), \quad B_{1,s} = B(1/2, t + 1), \quad B_{2,s} = B(1/2 - t, t + 3/2),$$
$$\mathfrak{c}_s^{\mathrm{ell}} = \frac{k_s(t + 1)B_{1,s}B_{2,s}}{\pi J_s} > 0.$$

There is a unique $\mathsf{H} = \mathsf{H}^T > 0$ solving

$$\mathsf{B} = 2t\mathfrak{c}_s^{\mathrm{ell}} m_{\mathrm{ell}} \int_{S^1} \theta\theta^T(\theta^T \mathsf{H}\theta)^{-t-1}\,\mathrm{d}\theta, \tag{2.5}$$

where the measure is ordinary arc length. Put

$$\ell_{\mathrm{ell}} = \mathfrak{c}_s^{\mathrm{ell}} m_{\mathrm{ell}} \int_{S^1} (\theta^T \mathsf{H}\theta)^{-t}\,\mathrm{d}\theta = \frac{\operatorname{tr}(\mathsf{BH})}{2t} > 0,$$
$$\rho_{\mathsf{H}}(x) = \frac{m_{\mathrm{ell}}(t + 1)}{\pi\sqrt{\det \mathsf{H}}}(1 - x^T \mathsf{H}^{-1} x)_+^t, \qquad V_{\mathsf{H}} = K_s * \rho_{\mathsf{H}}.$$

Then

$$U(x, Y) = Q_{\mathsf{B}}(x - z, Y) - \ell_{\mathrm{ell}} + \mathcal{E}_s V_{\mathsf{H}}(x - z, Y),$$
$$K = z + \{x : x^T \mathsf{H}^{-1} x \leq 1\}, \qquad \mathrm{d}\widehat{\mu} = e_s \rho_{\mathsf{H}}(x - z)\,\mathrm{d}x.$$

The correction is explicitly

$$\mathcal{E}_s V_{\mathsf{H}}(x, Y) = k_s \int_{\mathbb{R}^2} (|x - w|^2 + Y^2)^{-t} \rho_{\mathsf{H}}(w)\,\mathrm{d}w,$$

and decays as $O(|(x, Y)|^{-2t})$ uniformly in direction. The matrix H commutes with B and its eigenvalues are the squared semiaxis lengths. No eccentricity restriction is imposed. The unique parameters are $\mathsf{B}, z, m_{\mathrm{ell}}$, equivalently $\mathsf{B}, z, \ell_{\mathrm{ell}}$. For fixed B, $\mathsf{H}_m = m^{1/(1+t)}\mathsf{H}_1$ and $\ell_m = m^{1/(1+t)}\ell_1$. Intrinsic total pressure is $e_s m_{\mathrm{ell}}$, not m_{ell}.

2.7. **The fixed rank-one notation used in the proof.** In the rank-one sections $x = (X, Z)$, $a > 0$, $Q = a[Z^2 - Y^2/(1+\eta)]$, $V = U - Q$, and $P = u - aZ^2$. The full-limit condition is

$$R^{-2}U(R\,\cdot) \longrightarrow Q \quad \text{locally uniformly and strongly in } H^1_{\mathrm{loc}}(|Y|^\eta). \tag{2.6}$$

It follows from Proposition 4.1 after rotation when $\operatorname{rank} \mathsf{B} = 1$. The ordinary-tail auxiliary class, also used for every $0 < s < 1$ in the companion theorem, is

$$P \in C(\mathbb{R}^2) \cap \mathcal{T}_s, \qquad P \geq -aZ^2, \qquad \mu = \mathcal{A}_s P \geq 0 \text{ is Radon}, \qquad \operatorname{supp}\mu \subset \{P = -aZ^2\}. \tag{2.7}$$

This is an operator applied to the remainder P, never separately to the nonintegrable quadratic. In the intrinsic problem these tail hypotheses and $\widehat{\mu} = e_s\mu$ will be conclusions.

3. Local energy and comparison tools

Throughout the intrinsic arguments $1/2 < s < 1$, $\eta = 1 - 2s$, and $\rho(q) = |Y|^\eta$. We write $D = 3 + \eta$ and $L_\eta = \operatorname{div}(\rho\nabla)$. All measure pairings with energy tests use their quasi-continuous representatives. The following arguments do not require an ordinary tail or a canonical extension.

Lemma 3.1 (Local minimization from the intrinsic source)**.** *Suppose U is continuous, even in Y, and locally in $H^1(\rho)$, with*

$$u = U|_{Y=0} \geq 0, \qquad L_\eta U = -2\iota_*\widehat{\mu}, \qquad \widehat{\mu} \geq 0, \qquad \operatorname{supp}\widehat{\mu} \subset K := \{u = 0\}.$$

In every bounded ball Ω, U minimizes the weighted Dirichlet energy among functions with its outer trace and nonnegative thin trace. The obstacle constraint on energy competitors is understood through quasi-continuous trace representatives. In particular, for every admissible $h \in H^1_0(\Omega, \rho)$,

$$\int_\Omega \rho\big(|\nabla(U+h)|^2 - |\nabla U|^2\big) = \int_\Omega \rho|\nabla h|^2 + 4\int_{\Omega\cap\{Y=0\}} \widetilde{h}\ \mathrm{d}\widehat{\mu} \geq 0. \tag{3.1}$$

Proof. For smooth compactly supported ϕ the source identity gives

$$2\int \phi(x,0)\ \mathrm{d}\widehat{\mu}(x) = \int_\Omega \rho\nabla U \cdot \nabla\phi.$$

Thus the localized positive measure $\iota_*\widehat{\mu}$ is a continuous functional on $H^1_0(\Omega, \rho)$. Such a measure does not charge sets of zero weighted energy capacity: apply the displayed bound to nonnegative capacity cutoffs that are at least one near the set and whose energy norms tend to zero, first on compact subsets and then by inner regularity. The functional is consequently evaluated on the quasi-continuous representative of an energy test. Approximation in $H^1_0(\Omega, \rho)$ extends the identity to those tests. The thin trace of a nonnegative admissible energy function is nonnegative quasi-everywhere, and the continuous representative of U vanishes on K. Hence $\widetilde{h} \geq 0$ $\widehat{\mu}$-almost everywhere. Expansion of the square proves (3.1). This argument proves minimization for energy competitors, rather than only for smooth admissible perturbations. Restricting to even competitors is therefore also permitted. □

Lemma 3.2 (The two remainder phases)**.** *Under the assumptions of Lemma 3.1, put $V = U - Q_{\mathsf{B}}$, where $\mathsf{B} \geq 0$, and abbreviate $Q = Q_{\mathsf{B}}$. Then V_+ and V_- are globally weak L_η-subsolutions. For every compactly supported Lipschitz cutoff ζ,*

$$\int \rho\zeta^2|\nabla V|^2 \leq 4\int \rho V^2|\nabla\zeta|^2. \tag{3.2}$$

Moreover, for concentric balls centered at the origin,

$$\sup_{B_R} |V| \leq CR^{-D/2}\|V\|_{L^2(B_{2R},\rho)}. \tag{3.3}$$

The same statements hold for every positive rescaling of V.

Proof. The quadratic Q is weakly L_η-harmonic on all of $\mathbb{R}^3$: its tangential Laplacian and weighted normal derivative cancel, and its weighted normal flux tends to zero at the thin plane. Hence $L_\eta V = -2\iota_*\widehat{\mu} \le 0$. Convex truncation of the subsolution $-V$ gives $L_\eta V_- \ge 0$. On $K\times\{0\}$ one has $V = -x^T \mathsf{B} x \le 0$. For every $\varepsilon > 0$, continuity therefore makes $(V-\varepsilon)_+$ vanish on a neighborhood of each compact portion of the source support. On its complement V is harmonic, so convex truncation makes $(V-\varepsilon)_+$ a subsolution there. A partition of unity on these open sets proves the global subsolution inequality. As $\varepsilon \downarrow 0$, the truncations converge in local $H^1(\rho)$: their gradients are $1_{\{V>\varepsilon\}}\nabla V$, and dominated convergence applies. Thus $L_\eta V_+ \ge 0$ as well.

For a nonnegative energy subsolution v, testing with $\zeta^2 v$ gives

$$\int \rho\zeta^2|\nabla v|^2 \le 2\int \rho\zeta v|\nabla\zeta||\nabla v| \le \frac{1}{2}\int \rho\zeta^2|\nabla v|^2 + 2\int \rho v^2|\nabla\zeta|^2.$$

An additional bounded truncation, followed by a limit, justifies the test if needed. Apply the resulting estimate to both phases; their gradients have disjoint supports almost everywhere and their squared values add to V^2. This proves (3.2). The weight $|Y|^\eta$ belongs to A_2. Weighted local boundedness [13], applied separately to the two phases, gives (3.3), since $\int_{B_R}\rho = C_\eta R^D$. Continuity turns essential suprema into suprema. The source signs and all the estimates are preserved by positive rescaling. □

3.1. **Halfspace comparison with a subcritical negative part.** We now work with $1/2 < s < 1$ and $\eta = 1-2s \in (-1,0)$. The comparison principle below uses an explicit harmonic barrier, without a Poisson representation of the function being compared.

Lemma 3.3 (Halfspace maximum principle)**.** *Let w be continuous on $\overline{\mathbb{R}^3_+}$, locally in $H^1(\overline{\mathbb{R}^3_+}, Y^\eta)$, and weakly L_η-harmonic in $\mathbb{R}^3_+ = \{Y>0\}$. Suppose $w(x,0)\ge 0$ and, for some $\gamma < 2s$,*

$$w_-(q) \le C(1+|q|)^{\max\{\gamma,0\}}.$$

Then $w \ge 0$ in $\overline{\mathbb{R}^3_+}$.

Proof. For $(x,Y)\in B_1^+$ define

$$B(x,Y) = \int_{|\xi|=1} \frac{Y^{2s}}{(|x-\xi|^2+Y^2)^{1+s}}\,\mathrm{d}\sigma(\xi), \qquad \xi\in\mathbb{R}^2. \tag{3.4}$$

Each integrand is L_η-harmonic for $Y>0$, as is checked by differentiation, so B is positive and harmonic in B_1^+. It has zero trace on the open flat disk. Since $|x-\xi|\ge 1/2$ in $B^+_{1/2}$,

$$B(x,Y) \le CY^{2s} \quad \text{on } B^+_{1/2}. \tag{3.5}$$

There is a uniform $c>0$ with $B\ge c$ on the open spherical cap. To check the only delicate part, let $|x|^2+Y^2=1$ and let Y be small. With $\xi_0 = x/|x|$, one has $1-|x| = O(Y^2)$. On an arc $|\xi-\xi_0|\le c_0 Y$ of length comparable to Y, the denominator in (3.4) is at most CY^{2+2s}. Hence $B(x,Y)\ge c/Y$. Away from the rim the cap bound follows by positivity and compactness. This argument only uses approach to the rim along the cap.

The barrier need not have finite energy on a halfball containing the rim. We therefore implement comparison on truncated domains. Fix $R>0$, $\varepsilon>0$, and write

$$M_R = \sup_{\partial B_R\cap\{Y>0\}} w_-, \qquad B_R^*(q) = B(q/R).$$

The compact rim $\{|x|=R, Y=0\}$ has $w_-=0$. Continuity gives a neighborhood of it in which $w_- < \varepsilon$. Excise a smaller tubular neighborhood of the rim from B_R^+, choosing its boundary piecewise Lipschitz and inside that neighborhood. Denote the remaining domain by Ω. On Ω, B_R^* has finite weighted energy. Indeed, away from the excised rim its coefficients are smooth in x and Y^2, multiplied by Y^{2s}; near its flat boundary its normal derivative is $O(Y^{2s-1})$ and its tangential

derivatives are $O(Y^{2s})$. The corresponding normal energy density is $O(Y^{\eta+4s-2}) = O(Y^{2s-1})$, which is integrable. The remaining compact parts of $\overline{\Omega}$ present no singularity.

Choose $C_0 \geq 1/c$, where c is the cap lower bound. On the spherical part of $\partial\Omega$, $w_- \leq C_0 M_R B_R^*$. On the flat part $w_- = 0$, and on the artificial boundary $w_- < \varepsilon$. Consequently

$$\psi = (w_- - C_0 M_R B_R^* - \varepsilon)_+ \in H_0^1(\Omega, Y^\eta).$$

Here w_- is a subsolution by convex truncation of the harmonic function $-w$. Subtract the weak equation for $C_0 M_R B_R^*$ from its subsolution inequality and test by ψ. This gives $\int_\Omega Y^\eta |\nabla\psi|^2 \leq 0$, hence $\psi = 0$. For fixed $q \in B_R^+$ the excision can be made small enough to exclude neither q nor any prescribed compact subset. The resulting inequality is independent of the excision size. Shrinking the excision and then letting $\varepsilon \downarrow 0$ gives

$$w_-(q) \leq C_0 M_R B(q/R).$$

For $R > 2|q|$, (3.5) and the growth assumption therefore imply

$$w_-(q) \leq C_q (1+R)^{\max\{\gamma,0\}} R^{-2s} \longrightarrow 0.$$

This proves the assertion, including on the boundary by hypothesis. □

3.2. **Entire and zero-trace weighted Liouville statements.** We will use the following statements both in three ambient dimensions and after reduction to the (Z, Y) plane. Keeping their parity conditions distinct is important: a zero-trace solution in one halfspace may have a nonzero Y^{2s} term.

Lemma 3.4 (Even entire subquadratic Liouville theorem)**.** *Let $n \in \{1, 2\}$, $1/2 < s < 1$, and $\eta = 1 - 2s$. Suppose $H \in H^1_{\mathrm{loc}}(\mathbb{R}^{n+1}, |Y|^\eta)$ is an entire weak L_η-harmonic function, even in Y, and*

$$\sup_{B_R} |H| = o(R^2).$$

Then $H(x, Y) = b \cdot x + c$ for constants $b \in \mathbb{R}^n$ and $c \in \mathbb{R}$.

Proof. We first establish an estimate that does not require evenness. For any entire weak harmonic local-energy function and physical indices i, j, tangential difference quotients and Caccioppoli give, on successively smaller concentric balls,

$$\|\partial_i \partial_j H\|_{L^2(B_{3R/4}, |Y|^\eta)} \leq C R^{-2} \|H\|_{L^2(B_R, |Y|^\eta)}. \tag{3.6}$$

Here is the energy justification. Caccioppoli for H first bounds ∇H in L^2 on a smaller ball by $CR^{-1}\|H\|_{L^2(B_R)}$. Because the weight is independent of x, every tangential difference quotient is weakly harmonic where both translates are defined. Its L^2 norm is controlled by the corresponding first derivative of H on a slightly larger ball. Caccioppoli for these quotients then gives uniform local H^1 bounds. Passing weakly to their limits shows that each $\partial_i H$ is a harmonic local-energy function. Repeating this argument gives all second tangential derivatives and (3.6), with nested radii chosen between $3R/4$ and R. One more repetition makes the second derivatives themselves harmonic local-energy functions. This argument uses no differentiability of the weight in the normal direction.

Weighted local boundedness [13] for the positive and negative parts of these harmonic derivatives, followed by (3.6), proves

$$\begin{aligned} \sup_{B_{R/2}} |\partial_i \partial_j H| &\leq C R^{-(n+1+\eta)/2} \|\partial_i \partial_j H\|_{L^2(B_{3R/4}, |Y|^\eta)} \\ &\leq C R^{-2} \sup_{B_R} |H|. \end{aligned} \tag{3.7}$$

Harmonic functions have their locally bounded continuous representatives by the same weighted local theory. The growth hypothesis and exhaustion in (3.7) show that all tangential second derivatives vanish. In the upper halfspace, where the equation is smooth, it follows that

$$H(x,Y) = b_0(Y) + \sum_{i=1}^{n} x_i b_i(Y).$$

Substitution in $L_\eta H = 0$ and independence of the affine functions of x give

$$(Y^\eta b_i'(Y))' = 0, \qquad b_i(Y) = A_i + B_i Y^{1-\eta} = A_i + B_i Y^{2s}, \quad i = 0, \dots, n.$$

Evenness determines the lower-halfspace expression. However the B_i terms would create a plane source: direct integration in Y gives

$$L_\eta \left[|Y|^{2s} \left(B_0 + \sum_{i=1}^{n} B_i x_i \right) \right] = 4s \left(B_0 + \sum_{i=1}^{n} B_i x_i \right) \delta_{\{Y=0\}}.$$

This identity is legitimate in local energy, since the normal energy exponent is $\eta + 4s - 2 = 2s - 1 > -1$. The entire weak equation forces every $B_i = 0$, leaving precisely $A_0 + \sum_i A_i x_i$. □

Lemma 3.5 (Zero-trace subquadratic Liouville theorem)**.** *Let $n \in \{1, 2\}$, $1/2 < s < 1$, and $\eta = 1 - 2s$. Suppose D is continuous on $\overline{\mathbb{R}^{n+1}_+}$, locally in $H^1(\overline{\mathbb{R}^{n+1}_+}, Y^\eta)$, harmonic in $Y > 0$, and has zero thin trace. If $\sup_{B_R^+} |D| = o(R^2)$, then*

$$D(x,Y) = cY^{2s} \qquad (Y \geq 0)$$

for a constant c.

Proof. Define the odd reflection

$$\widetilde{D}(x,Y) = \begin{cases} D(x,Y), & Y \geq 0, \\ -D(x,-Y), & Y < 0. \end{cases}$$

Its two traces agree at zero. Gluing the weighted Sobolev functions across this common trace shows that $\widetilde{D} \in H^1_{\mathrm{loc}}(\mathbb{R}^{n+1}, |Y|^\eta)$; its local energy is the sum of the two reflected halfspace energies. This gluing can also be seen on almost every vertical line by weighted Cauchy–Schwarz, since $\int_0^1 Y^{-\eta}\, \mathrm{d}Y < \infty$, followed by integration in x.

For $\phi \in C_c^\infty(\mathbb{R}^{n+1})$ the reflected weak integral is

$$\int_{\mathbb{R}^{n+1}} |Y|^\eta \nabla \widetilde{D} \cdot \nabla \phi = \int_{Y>0} Y^\eta \nabla D \cdot \nabla \psi, \qquad \psi(x,Y) = \phi(x,Y) - \phi(x,-Y).$$

The test ψ has zero trace. To justify using it in the upper halfspace equation, multiply it by a smooth cutoff that vanishes for $Y \leq \varepsilon$ and is one for $Y \geq 2\varepsilon$. On a fixed compact set $|\psi| \leq CY$ and $|\nabla \psi| \leq C$, so the squared weighted H^1 norm of the difference is $O(\varepsilon^{1+\eta}) \to 0$. Further harmless compact cutoffs keep its support fixed. Approximation and the local energy of D show that the displayed integral is zero. Thus the odd reflection is an entire weak harmonic energy function; an even-only Liouville statement would not suffice at this step.

The proof of (3.7) did not use evenness. Apply it to $\widetilde{D}$ and use its subquadratic growth to obtain zero tangential second derivatives. The same coefficient ODE in $Y > 0$ therefore gives

$$D(x,Y) = A_0 + \sum_{i=1}^{n} A_i x_i + Y^{2s} \left(B_0 + \sum_{i=1}^{n} B_i x_i \right).$$

The zero trace forces all $A_i = 0$. For each $i \geq 1$, comparison of the values at (Re_i, R) and $(0, R)$ leaves $B_i R^{1+2s}$. Both points lie in a ball of radius $2R$, whereas $1 + 2s > 2$. The growth assumption therefore forces $B_i = 0$ for $i \geq 1$. The constant B_0 is allowed, and the conclusion follows with $c = B_0$.

The odd continuation of this remaining mode is source-free. Its even continuation is different: $L_\eta(c|Y|^{2s}) = 4sc\,\delta_{\{Y=0\}}$. Accordingly the present lemma leaves a vertical mode that must be removed by a separate argument when identifying a canonical even extension. □

3.3. **Ordinary-tail local energy and normalization.** The following extension statement is valid for every $0 < s < 1$. Here $\mathcal{T}_s$ is the absolute ordinary tail space defined in Section 2. We use it only for traces for which that tail has been assumed or proved.

Lemma 3.6 (Local energy and conormal identity)**.** *More generally, let $f \in C(\mathbb{R}^2) \cap \mathcal{T}_s$ and suppose $\mathcal{A}_s f = m$ is a locally finite signed Radon measure. Then $f \in H^s_{\mathrm{loc}}(\mathbb{R}^2)$ and $\mathcal{E}_s f \in H^1_{\mathrm{loc}}(\mathbb{R}^3, |Y|^\eta)$, $\eta = 1 - 2s$, with $L_\eta \mathcal{E}_s f = -2e_s \iota_* m$, where $e_s = 2^{1-2s}\Gamma(1-s)/\Gamma(s) > 0$ and $\iota(X, Z) = (X, Z, 0)$. In particular, for the obstacle state P and $V = \mathcal{E}_s P$,*

$$L_\eta V = -2e_s\, \iota_* \mu. \tag{3.8}$$

The conormal formula also holds distributionally for every ordinary-tail trace; whenever its extension has local energy it gives the corresponding weak source identity.

Proof. Take bounded nested sets $D_0 \Subset D$ and a compact cutoff ζ in D, equal to one near D_0, with support separated from D^c. For $u = f_\varepsilon$, test $\mathcal{A}_s u = m_\varepsilon$ by $\zeta^2 u$. On $D \times D$ use

$$\begin{aligned}&(u(x) - u(y))(\zeta(x)^2 u(x) - \zeta(y)^2 u(y))\\ &\quad = |\zeta(x)u(x) - \zeta(y)u(y)|^2 - u(x)u(y)|\zeta(x) - \zeta(y)|^2.\end{aligned}$$

The error is bounded by local suprema and $\int_0^1 r^{1-2s}\,\mathrm{d}r < \infty$. Each cross term has only one remote factor $u(y)$; separation of the sets and the common ordinary L^1 tail bound its absolute value. The source pairing is bounded by a compact supremum times the uniformly bounded local total variation of m_ε. This gives common local H^s bounds, and Fatou gives $f \in H^s_{\mathrm{loc}}$.

On a fixed extension footprint, split $f = \chi f + (1 - \chi)f$, with χ compact and equal to one on a larger footprint. The compact H^s trace has finite-energy extension [4, Section 3.2]. The separated part g satisfies

$$\mathcal{E}_s g = O(Y^{2s}), \qquad \nabla_x \mathcal{E}_s g = O(Y^{2s}), \qquad \partial_Y \mathcal{E}_s g = O(Y^{2s-1}),$$

by differentiated tail kernels. Its weighted normal energy is bounded by $C\int_0^\delta Y^{2s-1}\,\mathrm{d}Y < \infty$. This proves local energy without a global squared-tail premise.

For the exact conormal constant, put

$$G_Y(h) = -Y^{1-2s}\partial_Y \Pi_s(h, Y) = -\frac{s}{\pi}\left[\frac{2s}{(|h|^2 + Y^2)^{1+s}} - \frac{2(1+s)Y^2}{(|h|^2 + Y^2)^{2+s}}\right].$$

This even kernel has integral zero. For a compact smooth test ϕ, subtract $\phi(x)$ and symmetrize. Its second difference is $O(|h|^2)$, while $|G_Y(h)| \le C_s|h|^{-2-2s}$, uniformly for $0 < Y \le 1$. The local majorant is integrable since $s < 1$. At $h \neq 0$ the kernel tends to $-2s^2/(\pi|h|^{2+2s})$, so

$$G_Y * \phi \longrightarrow \frac{2s^2}{\pi c_{2,s}}\mathcal{A}_s\phi = e_s\mathcal{A}_s\phi, \qquad -\lim_{Y\downarrow 0} Y^\eta \partial_Y \mathcal{E}_s f = e_s \mathcal{A}_s f. \tag{3.9}$$

Gamma recurrence gives the constant. Convergence is uniform on compact sets, and outside the support of ϕ the common bound is $C_\phi(1+|x|)^{-2-2s}$. Thus it can be paired with every ordinary-tail f. Localization into the finite-energy compact part and differentiable separated tail passes the half-space integration by parts to the plane. Even reflection doubles the conormal source, proving (3.8). □

4. Automatic quadratic blowdown

No asymptotic polynomial or rank is prescribed in this section. The local intrinsic estimates of Caffarelli–Salsa–Silvestre [3, Lemma 4.1 and Proposition 4.3] apply to the local zero-obstacle minimizer formulation of Lemma 3.1. A local amplitude bound gives tangential $C^{1,\gamma}$ bounds for $0<\gamma<s$, trace semiconvexity, and a bounded one-sided weighted normal derivative. In particular $|U_Y|\le CY^{-\eta}$ in a smaller upper halfball after normalization. This is a one-sided flux bound, not continuity of the odd flux across the thin plane.

Proposition 4.1 (The unique automatic quadratic limit)**.** *Every solution of* (1.1) *has a unique symmetric matrix* $\mathsf{B}\ge 0$ *such that*

$$R^{-2}U(R\,\cdot)\longrightarrow Q_{\mathsf{B}}\quad \textit{locally uniformly and strongly in } H^1_{\mathrm{loc}}(|Y|^\eta).$$

There is $\delta=\delta(s)\in(0,1)$ *for which*

$$\sup_{B_R}|U-Q_{\mathsf{B}}|\le C_U(1+R^{2-\delta}),\qquad R\ge 1. \tag{4.1}$$

For each fixed $L\ge 1$,

$$\|R^{-2}U(R\,\cdot)-Q_{\mathsf{B}}\|_{L^\infty(B_L)}+\|R^{-2}U(R\,\cdot)-Q_{\mathsf{B}}\|_{H^1(B_L,|Y|^\eta)}\le C_{U,L}R^{-\delta}. \tag{4.2}$$

The matrix may be zero.

Proof. Compactness and strong energy convergence. Put $U_R(q)=R^{-2}U(Rq)$, $R\ge 1$. Quadratic growth bounds these functions uniformly on each fixed ball. Testing the equation with a cutoff times U_R is legitimate by Lemma 3.1; the source term vanishes on zero trace. Caccioppoli gives local energy bounds. The local estimates recalled above give uniform tangential bounds and $|(U_R)_Y|\le CY^{-\eta}$ on smaller upper halfballs. Normal integration gives $|U_R(x,Y)-U_R(x,0)|\le CY^{1-\eta}$; since $\eta<0$, the derivative bound also gives uniform vertical Lipschitz control on smaller halfballs. Hence any sequence tending to infinity has a subsequence, still denoted R_j, with $U_j=U_{R_j}\to F$ locally uniformly and weakly in local weighted H^1. The rescaled pressures are $\widehat{\mu}_j(E)=R_j^{-D}\widehat{\mu}(R_jE)$. A nonnegative compact cutoff ψ equal to one near a fixed thin compact set satisfies

$$2\int\psi(x,0)\,\mathrm{d}\widehat{\mu}_j=\int\rho\nabla U_j\cdot\nabla\psi.$$

The energy bound therefore controls local pressure mass. Extracting a further diagonal subsequence gives a nonnegative local weak-* limit ν. The equation passes to F, and where its trace is positive, uniform convergence makes the approximating pressures vanish locally. Thus $L_\eta F=-2\iota_*\nu$ with source on its zero trace. Evenness and nonnegativity pass, so F is also a local minimizer.

For $d_j=U_j-F$ and a cutoff $0\le\chi\le 1$, the perturbations $-\chi^2d_j$ of U_j and χ^2d_j of F have admissible thin traces, because the new traces are convex combinations. Add their variational inequalities to obtain

$$\int\rho\chi^2|\nabla d_j|^2\le -2\int\rho\chi d_j\nabla d_j\cdot\nabla\chi\le\tfrac12\int\rho\chi^2|\nabla d_j|^2+2\int\rho d_j^2|\nabla\chi|^2.$$

Local uniform convergence forces strong local energy convergence. It also gives weighted spherical L^2 convergence on every fixed sphere; the angular weight is integrable.

Weiss monotonicity. For an energy function on B_1 define

$$\mathcal{W}_2(F)=\int_{B_1}\rho|\nabla F|^2-2\int_{S^2}|\theta_Y|^\eta F^2\,\mathrm{d}S,\qquad \mathcal{W}(R)=\mathcal{W}_2(U_R).$$

Here and below $\|\cdot\|_{\mathcal{H}}$ denotes the norm of $\mathcal{H}=L^2(S^2,|\theta_Y|^\eta\,\mathrm{d}S)$. Put

$$E(R)=\int_{B_R}\rho|\nabla U|^2,\qquad H(R)=\int_{\partial B_R}\rho U^2.$$

We give the stationarity calculation, since it uses only local minimization. Testing the source equation by U times radial cutoffs is legitimate by Lemma 3.1; its measure term vanishes because $u\widehat{\mu} = 0$. Approximation of the indicator of B_R therefore gives, for almost every R,

$$E(R) = \int_{\partial B_R} \rho U \partial_r U, \qquad H'(R) = \frac{D-1}{R} H(R) + 2E(R). \tag{4.3}$$

For completeness, take the compactly supported radial vector field $\xi(q) = \chi(|q|)q$. Its small positive and negative domain variations preserve the thin plane, evenness, and nonnegativity of the thin trace. Differentiating the energy of $U \circ (\mathrm{Id} + t\xi)^{-1}$ at $t = 0$ gives

$$\int \rho \left((D-2)\chi |\nabla U|^2 + r\chi' \big(|\nabla U|^2 - 2|\partial_r U|^2 \big) \right) = 0.$$

The additional contribution $\eta\chi$ to the Jacobian is the homogeneity of the weight; this accounts for $D = 3 + \eta$. Approximating a ball cutoff now yields the weighted Rellich identity

$$RE'(R) = (D-2)E(R) + 2R \int_{\partial B_R} \rho |\partial_r U|^2. \tag{4.4}$$

These limiting calculations hold for almost every radius by radial integration of the local energy. No assumption on the frequency or on whether the center belongs to the free boundary is involved.

Since $\mathcal{W}(R) = R^{-D-2}E(R) - 2R^{-D-3}H(R)$, (4.3)–(4.4) give

$$R\mathcal{W}'(R) = 2\|\partial_\tau f(\tau)\|_{\mathcal{H}}^2, \qquad f(\tau, \theta) = e^{-2\tau} U(e^\tau \theta), \quad \tau = \log R. \tag{4.5}$$

The radial energy on compact annuli implies that f is locally H^1 with values in $\mathcal{H}$. Its absolutely continuous representative agrees with the continuous spherical traces of U. The same annular calculation and (4.3) show that $\mathcal{W}$ is locally absolutely continuous on $(0, \infty)$. Thus (4.5) gives a nondecreasing representative of $\mathcal{W}$ at every radius.

Quadratic growth and Caccioppoli bound $\mathcal{W}(R)$ for $R \geq 1$, so monotonicity gives a finite limit at infinity. For fixed $0 < r < t$, its dissipation on the rescaled annulus is $\mathcal{W}(R_j t) - \mathcal{W}(R_j r) \to 0$. In logarithmic radius this is the squared Hilbert norm of the radial derivative of $\varrho^{-2} U_j(\varrho\theta)$. Weak lower semicontinuity on annuli therefore gives $\partial_\varrho(\varrho^{-2} F(\varrho\theta)) = 0$. Thus F is homogeneous of degree two, with zero permitted.

Removal of the limiting source. Let $H_0(x, Y) = |x|^2 - 2Y^2/(1+\eta)$. It is an even entire weakly L_η-harmonic quadratic with thin trace $|x|^2$. For a smooth radial cutoff $\chi(|q|)$ supported away from zero, subtract the weak tests $H_0\chi$ in the equation for F and $F\chi$ in that for H_0. Energy approximation legitimizes the latter. The result is

$$2 \int |x|^2 \chi(|x|) \, \mathrm{d}\nu(x) = \int \rho (H_0 \nabla F - F \nabla H_0) \cdot \nabla \chi = 0.$$

Euler homogeneity cancels the radial integrand. Positivity shows that ν is supported at the origin. The distributional source scaling of a degree-two solution is $\nu(rE) = r^D \nu(E)$; local finiteness and $D > 0$ rule out an origin atom. Hence F is entire weakly harmonic. CSS's even harmonic Liouville theorem [3, Lemma 2.7] makes it a polynomial. Homogeneity and evenness give $F = x^T \mathsf{B} x + cY^2$; the equation yields $c = -\operatorname{tr}(\mathsf{B})/(1+\eta)$ and nonnegative thin trace yields $\mathsf{B} \geq 0$. This identifies every subsequential limit. Fix one such limit and abbreviate $Q = Q_{\mathsf{B}}$.

Strong energy and spherical convergence along the chosen sequence give $\mathcal{W}(R_j) \to \mathcal{W}_2(Q) = 0$. Monotonicity implies

$$\mathcal{W}(R) \leq 0 \quad (R > 0), \qquad \lim_{R \to \infty} \mathcal{W}(R) = 0. \tag{4.6}$$

Set $V = U - Q$. Its two phases obey Lemma 3.2 because $V = -x^T \mathsf{B} x \leq 0$ on the pressure support.

The negative-energy comparison. The external input is Carducci's negative-energy epiperimetric inequality [5, Theorem 5.2], at degree two. Precisely, if $c \in H^1(S^2, |\theta_Y|^\eta)$ is even and its homogeneous extension $z(r\theta) = r^2 c(\theta)$ has nonnegative thin trace, the normalization $\|c\|_{\mathcal{H}}^2 \leq 1$ permits a

same-boundary, even, nonnegative-thin-trace competitor ζ with

$$\mathcal{W}_2(\zeta) \le (1+\varepsilon)\mathcal{W}_2(z), \qquad \varepsilon = \varepsilon(2,1,\eta) > 0. \tag{4.7}$$

For arbitrary c, apply this statement to $c/\max\{1, \|c\|_{\mathcal{H}}\}$ and multiply the competitor back. The admissible class is closed under positive multiplication and the energy is quadratic, so the same ε works without a norm restriction. No closeness to Q is required. This is the established negative-energy inequality with factor $1+\varepsilon$.

At almost every R, annular Fubini gives $f(\log R) \in H^1(S^2, |\theta_Y|^\eta)$, so its degree-two cone

$$z_R(r\theta) = r^2 f(\log R, \theta)$$

is an energy competitor with nonnegative thin trace. If $\mathcal{W}_2(z_R) < 0$, local minimization and (4.7) give $\mathcal{W}(R) \le (1+\varepsilon)\mathcal{W}_2(z_R)$. If the cone energy is nonnegative, this same inequality follows already from (4.6). Consequently

$$\mathcal{W}(R) \le (1+\varepsilon)\mathcal{W}_2(z_R) \quad \text{for almost every } R. \tag{4.8}$$

We check the cone factor explicitly. With $T = \|\nabla_\theta f\|^2_{\mathcal{H}}$, $H_1 = \|f\|^2_{\mathcal{H}}$, $g = \partial_\tau f$, and $E_1 = \int_{B_1} \rho |\nabla U_R|^2$, polar integration gives

$$\mathcal{W}_2(z_R) = \frac{T - 2DH_1}{D+2}.$$

Since $\partial_r U_R = 2f + g$ on S^2, differentiation before using stationarity gives

$$\begin{aligned} \partial_\tau \mathcal{W} &= T + 4H_1 + \|g\|^2_{\mathcal{H}} - (D+2)E_1 \\ &= (D+2)\big(\mathcal{W}_2(z_R) - \mathcal{W}(R)\big) + \|g\|^2_{\mathcal{H}}. \end{aligned}$$

Combining with (4.5) proves

$$\mathcal{W}_2(z_R) = \mathcal{W}(R) + \frac{R\mathcal{W}'(R)}{2(D+2)}. \tag{4.9}$$

Substitution in (4.8) gives

$$R\mathcal{W}'(R) \ge -\varkappa \mathcal{W}(R), \qquad \varkappa = \frac{2(D+2)\varepsilon}{1+\varepsilon} > 0.$$

Thus $e(R) = -\mathcal{W}(R) \ge 0$ satisfies $Re'(R) \le -\varkappa e(R)$, and absolute continuity yields

$$e(R) \le e(R_0)(R/R_0)^{-\varkappa}, \qquad R \ge R_0 > 0. \tag{4.10}$$

Decreasing ε preserves the negative-cone inequality; fix it small enough that $\delta = \varkappa/2 \in (0,1)$.

Finite length in logarithmic radius. Integrate (4.5) to the already established zero limit at infinity. For $\tau \ge \log R_0$,

$$\int_\tau^\infty \|f'(t)\|^2_{\mathcal{H}} \, \mathrm{d}t = \frac{1}{2} e(e^\tau) \le Ce^{-\varkappa\tau}. \tag{4.11}$$

For every integer $j \ge 0$, Cauchy–Schwarz on the interval of length one gives

$$\int_{\tau+j}^{\tau+j+1} \|f'(t)\|_{\mathcal{H}} \, \mathrm{d}t \le Ce^{-\varkappa(\tau+j)/2}.$$

Summation proves finite total length and a unique limit $f_\infty \in \mathcal{H}$, with

$$\|f(\tau) - f_\infty\|_{\mathcal{H}} \le Ce^{-\delta\tau}.$$

The sum begins at arbitrary real τ, so the estimate controls the whole family. Uniform convergence on S^2 along the chosen R_j identifies $f_\infty = Q|_{S^2}$. In particular,

$$\|R^{-2}U(R\,\cdot) - Q\|_{\mathcal{H}} \le CR^{-\delta}. \tag{4.12}$$

If $e(R_0) = 0$, (4.11) instead makes f constant for $R \geq R_0$; the same subsequence identifies that constant with $Q|_{S^2}$. This proves exterior equality $U = Q$ in this case. No interior unique-continuation assertion is needed.

The bulk and energy rates. By homogeneity of Q, (4.12) reads

$$\int_{S^2} |V(r\theta)|^2 |\theta_Y|^\eta \, \mathrm{d}S \leq C r^{4-2\delta} \quad (r \geq R_0).$$

Polar integration, separating the fixed inner ball, gives

$$\int_{B_{2R}} \rho V^2 \leq C_0 + C \int_{R_0}^{2R} r^{D+3-2\delta} \, \mathrm{d}r \leq C(1 + R^{D+4-2\delta}).$$

Equation (3.3) proves (4.1). It also gives, on every fixed B_{2L}, $\|V_R\|_{L^\infty} = O(R^{-\delta})$ and hence the same weighted L^2 bound. Apply (3.2) to V_R with a cutoff equal to one on B_L and supported in B_{2L} to obtain its gradient bound $O(R^{-\delta})$. Increasing constants for bounded R proves (4.2). □

Remark 4.2. The competitor inequality is Carducci's established negative-energy result. Its use here does not require a prescribed rank or a nonzero target: amplitude normalization divides by the current sphere norm only through $\max\{1, \|f\|_{\mathcal{H}}\}$. The strict remainder power $2 - \delta$ need not lie below $2s$. Rank-one ordinary-tail entry will use a separate intrinsic argument.

Lemma 4.3 (Pressure-free states)**.** *An intrinsic solution has $\widehat{\mu} = 0$ if and only if it belongs to the polynomial family in Section 2.1.*

Proof. The entire weak equation, evenness, and quadratic growth permit CSS harmonic polynomial Liouville. The even harmonic polynomial has exactly the form $Q_{\mathsf{B}} + \boldsymbol{\ell} \cdot x + c$. Nonnegative trace forces $\mathsf{B} \geq 0$ and $\boldsymbol{\ell} \in \operatorname{range} \mathsf{B}$ by testing lines in the kernel and negative eigendirections. Minimization on the range gives $c \geq \boldsymbol{\ell}^T \mathsf{B}^+ \boldsymbol{\ell}/4$. Conversely these conditions give nonnegative trace, and direct weighted differentiation gives zero source and local energy. The literal polynomial coefficients are unique. □

5. The zero quadratic branch

The strict power remainder in Proposition 4.1 is sufficient to classify the case $\mathsf{B} = 0$. The trace classification will follow from Caffarelli–Ros-Oton–Serra [2, Theorem 4.1]. We first establish its extension interface: only tangential differences, and not necessarily U itself, are ordinary Poisson extensions.

Lemma 5.1 (Canonical extension below degree $2s$)**.** *Let F be continuous on $\overline{\mathbb{R}^3_+}$, locally in $H^1(\overline{\mathbb{R}^3_+}, Y^\eta)$, and weakly L_η-harmonic in $Y > 0$. If*

$$|F(x, Y)| \leq C(1 + |(x, Y)|^\vartheta), \qquad 0 \leq \vartheta < 2s,$$

then, writing $f(x) = F(x, 0)$, one has $F = \mathcal{E}_s f$ in $Y \geq 0$.

Proof. The weighted trace theorem gives $f \in H^s_{\mathrm{loc}}(\mathbb{R}^2)$. Its growth gives $f \in \mathcal{T}_s$. The Poisson integral is therefore absolute, continuous up to the plane, and has trace f. The kernel has mass one and finite ϑ-moment, since

$$\int_{\mathbb{R}^2} \frac{|z|^\vartheta}{(1 + |z|^2)^{1+s}} \, \mathrm{d}z < \infty.$$

After scaling the kernel and translating its center, this gives

$$|\mathcal{E}_s f(x, Y)| \leq C'(1 + |x|^\vartheta + Y^\vartheta).$$

For $\vartheta = 0$ the same assertion follows directly from its mass.

For completeness, local energy here needs only the already established local H^s trace and the absolute tail. On a fixed compact footprint, split $f = \chi f + (1-\chi)f$, where χ is compactly supported and equals one on a larger footprint. The extension of $\chi f \in H^s(\mathbb{R}^2)$ has finite weighted energy. For the separated part, kernel differentiation gives, locally as $Y \downarrow 0$,

$$\mathcal{E}_s((1-\chi)f) = O(Y^{2s}), \quad \nabla_x \mathcal{E}_s((1-\chi)f) = O(Y^{2s}), \quad \partial_Y \mathcal{E}_s((1-\chi)f) = O(Y^{2s-1}).$$

The normal energy is integrable because $\eta + 4s - 2 = 2s - 1 > -1$. This is the local argument in Lemma 3.6; it does not require a squared tail. Thus $F - \mathcal{E}_s f$ is a harmonic local-energy function with zero trace and growth strictly below $2s$. Lemma 3.3, applied to it and its negative, makes it zero. Its rim-excision comparison is applicable because both functions are continuous up to the plane. The strict growth inequality is essential: the zero-trace function Y^{2s} is not excluded at the endpoint. □

Lemma 5.2 (Regular extension and its intrinsic source)**.** *The profile F_s in the regular family 2.3 is continuous, even in Y, and locally in the weighted energy space. Writing $T = e \cdot x - h_{\mathrm{reg}}$ and $r = (T^2 + Y^2)^{1/2}$, its upper normal flux and negative conormal are*

$$\begin{aligned} \lim_{Y \downarrow 0} Y^\eta \partial_Y F_s(T,Y) &= -r_s(-T)_+^{1-s}, \\ -\lim_{Y \downarrow 0} Y^\eta \partial_Y F_s(T,Y) &= r_s(-T)_+^{1-s}, \qquad r_s = \frac{2^{1-2s} s(1+s)}{1-s}. \end{aligned} \tag{5.1}$$

Consequently, for every $\alpha > 0$,

$$L_\eta\big[\alpha F_s(e \cdot x - h_{\mathrm{reg}}, Y)\big] = -2\iota_*\big[\alpha r_s (h_{\mathrm{reg}} - e \cdot x)_+^{1-s}\, \mathrm{d}x\big]. \tag{5.2}$$

These functions satisfy the intrinsic hypotheses and have full quadratic blowdown zero.

Proof. The defining formula gives the trace T_+^{1+s} and homogeneity $1+s$. For $Y > 0$, direct differentiation yields

$$\partial_T F_s(T,Y) = (1+s)\left(\frac{r+T}{2}\right)^s, \tag{5.3}$$

$$\partial_Y F_s(T,Y) = -\frac{s(1+s)}{2^s(1-s)} Y (r+T)^{s-1}. \tag{5.4}$$

Indeed, differentiating the first identity once more gives $\partial_{TT} F_s = s(1+s)(r+T)^s/(2^s r)$; differentiating the second gives

$$\partial_{YY} F_s + \frac{\eta}{Y}\partial_Y F_s = -\frac{s(1+s)}{2^s}\frac{(r+T)^s}{r}.$$

Thus $L_\eta F_s = 0$ in the upper halfspace. The identity $Y^2 = (r-T)(r+T)$ also rewrites the normal derivative and flux as

$$\begin{aligned} \partial_Y F_s &= -\frac{s(1+s)}{2^s(1-s)} Y^{2s-1}(r-T)^{1-s}, \\ Y^\eta \partial_Y F_s &= -\frac{s(1+s)}{2^s(1-s)}(r-T)^{1-s}. \end{aligned}$$

The last expression converges uniformly on compact thin sets to (5.1), including at $T = 0$.

On bounded sets the tangential derivative is bounded and the normal derivative is $O(Y^{2s-1})$. Their weighted squares are integrable, since $\eta > -1$ and $\eta + 4s - 2 = 2s - 1 > -1$. Continuity at the origin follows from homogeneity; the same formulas give continuity along the rest of the plane. Integration by parts above $Y = \varepsilon$, followed by the compact-uniform flux limit and even reflection, proves (5.2). In particular there is no additional source on the boundary line $T = 0$.

The trace is nonnegative, its zero set is $\{T \le 0\}$, and the nonnegative source density is supported there. Local energy and this locally bounded density give the capacity-compatible

source pairing, hence the local variational inequality by Lemma 3.1. Homogeneity and translation give $|F_s(e\cdot x-h_{\mathrm{reg}},Y)|\le C(1+|(x,Y)|^{1+s})$. They also give convergence of the quadratic rescalings to zero locally uniformly and strongly in local weighted energy: the amplitude factor is $R^{s-1}\to 0$, and the remaining profile is translated in T by h_{reg}/R. Tangential translations are continuous in the local weighted energy space. This verifies the claimed intrinsic converse without applying $\mathcal{E}_s$ to the super-$2s$ trace T_+^{1+s}. □

Proposition 5.3 (Classification when the quadratic matrix is zero)**.** *Let U satisfy the intrinsic hypotheses of Proposition 4.1, and let its unique quadratic matrix be $\mathsf{B}=0$. Exactly one of the following occurs:*

1. *$U\equiv c$ for $c\ge 0$, with $\widehat{\mu}=0$;*
2. *$U=-c_0|Y|^{2s}$ for $c_0>0$, with $\mathrm{d}\widehat{\mu}=2sc_0\,\mathrm{d}x$;*
3. *$U(x,Y)=\alpha F_s(e\cdot x-h_{\mathrm{reg}},Y)$ for $\alpha>0$, $e\in S^1$, and $h_{\mathrm{reg}}\in\mathbb{R}$, with*

$$\mathrm{d}\widehat{\mu}=\alpha r_s(h_{\mathrm{reg}}-e\cdot x)_+^{1-s}\,\mathrm{d}x.$$

The parameters in each alternative are unique. Every displayed choice is an intrinsic solution with full quadratic blowdown zero. The second and third alternatives are precisely the catalogue's full-contact and regular families.

Proof. Convexity and growth of differences. Proposition 4.1, reducing its power gain if necessary, gives

$$\sup_{B_R}|U|\le C(1+R^p),\qquad p=2-\delta\in(1,2),\quad R\ge 1. \tag{5.5}$$

Local minimization is already known from Lemma 3.1. Apply the local semiconvexity and tangential regularity estimates of Caffarelli–Salsa–Silvestre [3, Lemma 4.1 and Proposition 4.3] to the uniformly bounded functions $R^{-p}U(R\cdot)$ on a fixed larger ball. For each thin unit vector e this yields

$$D_{ee}u\ge -CR^{p-2}\quad\text{in } B_{R/2}\cap\{Y=0\}.$$

Exhaustion gives $D^2u\ge 0$ in distributions, hence convexity. The same local estimates give $u\in C^{1,\gamma}_{\mathrm{loc}}$ for every $0<\gamma<s$, continuous tangential bulk derivatives, and

$$\sup_{B_R}|\nabla_x U|\le C(1+R^{\beta_0}),\qquad \beta_0=p-1\in(0,1). \tag{5.6}$$

Here and below constants may change when a fixed thin translation is made. In particular $K=\{u=0\}$ is closed and convex.

For $h\in\mathbb{R}^2$ set

$$D_hU(x,Y)=U(x,Y)-U(x-h,Y),\qquad d_hu(x)=u(x)-u(x-h).$$

The fundamental theorem of calculus along the segment of length $|h|$, using (5.6), shows that D_hU has $O_h(1+|(x,Y)|^{\beta_0})$ growth. Moreover,

$$\frac{|D_hU(x,Y)|}{|h|}\le C(1+|(x,Y)|^{\beta_0}),\qquad 0<|h|\le 1. \tag{5.7}$$

Since $\beta_0<1<2s$, all these traces and the derivatives of u belong to $\mathcal{T}_s$. Each difference is already a local-energy function, being the difference of two translates of U. It is harmonic above the plane. Lemma 5.1 therefore proves

$$D_hU=\mathcal{E}_s(d_hu)\qquad\text{for every } h\in\mathbb{R}^2. \tag{5.8}$$

No ordinary-tail assertion about u itself has been used.

The exact nonlocal equations. Use the generator sign

$$\mathcal{L}=-\mathcal{A}_s=-(-\Delta)^s,\qquad \mathcal{L}f(x)=\frac{c_{2,s}}{2}\int_{\mathbb{R}^2}\frac{f(x+z)+f(x-z)-2f(x)}{|z|^{2+2s}}\,\mathrm{d}z \tag{5.9}$$

when the integral is defined. This isotropic generator belongs to the elliptic stable class used in [2]. If $\widehat{\mu}_h(E) = \widehat{\mu}(E-h)$, then $\widehat{\mu}_h$ is the source measure of $U(x-h,Y)$. Comparing the source of (5.8) with the conormal identity in Lemma 3.6 gives, on $\mathbb{R}^2$ in distributions,

$$e_s\mathcal{A}_s(d_hu) = \widehat{\mu} - \widehat{\mu}_h, \qquad \mathcal{L}(d_hu) = e_s^{-1}(-\widehat{\mu} + \widehat{\mu}_h). \tag{5.10}$$

Consequently, on $O = \mathbb{R}^2 \setminus K = \{u > 0\}$,

$$\mathcal{L}\big(u - u(\,\cdot - h)\big) \geq 0 \qquad \text{for every } h \in \mathbb{R}^2. \tag{5.11}$$

In particular the sign is that of the negative fractional Laplacian.

Fix a unit vector e and $\phi \in C_c^\infty(O)$. The compact support of ϕ has positive distance from K, so both source measures in (5.10) vanish near it when $h = \tau e$ and $|\tau|$ is sufficiently small. The quotients $d_{\tau e}u/\tau$ converge locally uniformly to $\partial_e u$. By (5.7) they have a common absolute tail majorant of degree $\beta_0 < 2s$. Since $\mathcal{L}\phi$ is bounded locally and is $O(|x|^{-2-2s})$ at infinity, dominated convergence in the compact-test pairing gives

$$\mathcal{L}(\partial_e u) = 0 \quad \text{in } O \qquad (e \in S^1). \tag{5.12}$$

This passage defines neither $\mathcal{A}_s u$ nor $\mathcal{L}u$.

We check the classical interpretation needed for the classification theorem. Choose once and for all

$$2s - 1 < \gamma < s. \tag{5.13}$$

For fixed h, the local $C^{1,\gamma}$ estimate for d_hu bounds the symmetric numerator in (5.9) by $C|z|^{1+\gamma}$, uniformly for x on compact sets. Its local radial integral is bounded by

$$C\int_0^1 r^{\gamma-2s}\,\mathrm{d}r < \infty.$$

The absolute ordinary tail controls the other part of the integral. These estimates and dominated convergence show that $\mathcal{L}(d_hu)$ is a continuous classical function. Symmetric kernel truncation and Fubini identify this function with its distributional value. A continuous function representing a positive distribution is pointwise nonnegative, proving (5.11) classically.

For $v = \partial_e u$, local smoothness on O follows directly from the source-free extension equation. On a ball crossing the plane whose thin footprint lies in O, U satisfies the source-free weighted equation across the plane. Tangential difference quotients and Caccioppoli may be iterated there, and weighted local estimates give tangential derivatives of every order on smaller balls. Thus v has a smooth trace locally in O. Together with its absolute tail from (5.6), this makes its classical symmetric integral continuous on O. The same truncated kernel argument identifies it with (5.12). Both required equations therefore hold classically, and hence in the viscosity sense as well.

The proper nonempty contact case. Suppose K is nonempty and proper, and choose $x_0 \in \partial K$. Translate this point to the origin. Choose

$$\max\{0, \beta_0 - s\} < \alpha_0 < 1 - s. \tag{5.14}$$

The interval is nonempty because $\beta_0 < 1$. Moreover $0 < \alpha_0 < 1 - s < s$, $\beta_0 < s + \alpha_0$, and $1 + s + \alpha_0 < 2$. There is a finite $C_* > 0$ such that $\widetilde{u}(x) = C_*^{-1}u(x + x_0)$ satisfies

$$\|\nabla\widetilde{u}\|_{L^\infty(B_R)} \leq R^{s+\alpha_0} \quad (R \geq 1).$$

Indeed, (5.6) after translation is at most $C_{x_0}R^{\beta_0}$ for $R \geq 1$, so any $C_* \geq C_{x_0}$ suffices. Positive amplitude scaling preserves every equation and inequality just proved.

Thus $\widetilde{u}$ is C^1, nonnegative and convex, vanishes on the closed convex set $K - x_0$ with $0 \in \partial(K - x_0)$, and satisfies both its gradient equation and every translated-difference inequality outside that set. These are precisely the hypotheses of [2, Theorem 4.1], with the exponent (5.14)

and the isotropic generator (5.9). Its nonzero alternative applies, since K is proper. Undoing the translation and amplitude normalization gives

$$u(x) = \alpha(e \cdot x - h_{\mathrm{reg}})_+^{1+s}, \qquad \alpha > 0, \quad e \in S^1, \quad h_{\mathrm{reg}} = e \cdot x_0. \tag{5.15}$$

In particular the contact set is the complementary closed halfspace. No interior assumption on the initial convex set was needed.

Removal of the vertical ambiguity. Put $F(x, Y) = \alpha F_s(e \cdot x - h_{\mathrm{reg}}, Y)$. By (5.3), every $D_h F$ has growth $O_h(1 + |(x, Y)|^s)$, strictly below degree $2s$. It is a continuous harmonic local-energy function above the plane, by Lemma 5.2. Its trace equals $d_h u$. Lemma 5.1 applied separately to $D_h U$ and $D_h F$ therefore gives

$$D_h U = D_h F \qquad (h \in \mathbb{R}^2).$$

Consequently $U - F$ is independent of x. Its upper-halfspace equation is $(Y^\eta (U - F)')' = 0$, and its trace is zero, so

$$U - F = c|Y|^{2s}$$

after even reflection. This is also the only possible subquadratic zero-trace ambiguity in Lemma 3.5. Both U and F have zero source on the nonempty halfspace $e \cdot x > h_{\mathrm{reg}}$, whereas

$$L_\eta(c|Y|^{2s}) = 4sc\, \iota_*(\,\mathrm{d}x). \tag{5.16}$$

Testing on any thin ball in that halfspace forces $c = 0$. The source in the third alternative now follows from (5.2). In particular the vertical term was removed only after canonicality of differences and the positive halfspace had been established.

The exceptional contact sets. If $K = \mathbb{R}^2$, every $D_h U$ has zero trace. Its sub-$2s$ growth and Lemma 5.1 give $D_h U = 0$ for all h. Hence U is independent of x and, by its upper-halfspace equation and zero trace, is $c|Y|^{2s}$. Equation (5.16) and the nonnegative intrinsic pressure force $c \le 0$. Writing $c = -c_0$ gives $\mathrm{d}\widehat{\mu} = 2sc_0\,\mathrm{d}x$. The case $c_0 = 0$ is assigned to the constant alternative, and only $c_0 > 0$ to the full-contact family 2.2.

If $K = \varnothing$, the source vanishes throughout $\mathbb{R}^3$. The strict subquadratic bound (5.5) and Lemma 3.4 give $U = \boldsymbol{\ell} \cdot x + c$. Global nonnegativity of its trace forces $\boldsymbol{\ell} = 0$, and empty contact forces $c > 0$. Neither exceptional case has been submitted to a theorem requiring a boundary point of K.

Converses and uniqueness of parameters. The regular converse was proved in Lemma 5.2. Nonnegative constants have zero source. The full-contact functions are continuous and even; their normal weighted energy is a constant multiple of $\int_0^1 Y^{2s-1}\,\mathrm{d}Y < \infty$, and (5.16) gives exactly their stated source. Their zero trace gives complementarity and the local variational inequality. Their degree $2s < 2$ gives quadratic blowdown zero, both locally uniformly and strongly in local weighted energy; the same assertion for constants is immediate. Thus all three alternatives satisfy the intrinsic hypotheses.

For a regular solution the positivity halfspace determines its oriented normal e and intercept h_{reg} uniquely. Its trace at any point of this halfspace then determines α. The orientation points into positivity; changing e to $-e$ does not preserve the solution. Translation tangent to the boundary introduces no parameter. The coefficient c_0 of a full-contact solution is recovered from its value at $Y = 1$, and a constant is its own unique parameter. The strict positivity conditions on α and c_0 separate the three alternatives. □

6. Compact pressure and the positive-definite branch

The positive-definite quadratic limit makes the contact set compact. We first identify the potential of an arbitrary compact pressure measure; its literal continuous trace will justify the energy comparison that identifies an ellipse. Throughout this section,

$$\tfrac{1}{2} < s < 1, \qquad t = 1 - s \in (0, \tfrac{1}{2}), \qquad \eta = 1 - 2s = 2t - 1, \qquad e_s = 2\pi k_s.$$

The constants k_s and e_s retain their respective Riesz and conormal normalizations from the catalogue. In particular, the fractional pressure below is $\nu = e_s^{-1}\widehat{\mu}$.

6.1. **A compact-source representation before energy pairings.**

Lemma 6.1 (Compact-source representation)**.** *Let U satisfy the intrinsic hypotheses, and let $\mathsf{B} \geq 0$ be symmetric. Suppose*

$$V = U - Q_{\mathsf{B}}, \qquad \sup_{B_R} |V| = o(R^2), \qquad \operatorname{supp} \widehat{\mu} \text{ is compact.}$$

Then there are unique $\boldsymbol{\alpha} \in \mathbb{R}^2$ and $c_0 \in \mathbb{R}$ such that

$$\begin{aligned} U(x,Y) &= Q_{\mathsf{B}}(x,Y) + \boldsymbol{\alpha} \cdot x + c_0 + \Phi(x,Y), \\ \Phi(x,Y) &= \frac{1}{2\pi} \int_{\mathbb{R}^2} (|x-w|^2 + Y^2)^{-t} \, \mathrm{d}\widehat{\mu}(w). \end{aligned} \tag{6.1}$$

The integral is finite at every point, is continuous, and belongs locally to $H^1(\mathbb{R}^3, |Y|^\eta)$. Its trace is the literal potential

$$v(x) = \Phi(x,0) = K_s * \nu(x) = k_s \int_{\mathbb{R}^2} |x-w|^{-2t} \, \mathrm{d}\nu(w) < \infty \qquad (x \in \mathbb{R}^2). \tag{6.2}$$

Both v and Φ are bounded, and $\Phi(x,Y) = O(|(x,Y)|^{-2t})$ as $|(x,Y)| \to \infty$, uniformly in direction. Moreover

$$\mathcal{A}_s v = \nu, \qquad \Phi = \mathcal{E}_s v, \qquad V = \mathcal{E}_s P, \qquad P(x) = u(x) - x^{\mathsf{T}} \mathsf{B} x = \boldsymbol{\alpha} \cdot x + c_0 + v(x).$$

In particular, $|P(x)| \leq C(1 + |x|)$ and P has an absolute ordinary tail. No energy or density assumption on $\widehat{\mu}$ is required.

Proof. Compact support and local finiteness give $\widehat{\mu}(\mathbb{R}^2) < \infty$. In bulk dimension three,

$$S_\eta := \int_{S^2} |\omega_Y|^\eta \, \mathrm{d}\omega = \frac{4\pi}{1+\eta}, \qquad \Gamma_\eta(x,Y) = \frac{(|x|^2 + Y^2)^{-t}}{4\pi}.$$

The function Γ_η is L_η-harmonic away from its pole. Its weighted outward flux across a sphere is $-2tS_\eta/(4\pi) = -1$, since $2t = 1 + \eta$. Consequently $L_\eta \Gamma_\eta = -\delta_0$. Tangential translations preserve this identity. The potential $\Phi = 2\Gamma_\eta * (\iota_* \widehat{\mu})$ is initially defined as a nonnegative measurable function. Both the kernel and its gradient are locally integrable with the weight: their radial integrability exponents are respectively 1 and 0. Fubini therefore gives the distributional source identity

$$\int_{\mathbb{R}^3} |Y|^\eta \nabla\Phi \cdot \nabla\varphi = 2 \int_{\mathbb{R}^2} \varphi(x,0) \, \mathrm{d}\widehat{\mu}(x) \qquad (\varphi \in C_c^\infty(\mathbb{R}^3)).$$

This only uses weighted local $W^{1,1}$ integrability of the potential; weighted square-integrability of its gradient has not been assumed.

Choose an even nonnegative $\varrho \in C_c^\infty(\mathbb{R}^2)$ with integral one and put $\varrho_\varepsilon(x) = \varepsilon^{-2}\varrho(x/\varepsilon)$. Mollify only tangentially:

$$V_\varepsilon = \varrho_\varepsilon *_x V, \qquad \widehat{\mu}_\varepsilon = \varrho_\varepsilon * \widehat{\mu}, \qquad \Phi_\varepsilon = 2\Gamma_\eta * (\iota_* \widehat{\mu}_\varepsilon).$$

The density f_ε of $\widehat{\mu}_\varepsilon$ is smooth and compactly supported. The corresponding potential is continuous and locally bounded, because $2t < 2$. Its tangential derivatives are bounded on compact sets, since $2t + 1 < 2$. For $0 < Y \leq 1$, direct differentiation and polar integration give

$$\begin{aligned} |\partial_Y \Phi_\varepsilon(x,Y)| &\leq C \|f_\varepsilon\|_\infty Y \int_{\mathbb{R}^2} (|z|^2 + Y^2)^{-t-1} \, \mathrm{d}z \\ &\leq C_\varepsilon Y^{1-2t} = C_\varepsilon Y^{-\eta}. \end{aligned} \tag{6.3}$$

Thus the weighted tangential and normal energy densities are bounded respectively by $C_\varepsilon Y^\eta$ and $C_\varepsilon Y^{-\eta}$, both integrable at zero. It follows that $\Phi_\varepsilon \in H^1_{\mathrm{loc}}(|Y|^\eta)$. Its weak source is $-2\iota_*\widehat{\mu}_\varepsilon$, the same as that of V_ε.

The difference $h_\varepsilon = V_\varepsilon - \Phi_\varepsilon$ is therefore entire, even, and weakly L_η-harmonic in the local energy class. For fixed ε, $V_\varepsilon = o(R^2)$ on B_R and Φ_ε decays outside a ball containing its sources. Lemma 3.4 gives

$$V_\varepsilon(x,Y) - \Phi_\varepsilon(x,Y) = \boldsymbol{\alpha}_\varepsilon \cdot x + c_\varepsilon. \tag{6.4}$$

This is an entire weak harmonic equation, so no even zero-trace term $C|Y|^{2s}$ can occur.

Choose three noncollinear thin points outside a fixed ball containing $\operatorname{supp}\widehat{\mu}$ and every sufficiently small mollified support. At these points $V_\varepsilon \to V$ by continuity, and $\Phi_\varepsilon \to \Phi$ because their kernels are continuous a positive distance from the sources. The fixed invertible three-point evaluation matrix in (6.4) proves

$$(\boldsymbol{\alpha}_\varepsilon, c_\varepsilon) \longrightarrow (\boldsymbol{\alpha}, c_0).$$

Tangential translation is continuous in weighted local L^1, since the weight is independent of the tangential variables. Applied to the locally integrable kernel, and then integrated over the compact finite measure $\widehat{\mu}$, this gives $\Phi_\varepsilon \to \Phi$ in $L^1_{\mathrm{loc}}(|Y|^\eta)$. For example, on a fixed ball the translated kernel norms are uniformly bounded as the pole ranges over the fixed compact source set; a larger ball supplies a common integrable bound for the translation argument. Passing to the limit in (6.4) identifies Φ almost everywhere with $V - \boldsymbol{\alpha}\cdot x - c_0$, a continuous local-energy function. Off the thin plane the literal integral is smooth, so it agrees there pointwise with this representative.

For each fixed x, the nonnegative kernel in (6.1) increases as $Y \downarrow 0$. Monotone convergence, followed by continuity of the representative, therefore proves

$$\frac{1}{2\pi}\int |x-w|^{-2t}\,\mathrm{d}\widehat{\mu}(w) = \lim_{Y\downarrow 0}\Phi(x,Y) = V(x,0) - \boldsymbol{\alpha}\cdot x - c_0 < \infty.$$

This holds at every x, including every pressure-support point. It proves (6.2) before any measure pairing is used. The integral decays like $|(x,Y)|^{-2t}$ outside a large bulk ball; continuity on the remaining compact ball proves global boundedness. The decay also proves uniqueness of the affine coefficients in (6.1).

With our Fourier convention, $\widehat{K_s}(\xi) = |\xi|^{-2s}$. The Riesz inverse identity for a compact finite measure gives $\mathcal{A}_s v = \nu$ in ordinary distributions. This is also obtained by testing the kernel identity against a compactly supported smooth function; the bounded potential has the required absolute tail. Lemma 3.6 yields a continuous local-energy extension $\mathcal{E}_s v$ with entire source $-2e_s\iota_*\nu = -2\iota_*\widehat{\mu}$. Both Φ and $\mathcal{E}_s v$ are bounded. Their difference is entire, even, weakly harmonic, and has zero trace; another application of Lemma 3.4 makes it a bounded affine function with zero trace, hence zero. Finally, the Poisson kernel has a finite first moment because $2s > 1$, and preserves affine functions. Thus $\mathcal{E}_s P = \boldsymbol{\alpha}\cdot x + c_0 + \Phi = V$. The displayed linear growth gives the absolute ordinary tail of P. □

Lemma 6.2 (Energy of the initial measure)**.** *Let ν be a nonnegative compact finite measure whose literal potential $v = K_s * \nu$ is continuous. Then*

$$\mathfrak{E}(\nu) := \frac{1}{(2\pi)^2}\int_{\mathbb{R}^2} |\xi|^{-2s}|\widehat{\nu}(\xi)|^2\,\mathrm{d}\xi = \int_{\mathbb{R}^2} v\,\mathrm{d}\nu < \infty. \tag{6.5}$$

For two such measures ν_1, ν_2, their difference $\sigma = \nu_1 - \nu_2$ satisfies

$$\mathfrak{E}(\sigma) = \int (v_1 - v_2)\,\mathrm{d}\nu_1 - \int (v_1 - v_2)\,\mathrm{d}\nu_2 \geq 0, \tag{6.6}$$

and equality holds only if $\nu_1 = \nu_2$.

Proof. Use the preceding nonnegative even approximate identity and write $\nu_\varepsilon = \varrho_\varepsilon * \nu$. Associativity, justified by nonnegative integration, gives $K_s * \nu_\varepsilon = v * \varrho_\varepsilon$. The potentials converge uniformly on a common compact neighborhood of all supports, and ν_ε converges weakly to ν with unchanged mass. Hence

$$\int (v * \varrho_\varepsilon)\, \mathrm{d}\nu_\varepsilon \longrightarrow \int v\, \mathrm{d}\nu < \infty. \tag{6.7}$$

For these smooth compact densities the Riesz energy formula is

$$\int (v * \varrho_\varepsilon)\, \mathrm{d}\nu_\varepsilon = \frac{1}{(2\pi)^2} \int |\xi|^{-2s} |\widehat{\nu}(\xi)|^2 |\widehat{\varrho}(\varepsilon\xi)|^2\, \mathrm{d}\xi.$$

One direct verification uses

$$K_s(x) = \frac{1}{\Gamma(s)} \int_0^\infty r^{s-1} \frac{e^{-|x|^2/(4r)}}{4\pi r}\, \mathrm{d}r$$

and the Fourier transform of the heat kernel. Tonelli and Parseval then give the displayed formula with exactly the factor $(2\pi)^{-2}$. Fatou's lemma first proves finiteness of the unsmoothed Fourier integral in (6.5). Only after this conclusion do we use $|\widehat{\varrho}| \le 1$ and dominated convergence to identify its value with (6.7).

The same argument applies to $\nu_1 + \nu_2$, whose continuous literal potential is $v_1 + v_2$. Tonelli identifies the two cross-pairings,

$$\int v_1\, \mathrm{d}\nu_2 = \iint K_s(x-y)\, \mathrm{d}\nu_1(y)\, \mathrm{d}\nu_2(x) = \int v_2\, \mathrm{d}\nu_1.$$

They are finite by continuity on compact supports. Polarization of the three finite Fourier energies gives (6.6), with $\mathfrak{E}(\sigma) = (2\pi)^{-2} \int |\xi|^{-2s} |\widehat{\sigma}(\xi)|^2\, \mathrm{d}\xi$. If it vanishes, the continuous Fourier transform of the finite signed measure σ vanishes almost everywhere, hence everywhere. Uniqueness of the Fourier transform gives $\sigma = 0$. □

6.2. **The angular ellipse identity and its global gap.** For the remainder of this section all circle integrals use ordinary arc length. Recall the positive constants

$$\begin{gathered} J = 2B(\tfrac{1}{2} - t, \tfrac{1}{2}), \qquad B_1 = B(\tfrac{1}{2}, t+1), \qquad B_2 = B(\tfrac{1}{2} - t, t + \tfrac{3}{2}), \\ \mathfrak{c}_s^{\mathrm{ell}} = \frac{k_s (t+1) B_1 B_2}{\pi J}. \end{gathered} \tag{6.8}$$

Here $B(\cdot,\cdot)$ is the beta function; it is unrelated to the quadratic matrix B.

Lemma 6.3 (Angular identity and exact contact)**.** *For $\mathsf{H} = \mathsf{H}^{\mathsf{T}} > 0$ and $m_{\mathrm{ell}} > 0$, set*

$$E_{\mathsf{H}} = \{x : x^{\mathsf{T}} \mathsf{H}^{-1} x \le 1\}, \qquad h(\theta) = (\theta^{\mathsf{T}} \mathsf{H} \theta)^{1/2}, \qquad \rho_{\mathsf{H}}(x) = \frac{m_{\mathrm{ell}} (t+1)}{\pi \sqrt{\det \mathsf{H}}} (1 - x^{\mathsf{T}} \mathsf{H}^{-1} x)_+^t.$$

Then $\int \rho_{\mathsf{H}} = m_{\mathrm{ell}}$. Define

$$F(v) = \int_{-1}^1 |v - r|^{-2t} (1 - r^2)^{t + 1/2}\, \mathrm{d}r, \qquad Q(v) = B_2 (1 - 2tv^2).$$

The interval identity and its strict exterior version are

$$F(v) = Q(v) \quad (|v| \le 1), \qquad F(v) > Q(v) \quad (|v| > 1), \qquad F \in C^1(\mathbb{R}). \tag{6.9}$$

The potential $v_{\mathsf{H}} = K_s * \rho_{\mathsf{H}}$ has the representation

$$v_{\mathsf{H}}(x) = \frac{k_s m_{\mathrm{ell}} (t+1) B_1}{\pi J} \int_{S^1} h(\theta)^{-2t} F\!\left(\frac{x \cdot \theta}{h(\theta)}\right)\, \mathrm{d}\theta. \tag{6.10}$$

In particular, with

$$\begin{aligned}\ell_{\mathsf{H}} &= \mathfrak{c}_s^{\mathrm{ell}} m_{\mathrm{ell}} \int_{S^1} (\theta^{\mathsf{T}} \mathsf{H} \theta)^{-t} \, \mathrm{d}\theta, \\ \mathsf{B}_{\mathsf{H}} &= 2t \mathfrak{c}_s^{\mathrm{ell}} m_{\mathrm{ell}} \int_{S^1} \theta\theta^{\mathsf{T}} (\theta^{\mathsf{T}} \mathsf{H} \theta)^{-t-1} \, \mathrm{d}\theta,\end{aligned} \tag{6.11}$$

one has

$$v_{\mathsf{H}}(x) + x^{\mathsf{T}} \mathsf{B}_{\mathsf{H}} x - \ell_{\mathsf{H}} \geq 0, \qquad \{v_{\mathsf{H}} + x^{\mathsf{T}} \mathsf{B}_{\mathsf{H}} x - \ell_{\mathsf{H}} = 0\} = E_{\mathsf{H}}. \tag{6.12}$$

Proof. The equality in (6.9) is the known one-dimensional identity of Carrillo–Huang [6, equation (16)], with their exponent $2t \in (0,1)$ and radius one. Its constant is

$$B_2 = \frac{\Gamma(1/2 - t)\Gamma(t + 3/2)}{\Gamma(2)} = \frac{(2t+1)\pi}{2\cos(\pi t)}.$$

We give the endpoint and exterior argument because the global inequality, rather than only its interior polynomial, is needed. Write $g(r) = (1 - r^2)_+^{t+1/2}$. This function is absolutely continuous with compact support, and differentiation of the convolution can be transferred to its locally integrable derivative:

$$F'(v) = \int_{-1}^{1} |v - r|^{-2t} g'(r) \, \mathrm{d}r.$$

There is no boundary term, since $g(\pm 1) = 0$. Away from the endpoints, localization makes continuity of this expression a consequence of translation continuity of the locally integrable kernel against a smooth factor. At $r = 1$, the derivative of the density has order $(1 - r)^{t-1/2}$. For $0 < \delta < 1/4$, uniformly when $|v - 1| \leq \delta$,

$$\int_{1-2\delta}^{1} |v - r|^{-2t} |g'(r)| \, \mathrm{d}r \leq C\delta^{1/2-t}.$$

To see this, put $1 - r = \delta u$ and split at distances comparable to $|v - 1|/\delta$ from the two possible singularities. Their combined endpoint exponent is $-t - 1/2 > -1$. The same estimate holds at -1. The remaining integrals converge uniformly as $v \to \pm 1$, so $F \in C^1$ there as well. Thus F and Q, and their first derivatives, agree at both endpoints. For $v > 1$,

$$F''(v) = 2t(2t+1) \int_{-1}^{1} (v - r)^{-2t-2} (1 - r^2)^{t+1/2} \, \mathrm{d}r > 0, \qquad Q'' = -4tB_2 < 0.$$

The matching value and derivative at one imply $F - Q > 0$ on $(1, \infty)$ by two integrations. Reflection gives the other side.

For $z \neq 0$, rotation and beta integration give the kernel decomposition

$$|z|^{-2t} = \frac{1}{J} \int_{S^1} |z \cdot \theta|^{-2t} \, \mathrm{d}\theta, \qquad \int_{S^1} |\theta_1|^{-2t} \, \mathrm{d}\theta = 4 \int_0^{\pi/2} (\cos\phi)^{-2t} \, \mathrm{d}\phi = J. \tag{6.13}$$

The angular singularities are integrable exactly because $2t < 1$. Let $T = \mathsf{H}^{1/2}$. The substitution $x = Ty$ gives

$$\rho_{\mathsf{H}}(x) \, \mathrm{d}x = \frac{m_{\mathrm{ell}}(t+1)}{\pi} (1 - |y|^2)_+^t \, \mathrm{d}y.$$

Polar integration on the unit disk gives mass m_{ell}. For a fixed $\theta \in S^1$, put $e = T\theta/h(\theta)$ and use the orthonormal coordinates $y = ue + we^{\perp}$. Then $x \cdot \theta = h(\theta)u$. Testing against any bounded continuous function of this projection and integrating first in w proves that its density is

$$m_\theta(v) = \frac{m_{\mathrm{ell}}(t+1)B_1}{\pi h(\theta)} \left(1 - \frac{v^2}{h(\theta)^2}\right)_+^{t+1/2}. \tag{6.14}$$

Indeed the transverse integral is

$$\int_{-\sqrt{1-u^2}}^{\sqrt{1-u^2}} (1-u^2-w^2)^t \, \mathrm{d}w = B_1(1-u^2)^{t+1/2}, \qquad |u| \le 1,$$

by $w = \sqrt{1-u^2}\, r$. This proves the projection formula without a choice of principal axes for the ellipse.

Apply (6.13) to the nonnegative potential and use Tonelli, followed by (6.14). In the resulting one-dimensional integral set $v = h(\theta)r$. The factor $1/h$ in the projected density cancels the Jacobian, leaving h^{-2t} and exactly (6.10). With $G = F - Q \ge 0$, it follows that

$$v_{\mathsf{H}}(x) + x^{\mathsf{T}}\mathsf{B}_{\mathsf{H}}x - \ell_{\mathsf{H}} = \frac{\mathfrak{c}_s^{\mathrm{ell}} m_{\mathrm{ell}}}{B_2} \int_{S^1} h(\theta)^{-2t} G\Big(\frac{x\cdot\theta}{h(\theta)}\Big) \, \mathrm{d}\theta. \tag{6.15}$$

If $x \in E_{\mathsf{H}}$, Cauchy–Schwarz gives $|x\cdot\theta| \le h(\theta)$ for every θ, so the right side vanishes. If $x \notin E_{\mathsf{H}}$, choose $\theta = \mathsf{H}^{-1}x/|\mathsf{H}^{-1}x|$. Then

$$\frac{x\cdot\theta}{h(\theta)} = (x^{\mathsf{T}}\mathsf{H}^{-1}x)^{1/2} > 1.$$

The strict inequality persists on an open arc. The integrand in (6.15) is strictly positive on that arc, which proves the exact equality set in (6.12). □

6.3. **The coefficient map on the positive-definite cone.**

Lemma 6.4 (Existence and uniqueness of the shape matrix)**.** *For every symmetric* $\mathsf{B} > 0$ *and* $m_{\mathrm{ell}} > 0$*, there is a unique symmetric* $\mathsf{H} > 0$ *satisfying*

$$\mathsf{B} = 2t\mathfrak{c}_s^{\mathrm{ell}} m_{\mathrm{ell}} \int_{S^1} \theta\theta^{\mathsf{T}} (\theta^{\mathsf{T}}\mathsf{H}\theta)^{-t-1} \, \mathrm{d}\theta. \tag{6.16}$$

The matrices H *and* B *commute, and scalar* B *gives scalar* H*. Its associated level is*

$$\ell_{\mathrm{ell}} = \mathfrak{c}_s^{\mathrm{ell}} m_{\mathrm{ell}} \int_{S^1} (\theta^{\mathsf{T}}\mathsf{H}\theta)^{-t} \, \mathrm{d}\theta = \frac{\operatorname{tr}(\mathsf{BH})}{2t} > 0. \tag{6.17}$$

For fixed B*, writing* H_1, ℓ_1 *for the mass-one parameters, one has*

$$\mathsf{H}_{m_{\mathrm{ell}}} = m_{\mathrm{ell}}^{1/(1+t)} \mathsf{H}_1, \qquad \ell_{m_{\mathrm{ell}}} = m_{\mathrm{ell}}^{1/(1+t)} \ell_1, \qquad m_{\mathrm{ell}} = (\ell_{\mathrm{ell}}/\ell_1)^{1+t}. \tag{6.18}$$

Thus mass and positive level are equivalent parameters, with no restriction on eccentricity.

Proof. On the closed cone of nonnegative symmetric matrices consider

$$\mathfrak{F}(\mathsf{H}) = \operatorname{tr}(\mathsf{BH}) + 2\mathfrak{c}_s^{\mathrm{ell}} m_{\mathrm{ell}} \int_{S^1} (\theta^{\mathsf{T}}\mathsf{H}\theta)^{-t} \, \mathrm{d}\theta,$$

with the extended-value convention at vanishing quadratic forms. Fatou gives lower semicontinuity. Since $\operatorname{tr}(\mathsf{BH}) \ge \lambda_{\min}(\mathsf{B}) \operatorname{tr}\mathsf{H}$, the functional is coercive. It is finite on the positive-definite cone and infinite at zero, so it attains a minimum at a nonzero nonnegative matrix. At a rank-one matrix $\mathsf{H} = \lambda_{\mathsf{H}} f f^{\mathsf{T}}$, $\lambda_{\mathsf{H}} > 0$, its angular integral is finite because it has singularity $|\phi|^{-2t}$ near a null direction.

Such a rank-one matrix cannot minimize. Let e be a unit null vector, and write $r(\theta) = \theta^{\mathsf{T}}\mathsf{H}\theta$ and $p(\theta) = (\theta\cdot e)^2$. For almost every θ, the nonnegative quotients

$$\frac{r(\theta)^{-t} - (r(\theta) + \varepsilon p(\theta))^{-t}}{\varepsilon} \;\uparrow\; tp(\theta)r(\theta)^{-t-1} \qquad (\varepsilon \downarrow 0).$$

The monotonicity follows from convexity of $r \mapsto r^{-t}$. Near $\theta = e$ the limiting integrand is comparable to $|\phi|^{-2t-2}$, whose integral diverges. Monotone convergence therefore gives

$$\lim_{\varepsilon\downarrow 0} \frac{\mathfrak{F}(\mathsf{H} + \varepsilon e e^{\mathsf{T}}) - \mathfrak{F}(\mathsf{H})}{\varepsilon} = e^{\mathsf{T}}\mathsf{B}e - \infty = -\infty.$$

A sufficiently small inward perturbation decreases the functional. The minimizer is consequently positive definite.

Differentiation there in every symmetric direction gives (6.16). The function $r \mapsto r^{-t}$ is strictly convex on $(0,\infty)$. If two symmetric matrices differ, their quadratic forms differ on a nonempty open set of circle directions. Integration thus proves strict convexity on the positive-definite cone. Every solution of (6.16) is a global minimizer, by the supporting line inequality for the same convex integrand, integrated over the circle; this inequality is valid also for singular competitors. Hence the positive-definite solution is unique.

In an eigenbasis of H, reflection in either coordinate axis makes the off-diagonal integral in (6.16) vanish. Thus B commutes with H. The equation is covariant under orthogonal conjugation, so uniqueness makes H invariant under every orthogonal symmetry of B. In particular a scalar B yields a scalar H. Multiplication of (6.16) by H and taking the trace gives (6.17). Finally its right side is homogeneous of degree $-t-1$ in H and degree one in m_{ell}. Uniqueness gives the first identity in (6.18); substitution into (6.17) gives the other two. □

6.4. **Identification, converse, and exact conormal.**

Proposition 6.5 (The positive-definite branch)**.** *Suppose U satisfies the intrinsic hypotheses and has full quadratic blowdown Q_{B} with $\mathsf{B} > 0$. If $\widehat{\mu} = 0$, then*

$$U(x,Y) = Q_{\mathsf{B}}(x-z,Y) + c, \qquad z \in \mathbb{R}^2, \quad c \geq 0,$$

with unique z, c. Its contact is empty when $c > 0$ and is $\{z\}$ when $c = 0$. If $\widehat{\mu} \neq 0$, then U is exactly the ellipse in Section 2.6, with unique data $\mathsf{B} > 0$, $z \in \mathbb{R}^2$, and $m_{\mathrm{ell}} > 0$, or equivalently $\mathsf{B} > 0$, $z \in \mathbb{R}^2$, and $\ell_{\mathrm{ell}} > 0$. More explicitly, for the unique H in (6.16)*,*

$$\begin{aligned} U(x,Y) &= Q_{\mathsf{B}}(x-z,Y) - \ell_{\mathrm{ell}} + \Phi_{\mathsf{H}}(x-z,Y), \\ \Phi_{\mathsf{H}}(x,Y) &= k_s \int_{\mathbb{R}^2} (|x-w|^2 + Y^2)^{-t} \rho_{\mathsf{H}}(w)\, \mathrm{d}w = \mathcal{E}_s v_{\mathsf{H}}(x,Y), \\ K &= z + E_{\mathsf{H}}, \qquad \mathrm{d}\widehat{\mu}(x) = e_s \rho_{\mathsf{H}}(x-z)\, \mathrm{d}x. \end{aligned} \tag{6.19}$$

Every displayed parameter choice gives an intrinsic solution with this full quadratic blowdown, locally uniformly and strongly in weighted H^1. The potential Φ_{H} in (6.19) *decays as $O(|(x,Y)|^{-2t})$, uniformly in direction.*

Proof. Let $\lambda_{\min}(\mathsf{B}) > 0$. Full locally uniform blowdown implies, for all sufficiently large $|x|$,

$$u(x) \geq \tfrac{1}{2}\lambda_{\min}(\mathsf{B})|x|^2.$$

Indeed the uniform error on a ball of radius $2|x|$ is $o(|x|^2)$. Thus $K = \{u = 0\}$ is compact and the pressure is finite with compact support. This is established before any equilibrium argument. Lemma 6.1 gives (6.1). Put

$$z = -\tfrac{1}{2}\mathsf{B}^{-1}\boldsymbol{\alpha}, \qquad c = c_0 - \tfrac{1}{4}\boldsymbol{\alpha}^{\mathsf{T}}\mathsf{B}^{-1}\boldsymbol{\alpha}.$$

After translating x by z and translating the measure with it,

$$u(x+z) = x^{\mathsf{T}}\mathsf{B}x + c + v(x), \qquad v = K_s * \nu, \qquad \nu = e_s^{-1}\widehat{\mu} \quad \text{in the translated coordinates.} \tag{6.20}$$

If $\nu = 0$, global nonnegativity is exactly $c \geq 0$, yielding the pressure-free assertion. Suppose henceforth that $m_{\mathrm{ell}} := \nu(\mathbb{R}^2) > 0$. Write $\ell = -c$ without assuming its sign. The trace sign and source support give the literal relations

$$g(x) := v(x) + x^{\mathsf{T}}\mathsf{B}x - \ell \geq 0, \qquad g = 0 \quad \text{on } \operatorname{supp}\nu. \tag{6.21}$$

By Lemma 6.2 this initial measure has finite energy. In particular its pairings with the continuous potentials below are well defined; no pressure density has yet been inferred.

Choose the unique shape matrix from Lemma 6.4 with this same B and this same mass m_{ell}. Set $\nu_* = \rho_{\mathsf{H}}\,\mathrm{d}x$, $v_* = K_s * \nu_*$, and $\ell_* = \ell_{\mathsf{H}}$. The bounded compact density has a continuous potential and finite energy. Lemma 6.3 says

$$g_*(x) := v_*(x) + x^{\mathsf{T}}\mathsf{B}x - \ell_* \geq 0, \qquad g_* = 0 \quad \text{on } \operatorname{supp}\nu_*.$$

Using (6.21) on the first support and this last identity on the second gives

$$\int (v - v_*)\,\mathrm{d}\nu = (\ell - \ell_*)m_{\mathrm{ell}} - \int g_*\,\mathrm{d}\nu \leq (\ell - \ell_*)m_{\mathrm{ell}},$$
$$\int (v - v_*)\,\mathrm{d}\nu_* = (\ell - \ell_*)m_{\mathrm{ell}} + \int g\,\mathrm{d}\nu_* \geq (\ell - \ell_*)m_{\mathrm{ell}}.$$

Their difference is $\mathfrak{E}(\nu - \nu_*)$ by (6.6). It is at most zero and at least zero, so strict Fourier positivity forces $\nu = \nu_*$. The literal continuous potentials agree everywhere. On their common nonempty support, the two complementarity identities then give $\ell = \ell_* = \ell_{\mathrm{ell}} > 0$. This proves the density and positive level as conclusions. The strict exterior gap in (6.12) proves that the entire contact set, including its boundary, is exactly E_{H} in the translated coordinates. Lemma 6.1 identifies the bulk correction and gives (6.19).

We verify the source constant and every part of the converse directly. For the continuous bounded compact density ρ_{H}, its radial potential is continuous, even, and smooth off the thin plane. Its tangential derivatives are locally bounded, since $2t + 1 < 2$, and the normal estimate (6.3) holds with $\|\rho_{\mathsf{H}}\|_\infty$. Hence it belongs to local weighted H^1. For $Y > 0$,

$$-Y^\eta \partial_Y \Phi_{\mathsf{H}}(x, Y) = 2tk_s Y^{2t} \int_{\mathbb{R}^2} (|x - w|^2 + Y^2)^{-t-1} \rho_{\mathsf{H}}(w)\,\mathrm{d}w \longrightarrow e_s \rho_{\mathsf{H}}(x). \tag{6.22}$$

The kernels here form an approximate identity whose total mass is

$$2tk_s \int_{\mathbb{R}^2} (1 + |z|^2)^{-t-1}\,\mathrm{d}z = 2tk_s \frac{\pi}{t} = 2\pi k_s = e_s.$$

The convergence holds at every point, and uniformly, by uniform continuity of ρ_{H}. The quadratic part has vanishing weighted normal derivative at zero because $1 + \eta > 0$. Even reflection therefore gives precisely $L_\eta U = -2e_s \iota_*(\rho_{\mathsf{H}}(\cdot - z)\,\mathrm{d}x)$. This also shows

$$\widehat{\mu}(\mathbb{R}^2) = e_s m_{\mathrm{ell}}, \tag{6.23}$$

separating intrinsic mass from fractional mass. The bounded trace has an absolute ordinary tail. Either the entire harmonic comparison used in Lemma 6.1 or Lemma 3.6 identifies its radial extension with $\mathcal{E}_s v_{\mathsf{H}}$.

The global angular inequality gives nonnegative trace, exact contact, and support of the positive source on that contact set. The source and complementarity yield the local variational inequality by Lemma 3.1. Continuity, evenness, local weighted energy, and at most quadratic growth have just been checked. Compact support gives uniform $|(x, Y)|^{-2t}$ decay outside a large ball. Thus dilation divided by R^2 converges locally uniformly to Q_{B}. The strong local weighted-energy assertion follows from Proposition 4.1; its unique quadratic limit is the already identified Q_{B}. The pressure-free polynomials satisfy the same properties by direct differentiation.

Finally, the quadratic matrix is uniquely recovered from the full blowdown. The decaying representation has unique affine coefficient $\boldsymbol{\alpha}$, so it gives the unique center $z = -\mathsf{B}^{-1}\boldsymbol{\alpha}/2$. Equation (6.23) recovers m_{ell}, and Lemma 6.4 recovers H and ℓ_{ell}. Conversely (6.18) recovers mass from level. The eigenvalues of H are the squared semiaxis lengths; using the matrix rather than its eigenvectors also removes any ambiguity for a disk. Thin orthogonal transformations conjugate both matrices and rotate z, while translations change only the center. The angular equation, density, and isotropic kernel are covariant under these operations, so no coordinate choice adds a parameter. □

Remark 6.6 (Relation to planar Riesz equilibria). The compact ellipse classification belongs to established planar anisotropic Riesz equilibrium theory. Carrillo–Shu [7, Theorem 3.1 and Remark 3.2] give the direct two-dimensional antecedent, including positive-definite quadratic confinement. Frank–Mateu–Mora–Rondi–Scardia–Verdera [16, Theorem 1.1 and Proposition 2.1] provide a broader confirmation, including its printed dimension-two case and exponent $2t \in (0,1)$ used here. To check the normalization, put $T = \mathsf{B}^{-1/2}$ and $\nu = m_{\mathrm{ell}} T_{\#}\omega$, with ω a probability measure. Then

$$\frac{1}{m_{\mathrm{ell}}}\left\{\frac{1}{2}\mathfrak{E}(\nu) + \int x^{\mathsf{T}}\mathsf{B}x \,\mathrm{d}\nu(x)\right\} = \iint W(z-w)\,\mathrm{d}\omega(z)\,\mathrm{d}\omega(w) + \int |z|^2\,\mathrm{d}\omega(z),$$

where

$$W(z) = \frac{m_{\mathrm{ell}}}{2} K_s(Tz).$$

Its Fourier transform is $(m_{\mathrm{ell}}/2)\sqrt{\det \mathsf{B}}\,|\mathsf{B}^{1/2}\xi|^{-2s}$, strictly positive away from zero. The compact-source argument establishes finite mass, second moment, and energy before this comparison with probability-energy conventions. The direct proof above records the intrinsic-to-equilibrium passage, exact constants, and parameter map; it does not assert novelty for ellipse equilibria.

7. Ordinary tails and the first growth bound

We first derive a pointwise bound from the equation. The ordinary tail alone would not control the supremum of an arbitrary continuous function. Throughout this section $0 < s < 1$ and $a > 0$, and P satisfies (2.7). Put

$$\tau(R) = \int_{|x|>R} |P(x)||x|^{-2-2s}\,\mathrm{d}x \quad (R \geq 1), \qquad M_f(R) = \sup_{B_R} |f|.$$

The ball in this notation has the dimension of the function f.

Proposition 7.1 (Growth from the ordinary tail)**.** *Let $V = \mathcal{E}_s P$ be the even Poisson extension. Then*

$$M_P(R) \leq C_s R^{2s}\tau(2R) = o(R^{2s}), \qquad R \geq 1, \tag{7.1}$$

and $M_V(R) = o(R^{2s})$. Both P_+ and P_- satisfy $\mathcal{A}_s P_\pm \leq 0$ in distributions.

Proof. The two trace phases. Bounded translations act boundedly and strongly continuously on the ordinary weighted L^1 space, by comparison of the weights and approximation by compactly supported functions. Positive compact mollifications therefore satisfy $P_\varepsilon \to P$ locally uniformly and in that space, and $\mathcal{A}_s P_\varepsilon = \mu_\varepsilon \geq 0$ pointwise. Smoothness controls local second differences and the ordinary tail controls the far integral. For a smooth convex Lipschitz function F the supporting-line inequality gives

$$\mathcal{A}_s F(v) \leq F'(v)\mathcal{A}_s v.$$

Indeed, integrate $F(v(x)) - F(v(y)) \leq F'(v(x))(v(x) - v(y))$ with a symmetric puncture and then remove the puncture. Apply this to $-P_\varepsilon$ and nondecreasing convex approximations of the positive part. Passing to the limit proves $\mathcal{A}_s P_- \leq 0$.

For the positive phase choose convex F_δ with $0 \leq F_\delta' \leq 1$, $F_\delta = 0$ on $(-\infty, \delta]$, and $F_\delta \to (\cdot)_+$. Fix the compact support J of a nonnegative test. If $\mu_\varepsilon(x) > 0$ there, a contact point k lies within ε of x. Since $P(k) = -ak_Z^2 \leq 0$, continuity on a slightly larger compact set gives $P_\varepsilon(x) \leq 2\omega_J(2\varepsilon)$, where ω_J is a modulus of continuity. At fixed δ and small ε this is below δ, so $F_\delta'(P_\varepsilon)\mu_\varepsilon = 0$ on J. The convex inequality proves $\mathcal{A}_s F_\delta(P_\varepsilon) \leq 0$ there. First let $\varepsilon \downarrow 0$ using the local and weighted convergence; then let $\delta \downarrow 0$ using $0 \leq F_\delta(P) \leq P_+$. This proves the other phase inequality without multiplying a Radon measure by an unspecified almost-everywhere representative.

Ball comparison and averaging. For continuous ordinary-tail data g, the solution in B_r with exterior data g is given by the positive fractional ball kernel [1, Theorem 2.10]:

$$H_{r,g}(x) = \frac{\sin(\pi s)}{\pi^2} \int_{|y|>r} \frac{(r^2-|x|^2)^s g(y)}{(|y|^2-r^2)^s |x-y|^2}\,\mathrm{d}y, \qquad |x|<r. \tag{7.2}$$

It has mass one and joins g continuously across the sphere. A continuous ordinary-tail subsolution w satisfies $w \le H_{r,w}$. To see this in the distributional class, mollify w globally and compare $w_\varepsilon - H_{r,w_\varepsilon}$. A positive interior maximum would have strictly positive fractional Laplacian, since the difference is zero outside the ball. This contradicts its nonpositive Laplacian. Passage to w uses local uniform convergence at the integrable exterior-sphere singularity $(|y|-r)^{-s}$ and weighted L^1 convergence away from it.

Fix $|x| \le R$ and average this comparison over origin-centered balls with $2R \le r \le 3R$. With $\varrho = |y|$, the averaged kernel vanishes for $\varrho \le 2R$ and is at most

$$C_s R^{2s} \varrho^{-2} \frac{1}{R} \int_{2R}^{\min(3R,\varrho)} (\varrho^2 - r^2)^{-s}\,\mathrm{d}r.$$

For $2R < \varrho \le 4R$, use $\varrho^2 - r^2 \ge 2R(\varrho - r)$; the last average is at most $C_s R^{-2s} \le C_s' \varrho^{-2s}$. For $\varrho \ge 4R$, use $\varrho^2 - r^2 \ge 7\varrho^2/16$. Thus the kernel is bounded by $C_s R^{2s} |y|^{-2-2s} \mathbf{1}_{\{|y|>2R\}}$. Apply it separately to P_+ and P_- to obtain (7.1). Here the near-sphere singularity was integrated in radius, rather than bounded by its supremum. Radius averaging of fractional ball kernels is also used in the derivative estimates of Fall [14, Lemma 3.1].

The ordinary extension. The extension kernel is

$$\Pi_s(x,Y) = \frac{s}{\pi} \frac{|Y|^{2s}}{(|x|^2+Y^2)^{1+s}}, \qquad V(x,Y) = \int_{\mathbb{R}^2} \Pi_s(x-y,Y) P(y)\,\mathrm{d}y. \tag{7.3}$$

The integral is absolute and its continuous trace is P: locally this is an approximate identity, while the separated tail is $O(|Y|^{2s})$. For $|(x,Y)| \le R$, split the data at $|y| = 2R$. Mass one bounds the inner part by $M_P(2R)$, and $|x-y| \ge |y|/2$ bounds the far part by $C_s R^{2s} \tau(2R)$. Hence

$$M_V(R) \le M_P(2R) + C_s R^{2s} \tau(2R) = o(R^{2s}). \tag{7.4}$$

No moment of order $2s$ of the Poisson kernel is used; that moment diverges. □

Lemma 7.2 (Ordinary-tail Liouville principle)**.** *Let $n = 1$ or 2. If $\int_{\mathbb{R}^n} |f(x)|(1+|x|)^{-n-2s}\,\mathrm{d}x < \infty$ and $(-\Delta)^s f = 0$ in distributions on $\mathbb{R}^n$, then f is affine almost everywhere for $s > 1/2$, and constant for $s \le 1/2$.*

Proof. This is the ordinary-tail theorem of Fall [14, Theorem 1.1]. We give the Fourier-cutoff argument needed for measure-potential remainders. The tail makes f tempered. For a Schwartz test ϕ, its symmetric fractional integral gives

$$\sup_x (1+|x|)^{n+2s} |(-\Delta)^s \phi(x)| \le C_s \mathfrak{p}(\phi),$$

where $\mathfrak{p}$ is a finite sum of Schwartz seminorms. Indeed, for $|x| \ge 2$ split at $|h| = |x|/2$. The inner part is bounded by the rapidly decaying local Hessian times $\int_0^{|x|/2} r^{1-2s}\,\mathrm{d}r$; the shifted outer terms are bounded by $C|x|^{-n-2s}\|\phi\|_1$, and the center term by $C|x|^{-2s}|\phi(x)|$. On bounded x, split at one and use bounded second derivatives. Compact cutoffs of ϕ converge in Schwartz seminorms, hence after the operator in this weighted supremum norm. Harmonicity therefore extends to all Schwartz tests. For $\zeta \in C_c^\infty(\mathbb{R}^n \setminus \{0\})$, test with the inverse Fourier transform of $|\xi|^{-2s}\zeta(\xi)$. It is Schwartz, and $\widehat{f}$ annihilates every test away from zero. The structure theorem for a distribution supported at a point makes f a polynomial. A nonzero polynomial of degree m has finite absolute ordinary tail exactly when $m < 2s$. This gives the asserted alternatives; equality $m = 2s$ would produce logarithmic divergence. No reciprocal multiplier is used at zero. □

Corollary 7.3 (Analytic embedding of the ordinary-tail class)**.** *For every $0 < s < 1$, a solution of* (2.7) *gives*

$$U = Q + \mathcal{E}_s P \in C(\mathbb{R}^3) \cap H^1_{\mathrm{loc}}(|Y|^\eta), \qquad \widehat{\mu} = e_s \mu,$$

which is even, satisfies (1.1)*, and has $R^{-2}U(R\cdot) \to Q$ locally uniformly and strongly in local weighted H^1. In particular, above half order it belongs to the full-quadratic-blowdown class without any classification result or subsequence-upgrade theorem.*

Proof. The two terms in $L_\eta Q$ are $2a|Y|^\eta$ and $-2a|Y|^\eta$. The normal weighted flux of Q tends to zero at the thin plane, since $1 + \eta > 0$, so $L_\eta Q = 0$ also in the whole-space weak sense. The trace of U is $P + aZ^2 \geq 0$. Lemma 3.6 gives its local energy and the source $-2e_s \iota_* \mu$, supported on zero trace. Continuity and evenness follow from the extension. Proposition 7.1 gives $\sup_{B_R} |U - Q| = o(R^{2s})$, and $2s < 2$ supplies the quadratic bound and the claimed locally uniform limit. To obtain its local energy convergence, put $E = \mathcal{E}_s P$. The negative source makes E_- a subsolution. On its source, $E = P = -aZ^2 \leq 0$; hence $(E - \varepsilon)_+$ vanishes locally near that source. Gluing its harmonic-region inequality and passing $\varepsilon \downarrow 0$ in local energy makes E_+ a subsolution as well. Testing the two phase inequalities by their product with ζ^2 and adding gives $\int |Y|^\eta \zeta^2 |\nabla E|^2 \leq 4 \int |Y|^\eta E^2 |\nabla \zeta|^2$. Apply this estimate to $R^{-2}E(R\cdot)$ on two nested fixed balls. Its uniform convergence to zero forces both its weighted L^2 norm and gradient norm to zero. This construction uses only the original tail and source hypotheses. □

8. Trace convexity and weighted local tools

Only the two physical coordinates are involved in the convexity conclusion. We first isolate a common defect argument. Ordinary-tail data and intrinsic quadratic data supply its bounded defect in different ways.

Lemma 8.1 (Removal of a bounded second-difference defect)**.** *Let $0 < s < 1$, $a > 0$, and $F = Q + E$ be continuous, even, locally in $H^1(|Y|^\eta)$, with nonnegative trace and a nonpositive Radon source supported on its zero trace. Suppose $E = o(R^2)$ uniformly on B_R. If, for every physical increment h, the function $q_h = F(\cdot + h) + F(\cdot - h) - 2F$ is bounded below globally, then $F|_{Y=0}$ is jointly convex.*

Proof. Write $\Sigma = \{F(x, 0) = 0\} \times \{0\}$ and fix h. On Σ, $q_h \geq 0$ because both translated traces are nonnegative. Off Σ the central source is absent and both translated sources are nonpositive; hence $L_\eta q_h \leq 0$ there. The bounded continuous local-energy function $v_h = (-q_h)_+$ is therefore a global weak subsolution. To justify the passage across contact, first use $(-q_h - \varepsilon)_+$. Continuity makes it zero in a neighborhood of every compact portion of Σ; on the complement it is a convex nonincreasing truncation of q_h. A partition of unity on this open cover gives its global subsolution inequality. As $\varepsilon \downarrow 0$, its gradients $-1_{\{q_h < -\varepsilon\}} \nabla q_h$ converge strongly in local weighted L^2 by dominated convergence and the almost-everywhere vanishing of ∇q_h on $\{q_h = 0\}$. This proves the asserted inequality for v_h without regularity of Σ.

Fix a thin center x_0 and a direction $\theta \in S^2$ with $\theta_Y \neq 0$. Balls of radius $c_\theta R$ about $(x_0, 0) + R\theta$ stay off the thin plane. Interior tangential second-derivative estimates, after scaling these source-free balls, give $D_x^2 E = o(1)$ on the fixed h-segments there, since $\sup_{B_{C_\theta R}} |E| = o(R^2)$. The coefficients on the scaled balls are smooth and uniformly elliptic for this fixed direction. The integral second-difference formula thus gives $q_h((x_0, 0) + R\theta) \to 2ah_Z^2 \geq 0$ and $v_h \to 0$. The weighted spherical submean inequality [3, Lemma 2.9, with reversed sign and zero forcing] yields

$$v_h(x_0, 0) \leq \frac{1}{\omega_\eta} \int_{S^2} v_h((x_0, 0) + R\theta) |\theta_Y|^\eta \, \mathrm{d}S, \qquad \omega_\eta = \int_{S^2} |\theta_Y|^\eta \, \mathrm{d}S.$$

It suffices to take admissible radii tending to infinity. The equator has zero weighted surface measure, the weight is integrable for $\eta > -1$, and the global bound for v_h is a dominating function.

Dominated convergence gives $v_h(x_0,0)=0$. Every physical second difference of the trace is nonnegative, and continuity proves joint convexity. □

Theorem 8.2 (Ordinary-tail trace convexity)**.** *For every $0<s<1$, a continuous solution of* (2.7) *in the ordinary-tail class has jointly convex gap $u=P+aZ^2$.*

Proof. Corollary 7.3 gives $F=Q+\mathcal{E}_sP$ in the class of Lemma 8.1, with $\mathcal{E}_sP=o(R^{2s})=o(R^2)$. The local semiconvexity theorem of Caffarelli–Salsa–Silvestre [3, Lemma 4.1], applied to the uniformly bounded quadratic rescalings $R^{-2}F(R\cdot)$ on B_1, gives $D_{ee}u\ge -C$ globally for each physical unit direction e. Here the zero obstacle has controlled C^2 norm and the rescaled solutions have a common local bound. Mollification and integration along lines yield

$$\delta_h u(x)=u(x+h)+u(x-h)-2u(x)\ge -C|h|^2. \tag{8.1}$$

For fixed h, both δ_hP and $\delta_hu=\delta_hP+2ah_Z^2$ have ordinary tails. Translation in the absolutely convergent Poisson integral gives

$$q_h=\delta_hF=\mathcal{E}_s(\delta_hP)+2ah_Z^2=\mathcal{E}_s(\delta_hu)\ge -C|h|^2.$$

This is a Poisson representation of the second difference, not of the quadratic gap itself. The common defect lemma now proves convexity. □

Remark 8.3. The harmonic correction Q is not convex in Y. Convexity is a conclusion for the physical trace only. Local semiconvexity supplies a bounded defect; scaling or bounded subharmonicity alone would not remove it.

8.1. **Bounded intrinsic second differences and trace convexity.**

Proposition 8.4 (Intrinsic trace convexity)**.** *Suppose U satisfies the intrinsic weak conditions* (1.1) *and the full locally uniform quadratic blowdown* (2.6)*. Then its trace $u(X,Z)=U(X,Z,0)$ is jointly convex in (X,Z).*

Proof. The local minimizer property follows from Lemma 3.1. The quadratic growth bound makes $U_R(q)=R^{-2}U(Rq)$ uniformly bounded on each fixed ball for $R\ge 1$. Applied on fixed nested balls to this zero-obstacle problem, the local estimates of Caffarelli–Salsa–Silvestre [3, Lemma 4.1 and Proposition 4.3] give, for every fixed $0<\alpha<s$,

$$D_{ee}(U_R|_{Y=0})\ge -C,\qquad [\nabla_x U_R]_{C^\alpha(B_2)}\le C_\alpha, \tag{8.2}$$

where e is any physical unit vector; using B_4 for the input makes the displayed interior balls harmless. The second estimate is tangential regularity in the full weighted ball. It is not an isotropic $C^{1,\alpha}$ assertion across $Y=0$.

The first estimate is invariant under the quadratic scaling. Exhaustion therefore gives $D_{ee}u\ge -C$ in distributions on $\mathbb{R}^2$. Mollifying u, integrating the resulting one-dimensional second-derivative inequality on lines, and then removing the mollification proves

$$u(x+h)+u(x-h)-2u(x)\ge -C|h|^2\qquad (x,h\in\mathbb{R}^2). \tag{8.3}$$

The second estimate in (8.2) scales to

$$[\nabla_x U]_{C^\alpha(B_{2R})}\le C_\alpha R^{1-\alpha}.$$

For a fixed physical increment h, put $q_h(q)=U(q+(h,0))+U(q-(h,0))-2U(q)$. If $|q|\le R$ and $|h|\le R$, the identity

$$q_h(q)=\int_0^1 h\cdot\left(\nabla_x U(q+(th,0))-\nabla_x U(q-(th,0))\right)\,\mathrm{d}t$$

gives $|q_h(q)|\le C_\alpha|h|^{1+\alpha}R^{1-\alpha}$. Choose R comparable to $1+|q|+|h|$ and absorb bounded radii to get

$$|q_h(q)|\le C_h(1+|q|^{1-\alpha}). \tag{8.4}$$

The function $q_h + C|h|^2$ is continuous, locally weighted energy, and harmonic in the open upper halfspace. Its thin trace is nonnegative by (8.3). Since $1-\alpha < 1 < 2s$, Lemma 3.3 applies to (8.4). Evenness then gives

$$q_h \geq -C|h|^2 \quad \text{on all of } \mathbb{R}^3. \tag{8.5}$$

This obtains the required bulk lower bound before any ordinary tail or extension formula is known.

Finally $U = Q + V$ and the full blowdown says $\sup_{B_R} |V| = o(R^2)$. Every physical second difference of U is globally bounded below by (8.5), so all hypotheses of the common bounded-defect argument, Lemma 8.1, are satisfied. That lemma glues the truncated negative defect across the central contact set and uses off-equator harmonic derivative decay with weighted spherical averaging. It gives nonnegative thin second differences and hence joint convexity of u. □

9. Intrinsic contact alternatives

Throughout this section $1/2 < s < 1$, $\eta = 1 - 2s \in (-1, 0)$, and U satisfies the intrinsic hypotheses (1.1) and the full quadratic blowdown (2.6). Thus, with $x = (X, Z)$ and $q = (X, Z, Y)$,

$$Q(q) = a\left(Z^2 - \frac{Y^2}{1+\eta}\right), \qquad V = U - Q, \qquad u = U|_{Y=0}, \qquad P = u - aZ^2, \qquad M_V(R) = o(R^2).$$

We write $K = \{u = 0\}$ and retain the intrinsic normalization $L_\eta U = -2\iota_* \widehat{\mu}$. The trace u is jointly convex by 8.4. No ordinary fractional pressure or representation of V is used in this section.

9.1. **Bounded contact.**

Proposition 9.1 (Exclusion of bounded nonempty contact)**.** *If K is bounded, then $K = \varnothing$, $\widehat{\mu} = 0$, and*

$$U = Q + bZ + c, \qquad c > \frac{b^2}{4a}.$$

In particular, P has an absolute ordinary fractional tail.

Proof. The closed bounded set K is compact. Lemma 6.1, which requires no rank condition or trace convexity, supplies

$$V = b_X X + b_Z Z + c + \Phi, \qquad \Phi(q) = \frac{1}{2\pi}\int_K |q - (\xi, 0)|^{-1-\eta}\, \mathrm{d}\widehat{\mu}(\xi). \tag{9.1}$$

The literal potential agrees with its continuous local-energy representative at every point, is nonnegative, and satisfies

$$0 \leq \Phi(q) \leq C\widehat{\mu}(K)|q|^{-1-\eta} \tag{9.2}$$

outside a fixed source ball. Hence continuity on the remaining compact set gives $|P(x)| \leq C(1+|x|)$. Since $2s > 1$, this proves the ordinary tail before the contact exclusion. The representation's mollification and monotone-trace proof do not assume finite energy of an arbitrary Radon potential.

We now use convexity to exclude nonempty bounded contact. For each fixed Z, nonnegativity of $u(X, Z)$ as $X \to \pm\infty$ forces $b_X = 0$. The convex function $X \mapsto u(X, Z)$ then has the same finite limit $aZ^2 + b_Z Z + c$ at both ends, and is constant. Thus its exterior potential in (9.1) is zero. Strict positivity of Γ_3 and $\widehat{\mu} \geq 0$ imply $\widehat{\mu} = 0$. The entire Liouville statement gives $U = Q + b_Z Z + c$ everywhere. Nonnegativity says $c \geq b_Z^2/(4a)$; equality would give the full contact line $Z = -b_Z/(2a)$, contrary to boundedness. The strict inequality and $K = \varnothing$ follow. This includes the initially empty case. □

9.2. **An invariant trace determines an invariant bulk.**

Proposition 9.2 (Intrinsic invariant branch)**.** *If u is independent of X, then U is independent of X and*

$$\widehat{\mu} = \mathrm{d}X \otimes \widehat{\nu}_0$$

for a finite nonnegative measure $\widehat{\nu}_0$ supported on the compact set $K_0 = \{Z : u(0,Z) = 0\}$. With

$$\alpha_0 = 2s - 1 = -\eta \in (0,1), \qquad S_{2,\eta} = \int_{S^1} |\theta_Y|^\eta \,\mathrm{d}S(\theta),$$

there are $b, c \in \mathbb{R}$ such that

$$P(X,Z) = bZ + c - \frac{2}{\alpha_0 S_{2,\eta}} \int_{K_0} |Z - t|^{\alpha_0} \,\mathrm{d}\widehat{\nu}_0(t). \tag{9.3}$$

In particular $|P(X,Z)| \le C(1 + |Z|)$, and P has an absolute ordinary tail. No density or absence of atoms is asserted here.

Proof. For $h \in \mathbb{R}$ consider $D_h(q) = U(q + he_X) - U(q)$. It is continuous, locally weighted H^1, harmonic in the open upper halfspace, and has zero thin trace. Since Q is X-invariant, D_h is a difference of translates of the subquadratic remainder V. The zero-trace Liouville statement 3.5, followed by even reflection, gives

$$D_h(q) = c(h)|Y|^{2s}.$$

The translation identity $D_{h+k}(q) = D_h(q + ke_X) + D_k(q)$ gives $c(h+k) = c(h) + c(k)$. Evaluation at a fixed point with $Y > 0$ proves continuity of c, so $c(h) = b_* h$ for some real b_*. Thus

$$U(X,Z,Y) = U(0,Z,Y) + b_* X|Y|^{2s}.$$

Evaluating the difference at $(R,0,R)$ and $(0,0,R)$ would give $|b_*|R^{1+2s}$, whereas quadratic growth bounds it by $C(1+R^2)$. Since $1 + 2s > 2$, $b_* = 0$. Bulk invariance has now been proved.

Write $U = U_0(Z,Y)$. Product integration gives $U_0 \in H^1_{\mathrm{loc}}(|Y|^\eta)$ in the reduced plane. Choose $\chi \in C_c^\infty(\mathbb{R})$, $\chi \ge 0$, with $\int \chi \,\mathrm{d}X = 1$, and define $\widehat{\nu}_0$ by integrating $\widehat{\mu}$ against $\chi(X)$. Tests $\chi(X)\psi(Z,Y)$ give

$$L_\eta^{(Z,Y)} U_0 = -2\iota_* \widehat{\nu}_0.$$

For an arbitrary compact smooth function $\chi_1(X)$, the left side of the full equation tested against $\chi_1(X)\psi(Z,Y)$ is $\int \chi_1 \,\mathrm{d}X$ times the reduced one. Tensor-product tests and density therefore give exactly $\widehat{\mu} = \mathrm{d}X \otimes \widehat{\nu}_0$. In particular no integration over the whole horizontal axis, nor finite planar mass, has been assumed. Source support puts $\widehat{\nu}_0$ on K_0. The full blowdown at $(0,Z,0)$ gives $u(0,Z) = aZ^2 + o(Z^2)$; hence K_0 is compact, and the locally finite $\widehat{\nu}_0$ is finite.

The reduced weighted dimension is $d = 2 + \eta \in (1,2)$. Its fundamental solution with flux -1 is the negative growing kernel

$$\Gamma_2(Z,Y) = -\frac{|(Z,Y)|^{\alpha_0}}{\alpha_0 S_{2,\eta}}, \qquad L_\eta^{(Z,Y)} \Gamma_2 = -\delta_0.$$

Set $\Phi_0(Z,Y) = 2\int \Gamma_2(Z - t, Y) \,\mathrm{d}\widehat{\nu}_0(t)$. The α_0-Hölder inequality for r^{α_0} proves continuity, and $\Phi_0 = O(1 + |(Z,Y)|^{\alpha_0})$. For a single pole the radial exponent of its local gradient energy is

$$d - 1 + 2(\alpha_0 - 1) = -1 - \eta > -1.$$

The local $H^1(|Y|^\eta)$ norms are uniformly bounded as the pole varies in the compact support of $\widehat{\nu}_0$. Minkowski's inequality therefore gives local energy of Φ_0, even if $\widehat{\nu}_0$ has atoms. Flux and Fubini give $L_\eta^{(Z,Y)} \Phi_0 = -2\iota_* \widehat{\nu}_0$.

Subtract Φ_0 from $U_0 - Q$. The difference is entire, even, source-free, locally weighted H^1, and subquadratic. The one-physical-variable case of 3.4 makes it $bZ + c$. Restriction to the thin line proves (9.3). Its linear bound is integrable against the planar ordinary-tail weight because $2s > 1$. Equivalently, integrating that weight in X gives a weight comparable to $(1 + |Z|)^{-1-2s}$. □

9.3. **The unique recession direction.**

Lemma 9.3 (Minimal normalized geometry)**.** *Suppose $u_X \not\equiv 0$. There is a unique horizontal orientation in which every escaping contact sequence tends in direction to e_X. In that orientation the sets $K \cap \{X \le M\}$ are compact for every finite M, and the minimum of X on K is finite and attained. After translating one minimizing contact point to the origin, the normalized contact satisfies*

$$K \subset \{X \ge 0\}, \qquad \mathcal{L} = \{(X,0,0) : X \ge 0\} \subset \iota(K), \qquad P(X,0) = 0 \quad (X \ge 0).$$

Moreover $u(x + te_X) \le u(x)$ for $t \ge 0$, $u_X = 0$ on K, and $R^{-1}\iota(K)$ eventually avoids every compact subset of $\mathbb{R}^3 \setminus \mathcal{L}$ as $R \to \infty$.

Proof. By Proposition 9.1, K is nonempty and unbounded. Convexity and continuity of $u \ge 0$ make K closed and convex. For an escaping contact sequence $k_j = (X_j, Z_j)$, the full blowdown gives

$$aZ_j^2 = -P(k_j) = o(|k_j|^2).$$

Every subsequential limiting direction is therefore horizontal. Such a direction e is a recession direction from every $k \in K$: for fixed $t \ge 0$ the segment points $k + t(k_j - k)/|k_j - k|$ belong to K for large j and converge to $k + te$. This also proves existence of a recession direction.

If both e_X and $-e_X$ were recession directions, K would contain a complete horizontal zero line. At any point x and for any subgradient $\xi \in \partial u(x)$, the supporting-plane inequality against points of that line forces $\xi_X = 0$. Applying these inequalities at the endpoints of a horizontal segment shows that u is constant on it, contradicting $u_X \not\equiv 0$. There is exactly one direction; reflect X once so it is e_X. An escaping sequence in $K \cap \{X \le M\}$ would have a limiting direction different from e_X, so these sublevels are bounded and hence compact. Taking a nonempty one shows that X is bounded below on K and its minimum is attained, say at (d, h).

In the already oriented coordinates make the change

$$\begin{aligned} U^{\#}(X,Z,Y) &= U(X + d, Z + h, Y), \\ P^{\#}(X,Z) &= P(X + d, Z + h) + 2ahZ + ah^2, \\ V^{\#}(X,Z,Y) &= V(X + d, Z + h, Y) + 2ahZ + ah^2. \end{aligned} \tag{9.4}$$

Then $u^{\#}(X,Z) = u(X + d, Z + h)$ and $U^{\#} = Q + V^{\#}$. The intrinsic source is simply translated. Fixed translations and the displayed affine remainder preserve the full quadratic blowdown and local-energy class. The transformed set lies in $X \ge 0$, contains the origin and hence, by recession, the exact positive horizontal ray. The transformation uses no extension formula. Its affine correction has an ordinary tail when needed later, since $2s > 1$.

Drop the superscript. For $R > t$, convexity against the contact point Re_X gives

$$u\big((1 - t/R)x + te_X\big) \le (1 - t/R)u(x).$$

Letting $R \to \infty$ gives the stated monotonicity. The local Signorini tangential regularity used in 8.4 makes u differentiable; at a point of K it attains its nonnegative minimum, so $u_X = 0$ there. Finally, for each fixed T, all $k = (X,Z) \in K \cap B_{TR}$ satisfy

$$aZ^2 = |P(k)| \le M_P(TR) = o(R^2)$$

uniformly as $R \to \infty$. Together with $X \ge 0$ this implies uniform collapse onto the positive ray on bounded rescaled sets, and hence the asserted support avoidance. Neither a full transverse projection nor a quantitative contact width has been used. □

10. The slit domain and selected-scale compactness

The rest of the noncylindrical analysis takes $1/2 < s < 1$. Put

$$\sigma = s - \tfrac{1}{2}, \quad \kappa = 2\sigma, \quad \lambda = 1 + \sigma, \quad \beta = 1 - \sigma, \quad \eta = 1 - 2s = -2\sigma, \quad b_G = 1 + \eta, \quad D = 3 + \eta.$$

For $q = (X, Z, Y)$ write $\rho(q) = |Y|^\eta$, $L_\eta = \operatorname{div}(\rho\nabla)$, and $\mathcal{L} = \{(X, 0, 0) : X \geq 0\}$. Weak subsolution means $\int \rho\nabla v \cdot \nabla\phi \leq 0$ for nonnegative compact smooth tests. We prove the common trace, spectral, and compactness tools before their state-growth and signed-ordering applications.

10.1. **The slit form domain.** The angular space is the even subspace of $H^1(S^2, |\theta_Y|^\eta)$ with zero weighted point trace at e_X. There is no Dirichlet condition at $-e_X$. The point trace is bounded: in a transverse chart with polar distance r the radial Cauchy–Schwarz estimate uses $\int_0^\varepsilon r^{-1-\eta}\,\mathrm{d}r < \infty$, because $\eta < 0$; an annular average controls the additive value. This also shows that the pole has positive capacity.

Lemma 10.1 (Trace and density)**.** *The kernel of the local trace on $\mathcal{L}$ is the weighted H^1 closure of smooth functions supported away from $\mathcal{L}$. On S^2 the kernel of the trace at e_X is the closure of smooth angular functions vanishing near e_X. Even approximations can be chosen for even functions.*

Proof. After a compact localization, the extension trace and the ordinary line trace give bounded maps

$$H^1(\mathbb{R}^3, |Y|^\eta) \longrightarrow H^s(\mathbb{R}^2) \longrightarrow H^{s-1/2}(\mathbb{R}).$$

For the first map, Fourier transformation in (X, Z) and the rescaling $Y \mapsto |\xi|Y$ reduce the homogeneous estimate to the one-dimensional weighted evaluation inequality; the inhomogeneous norm controls low frequencies. For the second map, Cauchy–Schwarz uses

$$\int_{\mathbb{R}} (1 + \xi_X^2 + \xi_Z^2)^{-s}\,\mathrm{d}\xi_Z = C_s(1 + \xi_X^2)^{1/2-s}.$$

This proves the asserted bounded trace into $H^\sigma(\mathbb{R})$.

For density, first truncate a compactly supported zero-trace function v at heights $\pm N$; truncation preserves the trace and converges in $H^1(\rho)$. For bounded v, remove B_δ by a radial cutoff. The extra energy is at most $C\|v\|_\infty^2\delta^{D-2} \to 0$, and the lost energy tends to zero by absolute continuity. Thus the tip adds no boundary condition.

Away from the tip put $r_\perp = (Z^2 + Y^2)^{1/2}$. A zero-point-trace transverse slice satisfies

$$\int_{r_\perp < r_0} |Y|^\eta \frac{|v|^2}{r_\perp^2}\,\mathrm{d}Z\,\mathrm{d}Y \leq \frac{4}{\eta^2}\int_{r_\perp < r_0} |Y|^\eta |\partial_{r_\perp} v|^2\,\mathrm{d}Z\,\mathrm{d}Y. \tag{10.1}$$

For a smooth slice vanishing at zero, integrate $(r^\eta)'|v|^2$ by parts; the outer term has the favorable sign since $\eta < 0$, and Cauchy–Schwarz proves the estimate. General slices follow by smooth approximation and subtracting their point values, using bounded point evaluation. Integration in X proves the cylinder version. Multiplying by a cutoff vanishing when $r_\perp < \varepsilon$ costs at most $C\int_{r_\perp < 2\varepsilon} \rho|v|^2/r_\perp^2$, which tends to zero by (10.1). Since the tip has already been removed, this cutoff can be patched to one on the negative axis. Weighted smooth density away from the ray finishes the approximation. Reflection averaging preserves evenness. The same argument in a chart proves the spherical assertion. The reverse inclusion follows from trace continuity. □

10.2. **Exact angular spectrum.** Use the upper-hemisphere coordinates

$$X = r\cos\vartheta, \quad Z = r\sin\vartheta\cos\varphi, \quad Y = r\sin\vartheta\sin\varphi, \quad 0 < \vartheta, \varphi < \pi.$$

The angular mass is $\sin^{1+\eta}\vartheta\sin^\eta\varphi\,\mathrm{d}\vartheta\,\mathrm{d}\varphi$. Even reflection gives natural weighted flux conditions at $\varphi = 0, \pi$.

Lemma 10.2 (Slit spectrum and Liouville windows)**.** *In the form domain of Lemma 10.1, the positive homogeneity degrees are*

$$\sigma + j \quad (j \geq 0), \qquad l + j \quad (l \geq 1,\ j \geq 0).$$

The principal mode is $H(q) = (|q| - X)^\sigma$; the only degree-one mode is Z. If a locally bounded, locally weighted H^1, even function U is weakly harmonic off $\mathcal{L}$, has zero ray trace, and grows at most $C(1 + |q|^Q)$, then

$$\begin{cases} U = AH + BZ, & 1 < Q < 1 + \sigma, \\ U = AH, & \sigma < Q < 1. \end{cases}$$

Proof. The transverse equation is

$$-\sin^{-\eta}\varphi\,\partial_\varphi(\sin^\eta\varphi\,\partial_\varphi G) = \nu G.$$

With natural flux at both endpoints its eigenvalues are $l(l+\eta)$, $l = 0, 1, \ldots$. In $u = \cos\varphi$ this is the Jacobi problem with weight $(1-u^2)^{(\eta-1)/2}$. Its polynomials are complete because both endpoint parameters exceed -1: continuous functions are dense for this finite weighted measure, and uniform polynomial approximation gives density of polynomials. In particular $G_0 = 1$ and $G_1 = \cos\varphi$. For this transverse mode the other equation is

$$-\sin^{-1-\eta}\vartheta\,\partial_\vartheta(\sin^{1+\eta}\vartheta F') + \frac{l(l+\eta)}{\sin^2\vartheta}F = \Lambda F.$$

For $l = 0$, the two endpoint behaviors are 1 and $\vartheta^{-\eta}$. At the positive pole zero trace selects the second; at the negative pole natural zero flux selects the first. Set $z = (1-\cos\vartheta)/2$ and $F = z^\sigma p(z)$. The remaining Jacobi weight is $z^\sigma(1-z)^{-\sigma}$, and

$$F_j = z^\sigma P_j^{(\sigma,-\sigma)}(1-2z), \qquad \Lambda_j = (\sigma+j)(\sigma+j+b_G).$$

For $l \geq 1$ finite energy selects $\sin^l\vartheta$ at each pole. The alternative $\vartheta^{-l-\eta}$ has divergent energy. Factoring $F = \sin^l\vartheta p(\cos\vartheta)$ gives Jacobi parameters $l + \eta/2$ at both ends, hence eigenvalues $(l+j)(l+j+b_G)$. Orthogonality and polynomial completeness give the full basis, with no extra endpoint realization.

Direct calculation gives $L_\eta H = 0$ off the ray: with $r = |q|$, $\Delta_\eta r = (2+\eta)/r$ and $|\nabla(r-X)|^2 = 2(r-X)/r$, so the remaining factor is $\eta + 2\sigma = 0$. Its weighted normal flux vanishes on the equator off the ray. Its positivity and degree identify the principal mode. The integer branch has only $l = 1, j = 0$ at degree one, giving Z.

Lemma 10.1 allows separated radial tests times every angular eigenfunction. Direct cutoff errors for the $l = 0$ branch are $O(\varepsilon^{-\eta})$ and for $l \geq 1$ are $O(\varepsilon^{\eta+2l})$, both tending to zero. Thus harmonicity off the ray suffices without pairing a nonsmooth eigenfunction with an uncontrolled source. Each radial coefficient obeys

$$c'' + (D-1)c'/r - \Lambda c/r^2 = 0, \qquad c = Ar^\alpha + Br^{-b_G-\alpha}.$$

Local boundedness excludes the negative power, and growth excludes $\alpha > Q$. Completeness leaves precisely the modes stated, in angular L^2 and hence almost everywhere. A distribution supported on the ray may remain; the argument does not assert unrestricted entire harmonicity. □

10.3. **Choice of scales and nonzero slit limits.**

Lemma 10.3 (Record and tail selection)**.** *Let M be continuous, finite and nondecreasing. Suppose $M(R)/R^Q \to 0$ and $0 < p < Q$. If $M(R)/R^p$ is unbounded, there are $R_j \to \infty$, $A_j = M(R_j)$, with $A_j/R_j^p \to \infty$ and*

$$M(R_j r)/A_j \leq r^p \qquad (R_0/R_j \leq r \leq 1), \tag{10.2}$$

$$M(R_j r)/A_j \leq r^Q \qquad (r \geq 1). \tag{10.3}$$

Proof. Choose record radii $t_j \to \infty$ with $L_j = M(t_j)/t_j^p = \max_{R_0 \le u \le t_j} M(u)/u^p \to \infty$. Choose $R_j \ge t_j$ maximizing $M(u)/u^Q$ on $[t_j, \infty)$; continuity and the zero limit ensure attainment. Then

$$\frac{A_j}{R_j^p} \ge L_j (R_j/t_j)^{Q-p} \to \infty.$$

For $R_0 \le u \le t_j$,

$$\frac{M(u)}{A_j} \le (u/t_j)^p (t_j/R_j)^Q = (u/R_j)^p (t_j/R_j)^{Q-p} \le (u/R_j)^p.$$

For $t_j \le u \le R_j$ the tail maximum gives $(u/R_j)^Q \le (u/R_j)^p$; for $u \ge R_j$ it gives $(u/R_j)^Q$. These are the desired inequalities. Also $A_j \to \infty$, so bounded original radii vanish after normalization. □

Lemma 10.4 (Compactness at selected scales)**.** *Let V be continuous, even in Y, and locally in $H^1(\rho)$, with both phases weak subsolutions and zero trace on $\mathcal{L}$. Suppose its dilated source supports eventually avoid every compact set disjoint from $\mathcal{L}$. For scales in Lemma 10.3 applied to $M(R) = \sup_{B_R} |V|$, a subsequence of $U_j(q) = V(R_j q)/A_j$ converges weakly in local $H^1(\rho)$ and strongly in local $L^2(\rho)$ to a nonzero even slit-harmonic U with zero ray trace. Almost everywhere,*

$$|U(q)| \le |q|^p \ (0 < |q| \le 1), \qquad |U(q)| \le |q|^Q \ (|q| \ge 1).$$

Proof. The selection bounds give uniform bounds on each fixed ball and $\sup_{B_1} |U_j| = 1$. Testing the phase subsolution inequalities by their product with a squared cutoff and adding gives

$$\int \zeta^2 \rho |\nabla U_j|^2 \le 4 \int \rho U_j^2 |\nabla \zeta|^2.$$

This yields local weighted H^1 bounds. The weight $|Y|^\eta$ is A_2. Weighted Rellich compactness here can be seen directly from the local uniform bounds, ordinary Rellich off $|Y| \le \delta$, and the bound on the remaining layer by its weighted mass $O(\delta^{1+\eta})$. Diagonal extraction gives the stated convergence.

The weighted local boundedness estimate [13, Theorem 2.3.1], applied to the two phases, gives

$$1 \le C \left(\int_{B_2} \rho U_j^2 \right)^{1/2}.$$

Strong convergence therefore implies $U \ne 0$. Concentration in the shrinking source layer cannot erase the normalization. By the assumed support avoidance, on compact sets away from $\mathcal{L}$ the rescaled source is absent for sufficiently large j; hence U is weakly harmonic there. The bounded composed trace in Lemma 10.1 preserves zero line trace under weak convergence. To identify it with angular pole trace, approximate on a slightly larger annulus by smooth functions. Polar Fubini gives convergence in $L^2(\,\mathrm{d}r; H^1_\eta(S^2))$; bounded angular point evaluation gives convergence in $L^2(\,\mathrm{d}r)$. The composed ambient trace has the same convergence, and both traces coincide for smooth approximants. Their limits agree. Thus almost every angular slice is in the slit form domain. Lemma 10.1 extends the equation to all compactly supported zero-trace form tests. Finally pass the selection bounds at countably many fixed radii and then use monotonicity. For each fixed $r > 0$, the inner bound applies once $R_j r \ge R_0$. Only after taking $j \to \infty$ is r sent to zero. □

Corollary 10.5 (Two-sided spectral windows)**.** *Suppose a slit-harmonic function in the form domain above satisfies* $\operatorname{ess\,sup}_{B_r} |v| \le C r^p$ *for* $0 < r \le 1$ *and* $\operatorname{ess\,sup}_{B_r} |v| \le C r^Q$ *for* $r \ge 1$*, with* $0 < p \le Q$*. Only positive homogeneities in* $[p, Q]$ *can occur. Below degree two these are exactly* $\sigma, 1, \lambda$*. The degree-*λ *space is simple and is spanned by*

$$\Phi_\lambda(q) = (r - X)^\sigma (X + \sigma r), \qquad \partial_X \Phi_\lambda = (1 - \sigma^2)(r - X)^\sigma, \quad r = |q|. \tag{10.4}$$

Proof. The radial equation in Lemma 10.2 has roots α and $-b_G-\alpha$. The inner bound removes the negative branch and every $\alpha<p$; the outer bound removes every $\alpha>Q$. Completeness yields the assertion, including equality at the endpoints. For the displayed generator, put $H=(r-X)^\sigma$. Then $\partial_X H=-\sigma H/r$ and $\Delta_\eta(rH)=2H/r$, since $2\sigma+b_G=1$. Thus $L_\eta(XH+\sigma rH)=0$ off the ray. The same positive-pole energy calculation as for H puts it in the slit domain, and the simple $\sigma+1$ branch identifies it. Direct differentiation gives the last formula. □

11. Automatic tail and canonical extension

We keep the intrinsic full-blowdown class and notation of Section 9. In the noninvariant case we use Lemma 9.3 to orient and normalize the contact. The slit tools in Section 10 are local weighted statements; their application below precedes ordinary-tail entry. Put $\sigma=s-1/2$ and $\lambda=1+\sigma$.

11.1. **Positivity before derivative energy.**

Lemma 11.1 (The positive horizontal derivative)**.** *In the normalized noninvariant case the bulk derivative*

$$W=-U_X=-V_X$$

is continuous, nonnegative and nonzero, belongs to $H^1_{\mathrm{loc}}(|Y|^\eta)$, and vanishes on $\iota(K)$. Moreover

$$L_\eta W=2\iota_*(\partial_X\widehat{\mu})\ge 0,\qquad \partial_X\widehat{\mu}\ \text{is a nonnegative Radon measure},\qquad \operatorname{supp}(\partial_X\widehat{\mu})\subset K. \tag{11.1}$$

For every compactly supported smooth cutoff ζ,

$$\int\zeta^2|Y|^\eta|\nabla W|^2\le 4\int|Y|^\eta W^2|\nabla\zeta|^2. \tag{11.2}$$

Initially $W(q)\le C(1+|q|)$.

Proof. The intrinsic local tangential estimates, scaled using quadratic growth, give continuous tangential derivatives in the closed halfspaces and $\sup_{B_R}|\nabla_{X,Z}U|\le C(1+R)$. For $0<h\le 1$ use the already local-energy quotients

$$W_h(q)=\frac{U(q)-U(q+he_X)}{h}.$$

Their thin traces are nonnegative by the monotonicity in Lemma 9.3; they are harmonic for $Y>0$, continuous up to the plane, and bounded by $C(1+|q|)$ uniformly in h. The halfspace comparison 3.3 applies because $1<2s$. Thus $W_h\ge 0$ in the upper halfspace and, by evenness, everywhere. Local tangential C^1 regularity makes $W_h\to W=-U_X$ locally uniformly. Hence $W\ge 0$. Its trace is nonzero by noninvariance and zero on K by contact regularity. No energy of W has been assumed in this positivity step.

Positive recession gives $W_h=0$ on $\iota(K)$. On its complement the untranslated source is absent and

$$L_\eta W_h=\frac{2}{h}\iota_*[\widehat{\mu}(\cdot+he_X)]\ge 0.$$

To glue this inequality across the zero set, let $\phi\ge 0$ be a compact smooth test and let χ_ε increase from zero to one between ε and 2ε. The energy test $\phi\chi_\varepsilon(W_h)$ is supported in the positive set of W_h, where the displayed inequality is valid. It gives

$$\int|Y|^\eta\chi_\varepsilon(W_h)\nabla W_h\cdot\nabla\phi+\int|Y|^\eta\phi\chi_\varepsilon'(W_h)|\nabla W_h|^2\le 0.$$

The second term is nonnegative and can be discarded. The Sobolev gradient vanishes almost everywhere on the zero set; dominated convergence as $\varepsilon\downarrow 0$ proves $\int|Y|^\eta\nabla W_h\cdot\nabla\phi\le 0$ globally. The tests are justified first in the open positive set by energy approximation. This argument does not use a pressure density or a measure-zero contact boundary.

Test the resulting nonnegative subsolution with $\zeta^2 W_h$. Cauchy–Schwarz yields

$$\int \zeta^2 |Y|^\eta |\nabla W_h|^2 \le 4 \int |Y|^\eta W_h^2 |\nabla \zeta|^2.$$

The right side is bounded uniformly for $h \le 1$. Weak local H^1 compactness and the local uniform limit give $W \in H^1_{\mathrm{loc}}$, its global subsolution inequality and (11.2). Differentiating the original source equation in distributions gives $L_\eta W = 2\iota_*(\partial_X \widehat{\mu})$. Testing with nonnegative products whose normal factor equals one on the plane shows that $\partial_X \widehat{\mu}$ is a positive planar distribution, hence a Radon measure. Distributional differentiation does not enlarge support, proving (11.1). □

11.2. **A spectral window for the positive derivative.**

Proposition 11.2 (Sublinear derivative growth)**.** *For every $\delta \in (\sigma, 1)$ there is C_δ such that*

$$0 \le W(q) \le C_\delta (1 + |q|^\delta). \tag{11.3}$$

Proof. Choose an outer exponent q_0 with

$$\sigma < \delta < 1 < q_0 < 1 + \sigma.$$

The initial linear bound gives $M_W(R)/R^{q_0} \to 0$. If $M_W(R)/R^\delta$ were unbounded, Lemma 10.3 would give $R_j \to \infty$, $A_j = M_W(R_j)$, and $A_j/R_j^\delta \to \infty$, with the inner r^δ and outer r^{q_0} envelopes. Set $W_j(q) = W(R_j q)/A_j$. We check the compactness hypotheses explicitly. The positive phase of W is its global weak subsolution from Lemma 11.1; its negative phase is zero. Its local energy was proved there. Its continuous trace is zero on the exact ray $\mathcal{L}$. Finally $\operatorname{supp} L_\eta W \subset \iota(K)$, and the qualitative collapse in Lemma 9.3 removes this support from every fixed compact set off $\mathcal{L}$ after scaling.

Caccioppoli and the two growth envelopes give uniform local weighted H^1 bounds. Lemma 10.4 therefore yields a weak local H^1 and strong local L^2 limit W_∞, even and slit-harmonic. It is nonnegative. The composed bounded trace

$$H^1(|Y|^\eta) \longrightarrow H^s(\mathbb{R}^2) \longrightarrow H^{s-1/2}(\mathbb{R})$$

preserves zero ray trace under weak convergence. On annuli its identification with angular positive-pole trace follows from the smooth-approximation argument of Lemma 10.4. Thus the limit has exactly the Friedrichs domain of Lemma 10.1; no Dirichlet condition is imposed at the negative pole, and the capacity-zero tip introduces no extra condition.

The normalization cannot disappear inside a shrinking source layer. Indeed the weighted local boundedness inequality for the nonnegative subsolutions gives

$$1 = \sup_{B_1} W_j \le C \left(\int_{B_2} |Y|^\eta W_j^2 \right)^{1/2}.$$

Strong convergence implies $W_\infty \not\equiv 0$. The inner and outer envelopes pass to the limit, with exponents δ and q_0. By Corollary 10.5, the principal degree σ is excluded by the inner exponent, and the next noninteger degree $1 + \sigma$ by the outer exponent. The only remaining degree is one, whose entire eigenspace is spanned by Z. Every nonzero multiple of Z changes sign, contrary to $W_\infty \ge 0$. This contradiction proves (11.3). No principal-pressure moment or ordinary-tail result has been used. □

11.3. **Recovering the transverse quadratic channel.**

Proposition 11.3 (Strict power improvement)**.** *For every $\delta \in (\sigma, 1)$ in the normalized noninvariant intrinsic class,*

$$M_V(R) + M_P(R) \le C_\delta (1 + R^{1+\delta}). \tag{11.4}$$

Consequently, in any fixed original coordinates and for every $\lambda < p < 2s$,

$$M_V(R) + M_P(R) \le C_p(1 + R^p).$$

In particular P has an absolute ordinary fractional tail.

Proof. Fix $\delta \in (\sigma, 1)$. The whole halfspace $X < 0$ is source-free, including across $Y = 0$. For every fixed aperture $c_0 > 0$, the source-free tangential derivative estimates and (11.3) give

$$\begin{aligned} |W_X(-T,Z,Y)| + |W_Z(-T,Z,Y)| &\le C_{\delta,c_0} T^{\delta-1}, \\ |W_{ZZ}(-T,Z,Y)| &\le C_{\delta,c_0} T^{\delta-2} \end{aligned} \qquad \begin{cases} T \ge 1, \\ |Z| + |Y| \le c_0 T. \end{cases} \tag{11.5}$$

For points near the plane use source-free balls centered on the plane with radius comparable to T; away from it use comparable interior balls. All these balls stay in $X < 0$, and their points have norm at most a fixed multiple of T. This gives constants uniform throughout each fixed-aperture cone. The same second tangential derivative estimate for V, together with $M_V(R) = o(R^2)$, gives

$$V_{ZZ}(-T,Z,Y) \longrightarrow 0 \quad (T \to \infty) \quad \text{for every fixed } (Z,Y). \tag{11.6}$$

It is the full blowdown that supplies this limit.

For a fixed transverse point, integrate $\partial_X V_{ZZ} = -W_{ZZ}$ from negative infinity, using (11.6). If $|Z| + |Y| \le c_0 T$, the whole path $(-t, Z, Y)$, $t \ge T$, stays in the same cone. Since $\delta < 1$, the integral converges and gives

$$\begin{aligned} |V_{ZZ}(-T,Z,Y)| &\le C_{\delta,c_0} \int_T^\infty t^{\delta-2}\,\mathrm{d}t \le \frac{C_{\delta,c_0}}{1-\delta} T^{\delta-1}, \\ |V_{XX}(-T,Z,Y)| &= |W_X(-T,Z,Y)| \le C_{\delta,c_0} T^{\delta-1}. \end{aligned} \tag{11.7}$$

Although the limiting value was fixed at each transverse point, the resulting bounds are uniform on the whole cone. This fixes the integration constant in the transverse quadratic channel; an integration of V_X from a finite slice would not do so.

The normalization gives $V(0,0,0) = 0$. Since $V_X = -W$,

$$|V(-T,0,0)| \le C_\delta(1 + T^{1+\delta}).$$

Starting at the fixed source-free point $(-1,0,0)$ and using $V_{XZ} = -W_Z$ also gives, for $T \ge 1$,

$$|V_Z(-T,0,0)| \le |V_Z(-1,0,0)| + C_\delta \int_1^T t^{\delta-1}\,\mathrm{d}t \le C_\delta(1 + T^\delta).$$

Fix $R \ge 1$ and use (11.7) at $X = -2R$. Two integrations in Z, starting at $Z = 0$, prove

$$|V(-2R,Z,0)| \le C_\delta R^{1+\delta}, \qquad |Z| \le R. \tag{11.8}$$

We next recover the normal variable without discarding a possible normal flux. The even weak solution is source-free on the entire negative slab, so its conormal flux on $Y = 0$ is zero there. More explicitly, the tangential derivatives in (11.7) are continuous and bounded on smaller negative slabs, and the weak equation gives $(Y^\eta V_Y)_Y = -Y^\eta(V_{XX} + V_{ZZ})$ for $Y > 0$. Its right side is integrable at zero. Thus the flux has a limit; evenness and the absence of a plane source force that limit to be zero, first in tangential distributions and hence pointwise in the integrated identity. Therefore, for $|Z|, Y \le R$,

$$Y^\eta V_Y(-2R,Z,Y) = -\int_0^Y t^\eta (V_{XX} + V_{ZZ})(-2R,Z,t)\,\mathrm{d}t. \tag{11.9}$$

Taking $c_0 = 1$ in the cone bounds, and enlarging it harmlessly on the edges, yields $|V_Y(-2R,Z,Y)| \le C_\delta Y R^{\delta-1}$. Normal integration and (11.8) prove

$$|V(-2R,Z,Y)| \le C_\delta R^{1+\delta}, \qquad |Z|, |Y| \le R,$$

where evenness treats $Y<0$. Finally integrate $V_X=-W$ from $-2R$ to any $|X|\le R$, keeping (Z,Y) fixed. The path has length at most $3R$ and lies in B_{4R}; (11.3) gives (11.4).

For $\lambda<p<2s$ choose $\delta=p-1$. Then $\sigma<\delta<2\sigma<1$, so the asserted strict power follows. Undoing (9.4) adds only affine growth and a fixed translation, harmless because $p>1$. Continuity handles the local part of the tail; at infinity its polar integral is bounded by

$$C\int_1^\infty r^{p-1-2s}\,\mathrm{d}r<\infty.$$

This argument has recovered a strict sub-$2s$ power from intrinsic data alone. □

11.4. **Tail entry and elimination of the vertical mode.**

Proposition 11.4 (Intrinsic tail and canonical extension)**.** *Every solution in the intrinsic full-blowdown class has*

$$P\in L^1_s(\mathbb{R}^2),\qquad V=\mathcal{E}_sP,\qquad \widehat{\mu}=e_s\mu,\qquad \mu=\mathcal{A}_sP,$$

where

$$e_s=\frac{2^{1-2s}\Gamma(1-s)}{\Gamma(s)},\qquad \mathcal{E}_sP(x,Y)=\frac{s}{\pi}\int_{\mathbb{R}^2}\frac{|Y|^{2s}P(\xi)}{(|x-\xi|^2+Y^2)^{1+s}}\,\mathrm{d}\xi\quad(Y\neq0).$$

The extension is continuous, even and locally weighted H^1. In the noninvariant case it has every strict power in Proposition 11.3. The measure μ is nonnegative and supported on K.

Proof. If K is bounded, Proposition 9.1 gives the tail. If $u_X\equiv0$, Proposition 9.2 gives it. All remaining cases are covered by the normalized geometry and Proposition 11.3. Thus the ordinary tail has been proved in every branch before an extension is constructed.

The original energy trace gives $P\in H^s_{\mathrm{loc}}(\mathbb{R}^2)$. The displayed Poisson integral is absolutely convergent by the tail. For its local energy, choose a smooth compact cutoff equal to one on a neighborhood of a fixed thin ball. Its product with P is compactly supported H^s data and has an energy extension. In a smaller cylinder the complementary integral has the form

$$|Y|^{2s}F(x,Y^2),$$

where F and all derivatives needed locally are bounded: the integration region is separated from x, and at infinity its kernels and derivatives are dominated by the ordinary-tail weight. The normal weighted energy exponent is $\eta+2(2s-1)=2s-1>-1$; the tangential energy is also integrable. Consequently $\mathcal{E}_sP\in H^1_{\mathrm{loc}}(|Y|^\eta)$. The local approximate-identity property and the same separated-tail estimate give continuity with trace P. This tail portion need not be ordinarily smooth across $Y=0$.

The kernel has mass one. For $|q|\le R$, split the integral at $|\xi|=2R$. Its near contribution is bounded in absolute value by $M_P(2R)=o(R^2)$. On the far part $|x-\xi|\ge|\xi|/2$, so its absolute value is at most

$$CR^{2s}\int_{|\xi|>2R}|P(\xi)||\xi|^{-2-2s}\,\mathrm{d}\xi=o(R^2).$$

Thus $M_{\mathcal{E}_sP}(R)=o(R^2)$. The difference $V-\mathcal{E}_sP$ is a continuous zero-trace, upper-halfspace harmonic local-energy function of subquadratic growth. The zero-trace Liouville statement 3.5 gives, by evenness,

$$V=\mathcal{E}_sP+c|Y|^{2s}\tag{11.10}$$

for some $c\in\mathbb{R}$. Zero trace alone does not remove this term.

For the Fourier multiplier $|\xi|^{2s}$ the conormal formula is $L_\eta\mathcal{E}_sP=-2e_s\iota_*(\mathcal{A}_sP)$, whereas $L_\eta(c|Y|^{2s})=4sc\iota_*(\,\mathrm{d}x)$. Hence

$$e_s\mathcal{A}_sP=\widehat{\mu}+2sc\,\mathrm{d}x.\tag{11.11}$$

To determine c, choose a fixed thin ball B about $(0,1)$ whose closure stays away from $Z=0$. The full quadratic blowdown and $a>0$ imply $u>0$ on RB for every sufficiently large R; consequently

$\widehat{\mu}(RB) = 0$. Take $\phi \in C_c^\infty(B)$ with $\int \phi \, \mathrm{d}x \neq 0$, and put $g = \mathcal{A}_s\phi$. The standard compact-test kernel estimate gives $|g(x)| \le C(1+|x|)^{-2-2s}$. Testing (11.11) with $\phi(x/R)$ and dividing by R^2 gives

$$e_s R^{-2-2s} \int_{\mathbb{R}^2} P(y) g(y/R) \, \mathrm{d}y = 2sc \int_{\mathbb{R}^2} \phi \, \mathrm{d}x. \tag{11.12}$$

For $R \ge 1$,

$$R^{-2-2s}|g(y/R)| \le \frac{C}{(R+|y|)^{2+2s}} \le \frac{C'}{1+|y|^{2+2s}},$$

and this expression tends to zero for each fixed y. The already proved absolute ordinary tail permits dominated convergence in (11.12). Its left side tends to zero, so $c = 0$. Equations (11.10) and (11.11) now give both canonicity and the exact pressure conversion. The sign and support assertions follow from those of $\widehat{\mu}$. No classification theorem has entered this proof. □

11.5. **The interface for derivative-pressure analysis.**

Corollary 11.5 (Canonical derivative and positive ordinary pressure)**.** *In the oriented noninvariant class put $w = -P_X = W|_{Y=0}$. Then $w \ge 0$ is continuous, nonzero, zero on K, locally H^s, and has an absolute ordinary tail. For every $\sigma < \delta < 1$,*

$$w(x) + W(x,Y) \le C_\delta(1 + |(x,Y)|^\delta), \qquad W = \mathcal{E}_s w.$$

There is a nonnegative Radon measure ν such that

$$\nu = \partial_X \mu = -\mathcal{A}_s w, \qquad \operatorname{supp}\nu \subset K, \qquad L_\eta W = 2e_s \iota_* \nu. \tag{11.13}$$

The energy inequality (11.2) *holds. These assertions do not require a density for either μ or ν.*

Proof. All pointwise assertions and the bulk energy follow from Lemmas 9.3 and 11.1, and Proposition 11.2. The energy trace gives $w \in H^s_{\mathrm{loc}}$, and its δ-growth with $\delta < 1 < 2s$ gives the ordinary tail.

It remains to identify the extension, rather than merely its trace. Let

$$\omega_h(x) = \frac{P(x) - P(x + he_X)}{h} = \frac{1}{h}\int_0^h w(x + te_X) \, \mathrm{d}t, \qquad 0 < h \le 1.$$

The functions ω_h converge locally uniformly to w and have a common bound $C_\delta(1+|x|^\delta)$. Canonicity from Proposition 11.4, translation invariance of the absolutely convergent Poisson integral and $Q_X = 0$ give

$$W_h = \mathcal{E}_s \omega_h.$$

Since the Poisson kernel has a finite δ moment for $\delta < 2s$, dominated convergence yields $W = \mathcal{E}_s w$ in the open halfspaces; continuity gives equality also on the plane. This quotient argument justifies differentiating the canonical extension without an unproved remote derivative bound.

By (11.1) and $\widehat{\mu} = e_s\mu$, the distribution $\nu = \partial_X \mu$ is a nonnegative Radon measure supported on K, and $L_\eta W = 2e_s\iota_*\nu$. The conormal identity for $W = \mathcal{E}_s w$ is $L_\eta W = -2e_s\iota_*(\mathcal{A}_s w)$, proving (11.13). Equivalently, the uniformly tail-controlled quotients above permit passage to distributions in $\mathcal{A}_s\omega_h = (\mu - \mu(\cdot + he_X))/h$. □

12. One-dimensional inverse kernels and interval potentials

This section collects the one-dimensional tools used in the cylindrical classification and, for $s > 1/2$, in the mesoscopic argument. Unless a smaller range is stated, $0 < s < 1$, $t = 1 - s$, and $\kappa = 2s - 1$. The operator $\mathcal{A}_s^{(1)}$ has Fourier multiplier $|\zeta|^{2s}$ and singular-integral constant

$$c_{1,s} = \frac{4^s \Gamma(s + 1/2)}{\sqrt{\pi}\,|\Gamma(-s)|}.$$

All distributional pairings below use compactly supported smooth tests. We write $L^1_s(\mathbb{R})$ for the ordinary absolute-tail space with weight $(1+z^2)^{-1/2-s}$.

Lemma 12.1 (Exact dimension reduction)**.** *Let $p \in C(\mathbb{R}) \cap L^1_s(\mathbb{R})$ and $P(X,Z) = p(Z)$. Then*

$$\int_{\mathbb{R}^2} \frac{|P(X,Z)|}{(1+X^2+Z^2)^{1+s}}\,\mathrm{d}X\,\mathrm{d}Z = \frac{\sqrt{\pi}\Gamma(s+1/2)}{\Gamma(s+1)} \int_{\mathbb{R}} \frac{|p(Z)|}{(1+Z^2)^{1/2+s}}\,\mathrm{d}Z, \tag{12.1}$$

and

$$\mathcal{A}^{(2)}_s P = \mathrm{d}X \otimes \mathcal{A}^{(1)}_s p \quad \text{in distributions.} \tag{12.2}$$

If the left side is a nonnegative Radon measure μ, then $\nu = \mathcal{A}^{(1)}_s p$ is a nonnegative Radon measure and $\mu = \mathrm{d}X \otimes \nu$ as actual measures. If, in addition, $\operatorname{supp}\mu \subset \{P = -aZ^2\}$*, then* $\operatorname{supp}\nu \subset \{p = -aZ^2\}$.

Proof. Tonelli and the beta integral give (12.1). For $\Phi \in C^\infty_c(\mathbb{R}^2)$ set $\overline{\Phi}(z) = \int_{\mathbb{R}} \Phi(X,z)\,\mathrm{d}X$. The symmetric second-difference formula, or the Fourier transform at $\xi_X = 0$, gives

$$\int_{\mathbb{R}} \mathcal{A}^{(2)}_s \Phi(X,z)\,\mathrm{d}X = \mathcal{A}^{(1)}_s \overline{\Phi}(z), \tag{12.3}$$

with the exact normalization

$$c_{2,s} \int_{\mathbb{R}} (q^2+r^2)^{-1-s}\,\mathrm{d}q = c_{1,s}|r|^{-1-2s}.$$

For completeness, the absolute X integral of $\mathcal{A}^{(2)}_s \Phi(X,z)$ is bounded when z is bounded and is at most $C(1+|z|)^{-1-2s}$ for large $|z|$. Indeed, outside a fixed disk the operator of the compact test is $O((X^2+z^2)^{-1-s})$; inside that disk it is bounded. The ordinary tail of p therefore justifies pairing (12.3) with p and applying Fubini. This proves (12.2) without a slice operation on a measure. Choose $\chi \in C^\infty_c(\mathbb{R})$ nonnegative with $\int \chi = 1$ and test μ against $\chi(X)\varphi(Z)$. Nonnegative φ gives positivity of the distribution ν, hence its Radon representation. The product identity now identifies the two Radon measures. Testing with φ supported off $\{p = -aZ^2\}$ proves the support assertion. □

Lemma 12.2 (Inverse kernels and safe composition)**.** *The following locally integrable kernels are fundamental solutions for $\mathcal{A}^{(1)}_s$ in the ordinary-tail distributional sense:*

$$E_s(z) = \begin{cases} c_s|z|^{2s-1}, & 0 < s < 1/2, \\ -\pi^{-1}\log|z|, & s = 1/2, \\ -d_s|z|^{2s-1}, & 1/2 < s < 1, \end{cases} \qquad c_s = \frac{\Gamma(1/2-s)}{4^s\sqrt{\pi}\Gamma(s)} > 0 \quad (s < 1/2), \tag{12.4}$$

where, in the upper range,

$$d_s = \frac{1}{2\Gamma(1+\kappa)\sin(\pi\kappa/2)} = -\frac{\Gamma(1/2-s)}{4^s\sqrt{\pi}\Gamma(s)} > 0. \tag{12.5}$$

For every $0 < s < 1$ one has the two full-line identities

$$\mathcal{A}^{(1)}_s E_s = \delta_0, \qquad -E_s'' = \mathcal{A}^{(1)}_{1-s}\delta_0. \tag{12.6}$$

If ν is a bounded compactly supported density, then $E_s * \nu$ is continuous, belongs to $L^1_s(\mathbb{R})$, and

$$\mathcal{A}^{(1)}_s (E_s * \nu) = \nu, \qquad \mathcal{A}^{(1)}_{1-s}\nu = -(E_s * \nu)''. \tag{12.7}$$

These conclusions also hold for an arbitrary compactly supported finite measure when $s > 1/2$. The potential is bounded and decays at infinity when $s < 1/2$, has growth $O(1+\log(2+|z|))$ when $s = 1/2$, and has growth $O(1+|z|^{2s-1})$ when $s > 1/2$.

Proof. We prove the identities separately in the three kernel regimes, keeping the zero frequency in the distributional calculation.

For $0 < s < 1/2$, the heat kernel $H_r(z) = (4\pi r)^{-1/2} e^{-z^2/(4r)}$ gives

$$E_s(z) = \frac{1}{\Gamma(s)} \int_0^\infty r^{s-1} H_r(z)\, \mathrm{d}r, \qquad z \neq 0. \tag{12.8}$$

The substitution $u = z^2/(4r)$ evaluates the integral and gives c_s in (12.4). If $\varphi \in C_c^\infty(\mathbb{R})$, then $E_s|\mathcal{A}_s^{(1)}\varphi|$ is integrable: the kernel is locally integrable and the product is $O(|z|^{-2})$ at infinity. Positivity of the heat kernel therefore justifies the first interchange in

$$\begin{aligned}\langle E_s, \mathcal{A}_s^{(1)}\varphi\rangle &= \frac{1}{2\pi\Gamma(s)} \int_0^\infty \int_{\mathbb{R}} r^{s-1} e^{-r\zeta^2} |\zeta|^{2s} \widehat{\varphi}(\zeta)\, \mathrm{d}\zeta\, \mathrm{d}r \\ &= \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{\varphi}(\zeta)\, \mathrm{d}\zeta = \varphi(0).\end{aligned}$$

The Fourier–heat interchange is absolutely convergent because the r integral of its absolute multiplier is $\Gamma(s)$ for $\zeta \neq 0$, leaving $|\widehat{\varphi}| \in L^1$. Pairing (12.8) instead with $-\varphi''$ is also legitimate. After integration in r the absolutely integrable Fourier expression is

$$\langle -E_s'', \varphi\rangle = \frac{1}{2\pi} \int_{\mathbb{R}} |\zeta|^{2-2s} \widehat{\varphi}(\zeta)\, \mathrm{d}\zeta = (\mathcal{A}_{1-s}^{(1)}\varphi)(0). \tag{12.9}$$

This proves both identities in the lower range without multiplying singular Fourier distributions.

At $s = 1/2$, use the renormalized limit

$$c_s(|z|^{2s-1} - 1) \longrightarrow -\pi^{-1} \log|z| \quad \text{as } s \uparrow 1/2, \tag{12.10}$$

since $(1-2s)c_s \to \pi^{-1}$. Subtracting c_s from the fundamental kernel does not change its pairing with $\mathcal{A}_s^{(1)}\varphi$, whose integral is zero. For fixed $s_0 \in (0, 1/2)$ and $s_0 \leq s < 1/2$, the mean-value integral for $(r^{-\varepsilon} - 1)/\varepsilon$ bounds the absolute value in (12.10) by

$$C \log|z| \quad (|z| \geq 1), \qquad C|\log|z||\, |z|^{-(1-2s_0)} \quad (0 < |z| \leq 1).$$

The second bound is locally integrable. Symmetric second differences show that $\mathcal{A}_s^{(1)}\varphi$ is uniformly bounded on compact sets, converges pointwise to $\mathcal{A}_{1/2}^{(1)}\varphi$, and has the uniform far bound $C(1 + |z|)^{-1-2s_0}$. Dominated convergence gives $\mathcal{A}_{1/2}^{(1)} E_{1/2} = \delta_0$. The same kernel convergence is local L^1 convergence. Its second derivatives converge in distributions; the right side of (12.9) converges by the rapid decay of $\widehat{\varphi}$. This proves the second identity at the endpoint as well.

For $s > 1/2$, one has $0 < \kappa < 1$ and the convergent cosine identity

$$|z|^\kappa = C_\kappa \int_{\mathbb{R}} (1 - \cos(\zeta z)) |\zeta|^{-1-\kappa}\, \mathrm{d}\zeta, \qquad C_\kappa = \frac{\Gamma(1+\kappa)\sin(\pi\kappa/2)}{\pi}. \tag{12.11}$$

Indeed, at $z = 1$ integration by parts gives

$$\int_{\mathbb{R}} (1 - \cos\zeta)|\zeta|^{-1-\kappa}\, \mathrm{d}\zeta = \frac{2}{\kappa} \int_0^\infty q^{-\kappa} \sin q\, \mathrm{d}q = \frac{2}{\kappa}\Gamma(1-\kappa)\cos(\pi\kappa/2).$$

The sine integral follows by Abel damping in the Gamma integral; integration by parts bounds the discarded tail uniformly. Reflection and duplication give its value C_κ^{-1} and $d_s C_\kappa = (2\pi)^{-1}$, as well as (12.5). For a real compact smooth test, $|z|^\kappa |\mathcal{A}_s^{(1)}\varphi(z)|$ is integrable, again with far exponent -2. Absolute integration of (12.11) against this test operator is therefore valid. Since $\int \mathcal{A}_s^{(1)}\varphi = 0$ and

$$\int_{\mathbb{R}} \cos(\zeta z) \mathcal{A}_s^{(1)}\varphi(z)\, \mathrm{d}z = |\zeta|^{2s} \operatorname{Re}\widehat{\varphi}(\zeta),$$

we obtain

$$\langle E_s, \mathcal{A}_s^{(1)}\varphi\rangle = d_s C_\kappa \int_{\mathbb{R}} \operatorname{Re}\widehat{\varphi}(\zeta)\, \mathrm{d}\zeta = \varphi(0).$$

Testing the cosine identity with φ'' instead gives (12.9), because $\int \varphi'' = 0$ and $\int \cos(\zeta z)\varphi''(z)\,\mathrm{d}z = -\zeta^2 \operatorname{Re}\widehat{\varphi}(\zeta)$. The interchange is controlled by the compact integral $\int |z|^\kappa |\varphi''(z)|\,\mathrm{d}z$. Complex tests follow by linearity. Thus no term at frequency zero has been discarded in either identity.

For a bounded compact density, continuity of the potential follows from translation continuity of the locally integrable power or logarithmic kernel, with a smooth far part. If $s > 1/2$ and ν is merely a finite compact measure, the estimate

$$\big||z-y|^\kappa - |z'-y|^\kappa\big| \le |z-z'|^\kappa$$

gives continuity directly. The asserted far bounds follow from compact support and show ordinary-tail integrability. The absolute pairings $\int |E_s(z-y)\mathcal{A}_s^{(1)}\varphi(z)|\,\mathrm{d}z$ are uniformly bounded for y in a fixed compact set. Fubini against ν proves the first identity in (12.7); translated versions of (12.9), integrated against ν, prove the second. In particular, composition has been proved on tests on the whole line, including the origin, rather than inferred from a formal product of low-frequency multipliers. □

Lemma 12.3 (Bounded exterior-Dirichlet uniqueness)**.** *Let $0 < t < 1$ and let I be a bounded open interval. If $v \in L^\infty(\mathbb{R})$, $v = 0$ almost everywhere on $\mathbb{R} \setminus I$, and $\mathcal{A}_t^{(1)} v = 0$ in $\mathcal{D}'(I)$, then $v = 0$ almost everywhere. Consequently the zero-exterior equation with a prescribed distributional right side has at most one bounded solution. Moreover, with*

$$A_s^* = \frac{2^{1-2s}\Gamma(3/2-s)\Gamma(2-s)}{\sqrt{\pi}}, \qquad t = 1-s, \tag{12.12}$$

the unique bounded zero-exterior solution of $\mathcal{A}_t^{(1)}\nu = 2a$ on $(h-L, h+L)$ is

$$\nu(z) = \frac{a}{A_s^*}[L^2 - (z-h)^2]_+^{1-s}. \tag{12.13}$$

No boundary trace or finite-energy assumption is imposed.

Proof. We first record the exact barrier identities

$$\mathcal{A}_t^{(1)} B_t = 0, \qquad \mathcal{A}_t^{(1)} T_t = \mathfrak{d}_t := \frac{4^t\Gamma(t+1/2)\Gamma(t+1)}{\sqrt{\pi}}, \qquad |x| < 1, \tag{12.14}$$

where $T_t(x) = (1-x^2)_+^t$, and $B_t(x) = (1-x^2)^{t-1}$ for $|x| < 1$ and is zero outside. Endpoint values of B_t are immaterial. These are the $d = 1$, $\alpha = 2t$, $p = t-1, t$ cases of the power formula in [8, Theorem 1 and Table 1]. That formula uses the negative of our positive operator. In the singular case its beta factor is $\Gamma(-t)\Gamma(t)/\Gamma(0) = 0$, whereas the remaining factor ${}_2F_1(t+1/2, 1; 1/2; x^2)$ is finite at every $|x| < 1$. Thus the reciprocal-Gamma zero produces the first identity without an indeterminate product. For $p = t$ the second upper hypergeometric parameter is zero, and changing the sign of the beta factor yields $\mathfrak{d}_t$. Both choices have $p > -1$.

These pointwise identities are also distributional in the open interval, even for the singular barrier. To check this, let $g \in L^1(\mathbb{R})$ be compactly supported and C^2 locally on an open set containing the support of a compact smooth test φ. For the singular integral truncated at $|h| = \delta$, Fubini and symmetry give $\int g\mathcal{A}_{t,\delta}^{(1)}\varphi = \int \varphi\mathcal{A}_{t,\delta}^{(1)} g$. The left side converges by the uniform global bound for the truncated operator of φ. On the right, local second differences are bounded by $C|h|^2$, uniformly over the test support; the integral of $|h|^{1-2t}$ is finite. The separated far part is controlled by $\|g\|_1$. This proves the distributional identity. It applies to B_t, since its endpoint singularities have exponent $t - 1 > -1$. Scaling gives

$$B_{t,R}(x) = (R^2 - x^2)^{t-1}\mathbf{1}_{\{|x|<R\}}, \quad \mathcal{A}_t^{(1)} B_{t,R} = 0 \quad (|x| < R),$$

and $\mathcal{A}_t^{(1)}(R^2 - x^2)_+^t = \mathfrak{d}_t$ there.

For uniqueness, rescale I to $(-1,1)$ and set $M=\|v\|_\infty$. Let ρ_ε be a nonnegative smooth mollifier of mass one supported in $(-\varepsilon,\varepsilon)$, $0<\varepsilon<1/2$, and $v_\varepsilon=v*\rho_\varepsilon$. Then $|v_\varepsilon|\le M$, its support lies in $[-1-\varepsilon,1+\varepsilon]$, and

$$\mathcal{A}_t^{(1)}v_\varepsilon(x)=0\quad(|x|<1-\varepsilon). \tag{12.15}$$

Indeed $z\mapsto\rho_\varepsilon(x-z)$ is then a test supported in I. Commutation follows directly by inserting the convolution in the symmetric integral: its absolute integrand is controlled by

$$\|v\|_1\min\{\|\rho_\varepsilon''\|_\infty|h|^2,4\|\rho_\varepsilon\|_\infty\}|h|^{-1-2t},$$

which is integrable for fixed ε. Thus (12.15) is a pointwise classical identity for the smooth mollification.

Put $r=1-\varepsilon$, $R=1+\varepsilon$ and

$$H_\varepsilon(x)=M(4\varepsilon)^{1-t}B_{t,R}(x).$$

On $r\le|x|<R$ one has $0<R^2-x^2\le R^2-r^2=4\varepsilon$, so $H_\varepsilon\ge M$. Outside $[-R,R]$ both the mollification and barrier vanish. Hence $w=\pm v_\varepsilon-H_\varepsilon\le 0$ almost everywhere outside $(-r,r)$, including its endpoints, and w is smooth on a neighborhood of $[-r,r]$ and t-harmonic in $(-r,r)$. If w took a positive value, it would attain a positive global essential maximum m at an interior point x_0. Its first derivative there is zero, so the singular integral at x_0 is absolutely convergent near x_0 by a second-order Taylor bound. The negative singularities at $\pm R$ are integrable and separated from x_0. Thus

$$0=\mathcal{A}_t^{(1)}w(x_0)=c_{1,t}\int_{\mathbb{R}}\frac{m-w(y)}{|x_0-y|^{1+2t}}\,\mathrm{d}y\ge c_{1,t}m\int_{|y|>R}|x_0-y|^{-1-2t}\,\mathrm{d}y>0,$$

a contradiction. We have proved

$$|v_\varepsilon(x)|\le M\left(\frac{4\varepsilon}{(1+\varepsilon)^2-x^2}\right)^{1-t}\qquad(|x|<1-\varepsilon). \tag{12.16}$$

The right side tends uniformly to zero on every compact subset of $(-1,1)$. Approximate-identity convergence in local L^1 therefore gives $v=0$ almost everywhere. Translation and dilation prove the assertion on every bounded interval, and subtraction gives uniqueness for any fixed right side. Finally $\mathfrak{d}_{1-s}=2A_s^*$; the scaled torsion identity proves existence of (12.13) and the uniqueness just proved identifies it. The argument uses $t>0$ for integrability of the barrier and $t<1$ for comparison and the vanishing coefficient in (12.16); it has no restriction $t<1/2$. □

Lemma 12.4 (Interval beta identity)**.** *If $1/2<s<1$, then, for $L>0$ and $|z|\le L$,*

$$\int_{-L}^{L}|z-y|^{\kappa}(L^2-y^2)^t\,\mathrm{d}y=B(s,2-s)(L^2+\kappa z^2), \tag{12.17}$$

and

$$d_s\kappa B(s,2-s)=A_s^*=\frac{2^{1-2s}\Gamma(3/2-s)\Gamma(2-s)}{\sqrt{\pi}}. \tag{12.18}$$

Proof. It suffices to take $L=1$. Write $k=\kappa$, $K(r)=|r|^k$, $w(y)=(1-y^2)^t$ on $(-1,1)$, and $I(z)=\int_{-1}^{1}K(z-y)w(y)\,\mathrm{d}y$. The first derivative kernel is locally integrable, so I is C^1 in the open interval. Integrating the distributional derivative $\partial_y[(1-y^2)^{t+1}\partial_yK(z-y)]$ gives

$$0=2(t+1)\int_{-1}^{1}yw(y)K'(z-y)\,\mathrm{d}y+\int_{-1}^{1}(1-y^2)w(y)K''(z-y)\,\mathrm{d}y.$$

The second integral is distributional in z. The endpoint terms vanish because $(1-y^2)^{t+1}$ and its first derivative vanish at both endpoints. One rigorous implementation first smooths K and approximates the weight compactly inside the interval. On a compact interior z interval the kernel is smooth near the two endpoints, and the second derivative of the weight is integrable

there because $t>0$; thus the approximations pass to the displayed distributional identity. The homogeneity identities

$$rK'=kK,\qquad rK''=(k-1)K',\qquad r^2K''=k(k-1)K$$

and $y=z-(z-y)$ reduce it to

$$(1-z^2)I''+2(k+t)zI'-k(k+2t+1)I=0.$$

Since $k+2t=1$, this is

$$(1-z^2)I''+(k+1)zI'-2kI=0. \tag{12.19}$$

The leading coefficient is nonzero inside the interval. Ordinary regularity for a linear distributional ODE gives classical smoothness there. The initial values are $I(0)=B((k+1)/2,t+1)=B(s,2-s)$ and $I'(0)=0$ by evenness. The polynomial $B(s,2-s)(1+kz^2)$ has these initial values and solves (12.19), so ODE uniqueness proves the identity. Continuity extends it to the endpoints. Scaling gives the factor L^2 since $\kappa+2t+1=2$. Finally

$$\Gamma(3/2-s)=-\frac{\kappa}{2}\Gamma(1/2-s),\qquad B(s,2-s)=\Gamma(s)\Gamma(2-s),$$

and (12.5) give (12.18). This calculation never integrates the nonintegrable classical second derivative of the raw kernel at an interior point. □

Proposition 12.5 (The centered interval profiles)**.** *For $0<s<1$ and $L>0$, put $C_1=a/A_s^*$ and*

$$\nu_L(z)=C_1(L^2-z^2)_+^{1-s}. \tag{12.20}$$

The centered potential p_L is given by the following three formulas:

$$p_L(z)=c_sC_1\int_{-L}^{L}|z-y|^{2s-1}(L^2-y^2)^{1-s}\,\mathrm{d}y-\frac{aL^2}{1-2s},\qquad 0<s<1/2, \tag{12.21}$$

$$p_L(z)=-\frac{2a}{\pi}\int_{-L}^{L}\log|z-y|\sqrt{L^2-y^2}\,\mathrm{d}y+aL^2\log(L/2)-\frac{aL^2}{2},\qquad s=1/2, \tag{12.22}$$

$$p_L(z)=-d_sC_1\left[\int_{-L}^{L}|z-y|^{\kappa}(L^2-y^2)^t\,\mathrm{d}y-B(s,2-s)L^2\right],\qquad 1/2<s<1. \tag{12.23}$$

They satisfy

$$\mathcal{A}_s^{(1)}p_L=\nu_L,\qquad p_L(0)=0,\qquad -az^2\le p_L(z)\le 0,\qquad \{p_L=-az^2\}=[-L,L],$$
$$p_L\in C^1(\mathbb{R})\cap L^1_s(\mathbb{R}),\qquad p_L+az^2 \text{ is convex.}$$

Their pressure mass is

$$m_{s,L}:=\nu_L(\mathbb{R})=\frac{a}{A_s^*}B(1/2,2-s)L^{3-2s}. \tag{12.24}$$

In particular, for $s>1/2$, prescribed mass $M>0$ determines

$$D_s=\frac{B(1/2,2-s)}{A_s^*},\qquad L=\left(\frac{M}{aD_s}\right)^{1/(2\beta)},\qquad C_1=\frac{a}{A_s^*},\qquad \beta=\frac{3}{2}-s. \tag{12.25}$$

The centered profiles tend locally uniformly to zero as $L\downarrow 0$.

Proof. Let $V_L=E_s*\nu_L$. Lemmas 12.2 and 12.3 give $V_L''=-2a$ on $(-L,L)$. Since V_L is continuous and even, its interior polynomial is

$$V_L(z)=V_L(0)-az^2\quad(|z|\le L). \tag{12.26}$$

Thus $p_L=V_L-V_L(0)$ has the prescribed contact. This derives the interior quadratic also for the singular lower-order kernels without differentiating them twice as ordinary integrals. For $s>1/2$,

Lemma 12.4 gives the same quadratic by its independent ODE calculation and yields (12.23). For $s < 1/2$, direct beta integration gives

$$\int_{-L}^{L} |y|^{2s-1}(L^2-y^2)^{1-s}\,\mathrm{d}y = B(s,2-s)L^2, \qquad c_s(1-2s)B(s,2-s) = A_s^*.$$

The second identity follows by substituting the Gamma definitions and $\Gamma(3/2-s) = (1/2-s)\Gamma(1/2-s)$. It gives $V_L(0) = aL^2/(1-2s)$ and proves (12.21).

At $s = 1/2$, $A_{1/2}^* = 1/2$. If $I_L(z) = \int_{-L}^{L} \log|z-y|\sqrt{L^2-y^2}\,\mathrm{d}y$, then

$$I_L(0) = \frac{\pi L^2}{2}\log(L/2) - \frac{\pi L^2}{4}. \tag{12.27}$$

Indeed, differentiating the beta integral in its first parameter gives

$$\int_{-1}^{1} \log|u|\sqrt{1-u^2}\,\mathrm{d}u = \frac{1}{2}B(1/2,3/2)\big(\psi(1/2)-\psi(2)\big) = -\frac{\pi}{2}\log 2 - \frac{\pi}{4},$$

where $\psi = \Gamma'/\Gamma$; scaling yields (12.27). Since $V_L = -(2a/\pi)I_L$, the normalization $p_L = V_L - V_L(0)$ gives exactly (12.22). For $z > L$,

$$I_L'(z) = \int_{-L}^{L} \frac{\sqrt{L^2-y^2}}{z-y}\,\mathrm{d}y = \pi(z - \sqrt{z^2-L^2}). \tag{12.28}$$

To evaluate it set $y = L\cos\theta$ and use $\int_0^\pi (z - L\cos\theta)^{-1}\,\mathrm{d}\theta = \pi/\sqrt{z^2-L^2}$, obtained by $\tan(\theta/2)$. The decomposition

$$\frac{L^2\sin^2\theta}{z-L\cos\theta} = z + L\cos\theta + \frac{L^2-z^2}{z-L\cos\theta}$$

then proves (12.28). Integrating from L, where $p_L(L) = -aL^2$, and using evenness gives the equivalent endpoint formula

$$p_L(z) = \begin{cases} -az^2, & |z| \le L, \\ -az^2 + a|z|\sqrt{z^2-L^2} - aL^2\log\left(\dfrac{|z|+\sqrt{z^2-L^2}}{L}\right), & |z| > L. \end{cases} \tag{12.29}$$

It remains to justify the endpoint gluing and the strict exterior obstacle inequality in all orders. If $s \neq 1/2$, write $F_L(z) = \int_{-L}^{L} |z-y|^\kappa (L^2-y^2)^t\,\mathrm{d}y$. For $z > L$ ordinary differentiation is valid. As $z \downarrow L$, the first derivative integrand is dominated by an integrable multiple of $(L-y)^{\kappa-1+t} = (L-y)^{s-1}$ near L; the other endpoint is harmless. The substitution $y = L - 2Lu$ gives

$$F_L'(L+) = \kappa\int_{-L}^{L}(L-y)^{\kappa-1}(L^2-y^2)^t\,\mathrm{d}y = 2L\kappa B(s,2-s). \tag{12.30}$$

The normalizations above and (12.18) therefore give $p_L'(L+) = -2aL$. Continuity of p_L, followed by the fundamental theorem of calculus on the right, proves differentiability at L with that value. The interior derivative is already $-2aL$. Reflection handles $-L$. For $z > L$,

$$F_L''(z) = \kappa(\kappa-1)\int_{-L}^{L}(z-y)^{\kappa-2}(L^2-y^2)^t\,\mathrm{d}y.$$

Multiplication by c_sC_1 for $s < 1/2$ or $-d_sC_1$ for $s > 1/2$ shows $p_L''(z) > 0$ in both ranges. At $s = 1/2$, (12.28) gives the same matching derivative, and directly

$$p_L''(z) = \frac{2a}{\pi}\int_{-L}^{L}\frac{\sqrt{L^2-y^2}}{(z-y)^2}\,\mathrm{d}y > 0 \quad (z > L).$$

Hence $g_L = p_L + az^2$ has $g_L(L) = g_L'(L) = 0$ and $g_L'' > 2a$ outside the interval. It is strictly positive there and is globally convex after even reflection. Also $p_L' < 0$ on (L,∞): this follows from the

sign of the first derivative kernel in each power regime, and from (12.28) at the endpoint. Together with its interior values and evenness this proves $p_L \le 0$.

The fundamental identity in Lemma 12.2 gives $\mathcal{A}_s^{(1)} p_L = \nu_L$, so there are no additional endpoint atoms. Beta integration of (12.20) gives (12.24). The tails are, more precisely,

$$p_L(z) = -\frac{aL^2}{1-2s} + c_s m_{s,L} |z|^{2s-1} + o(|z|^{2s-1}), \qquad s < 1/2, \tag{12.31}$$

$$p_L(z) = -aL^2 \log(2|z|/L) - \frac{aL^2}{2} + o(1), \qquad s = 1/2, \tag{12.32}$$

$$p_L(z) = O(1 + |z|^{2s-1}), \qquad s > 1/2. \tag{12.33}$$

The first follows uniformly over the compact pressure support, the second by expanding (12.29), and the third from the compact potential. Each is an ordinary absolute tail. Finally the lower-order profiles are bounded in absolute value by CL^2 using the L^2 scaling of the bounded dimensionless profile. At $s = 1/2$, (12.29) gives $O_R(L^2(1 + |\log L|))$ on $|z| \le R$. For $s > 1/2$ the explicit integral is bounded there by $C_R L^{3-2s} + CL^2$ for $L \le 1$. These bounds prove local uniform convergence to zero. Since $3 - 2s = 2\beta > 0$, the mass determines L uniquely and gives (12.25). □

Lemma 12.6 (Equal-mass energy)**.** *Suppose $1/2 < s < 1$. If ν is a compactly supported finite measure and $p \in C(\mathbb{R}) \cap L^1_s(\mathbb{R})$ has sublinear growth and satisfies $\mathcal{A}_s^{(1)} p = \nu$, then $p = E_s * \nu + c$ for a constant c. For every compactly supported finite signed measure θ with $\theta(\mathbb{R}) = 0$,*

$$\int_{\mathbb{R}} (E_s * \theta)\, \mathrm{d}\theta = \frac{1}{2\pi} \int_{\mathbb{R}} |\zeta|^{-2s} |\widehat{\theta}(\zeta)|^2 \, \mathrm{d}\zeta. \tag{12.34}$$

Both sides are finite.

Proof. Lemma 12.2 makes $p - E_s * \nu$ an entire harmonic ordinary-tail function. Lemma 7.2 makes it affine; its sublinear growth eliminates the slope. For the energy identity, insert (12.11) in the compact double integral. Fubini for the signed measures is justified by the absolute bound

$$C \iint |x - y|^{\kappa} \, |\theta|(\mathrm{d}x) |\theta|(\mathrm{d}y) < \infty.$$

The constant term vanishes because $\theta(\mathbb{R}) = 0$, and the integrated cosine is $|\widehat{\theta}(\zeta)|^2$. Thus

$$\iint |x - y|^{\kappa} \, \theta(\mathrm{d}x) \theta(\mathrm{d}y) = -C_\kappa \int_{\mathbb{R}} |\widehat{\theta}(\zeta)|^2 |\zeta|^{-1-\kappa} \, \mathrm{d}\zeta.$$

Multiplication by $-d_s$ and $d_s C_\kappa = (2\pi)^{-1}$ prove (12.34). Compact support and zero mass give $|\widehat{\theta}(\zeta)| \le C \min\{|\zeta|, 1\}$. The integral is finite at zero and infinity because $1 < 2s < 2$. Adding a constant to the potential leaves its pairing with θ unchanged. □

Proposition 12.7 (Equal-mass interval uniqueness)**.** *Let $1/2 < s < 1$. Suppose $p_1, p_2 \in C(\mathbb{R}) \cap L^1_s(\mathbb{R})$ satisfy the one-dimensional obstacle problem with obstacle $-az^2$, with compact nonnegative pressures ν_1, ν_2 of the same total mass. If*

$$|p_i(z)| \le C(1 + |z|^{\kappa}), \qquad p_1(0) = p_2(0) = 0,$$

then $p_1 = p_2$.

Proof. Lemma 12.6 writes $p_i = E_s * \nu_i + c_i$. Set $w = p_1 - p_2$ and $\theta = \nu_1 - \nu_2$. The zero-mass identity gives

$$\int w \, \mathrm{d}\theta = \frac{1}{2\pi} \int_{\mathbb{R}} |\zeta|^{-2s} |\widehat{\theta}(\zeta)|^2 \, \mathrm{d}\zeta \ge 0.$$

All pairings are well defined because the potential kernel is continuous on the compact pressure supports. On $\operatorname{supp} \nu_1$, $p_1 = -az^2 \le p_2$, so $w \le 0$; on $\operatorname{supp} \nu_2$, $w \ge 0$. Complementarity gives

$\int w\,\mathrm{d}\theta \le 0$. The energy therefore vanishes. Its nonnegative Fourier integrand and continuity of $\widehat{\theta}$ imply $\widehat{\theta} = 0$, hence $\theta = 0$. The difference w is constant and its value at zero is zero. □

13. Cylindrical reduction and classification

The preliminary tail estimate and convexity already force cylindricality at and below order $1/2$. At higher orders the cylindrical branch admits an affine ambiguity, which is determined by contact with the fixed obstacle. We prove the reduction, compactness of contact, and boundedness of pressure before solving its one-dimensional torsion problem.

Theorem 13.1 (The cylindrical branch at every order)**.** *Let $0 < s < 1$ and $a > 0$. Suppose $P \in C(\mathbb{R}^2)$ has the ordinary absolute fractional tail, satisfies $P \ge -aZ^2$, and has nonnegative Radon pressure $\mu = \mathcal{A}_s^{(2)} P$ supported on $K = \{P = -aZ^2\}$. If $s > 1/2$, assume in addition that $\partial_X P = 0$ in distributions; if $s \le 1/2$, this independence is automatic. Then exactly one of the following alternatives occurs, with p_L as in Proposition 12.5.*

(1) *If $0 < s \le 1/2$, either $P(X,Z) = c$ for a unique $c \ge 0$ and $\mu = 0$, or, for a unique $L > 0$,*

$$P(X,Z) = p_L(Z), \qquad K = \mathbb{R} \times [-L, L], \qquad \mu = \mathrm{d}X \otimes \frac{a}{A_s^*}(L^2 - Z^2)_+^{1-s}\,\mathrm{d}Z. \tag{13.1}$$

The constant $c > 0$ has empty contact; $c = 0$ has the pressure-free contact line $Z = 0$.

(2) *If $1/2 < s < 1$, either*

$$P(X,Z) = bZ + c, \qquad c \ge \frac{b^2}{4a}, \qquad \mu = 0,$$

with unique $b, c \in \mathbb{R}$, or, for unique $h \in \mathbb{R}$, $L > 0$,

$$P(X,Z) = p_L(Z - h) - 2ahZ + ah^2, \tag{13.2}$$

$$K = \mathbb{R} \times [h - L, h + L], \qquad \mu = \mathrm{d}X \otimes \frac{a}{A_s^*}[L^2 - (Z - h)^2]_+^{1-s}\,\mathrm{d}Z. \tag{13.3}$$

For the affine family, contact is empty when $c > b^2/(4a)$ and equals $\mathbb{R} \times \{-b/(2a)\}$ when equality holds.

Every displayed profile satisfies the hypotheses. Positive pressure has mass per unit horizontal length $m_{s,L}$ in (12.24)*, not finite total planar mass. Every higher-order cylindrical profile has at most linear growth and zero degree-$s + 1/2$ blow-down. There is no positive-pressure point-contact state in the reduced one-dimensional problem.*

Proof. Horizontal independence at lower order. Theorem 8.2 gives convexity of $u = P + aZ^2$, and Proposition 7.1 gives $M_P(R) = \sup_{|x| \le R} |P(x)| = o(R^{2s})$. If $s \le 1/2$, fix Z. The finite function $f(X) = P(X,Z)$ is convex, and $|f(X)| = o(|X|)$ at both ends. At $s = 1/2$ this uses the strict little-o statement. Any positive secant slope of a convex function forces at least linear growth at $+\infty$; any negative secant slope forces at least linear growth at $-\infty$. Both contradict absolute sublinearity. Thus every secant has zero slope and f is constant. This holds for every Z.

At higher order $\partial_X P = 0$ is assumed. Convolving with a compact smooth approximate identity makes each mollification independent of X; local uniform convergence gives $P(X,Z) = p(Z)$ everywhere. We therefore have a continuous cylindrical function in either range. Lemma 12.1 supplies the exact ordinary tail of p and the actual product measure

$$\mu = \mathrm{d}X \otimes \nu, \qquad \nu = \mathcal{A}_s^{(1)} p \ge 0, \qquad \operatorname{supp}\nu \subset J := \{p = -az^2\}. \tag{13.4}$$

In particular, the reduced pressure has not been obtained by restricting an arbitrary planar measure to a line.

Compact contact and the higher-order linear bridge. Convexity of u gives convexity of the nonnegative function $u(z) = p(z) + az^2$. Its zero set J is closed and convex. At contact,

$az^2 = |p(z)| \le M_P(|z|) = o(|z|^{2s})$; since $2s < 2$, contact cannot contain an unbounded sequence. Thus J is empty, a point, or a compact interval. Equivalently, one can exclude unbounded one-dimensional contact directly from convexity: an unbounded closed convex J contains a half-line, on which $p = -az^2$ violates its ordinary absolute tail. In all cases ν is supported on a compact set, so $m := \nu(\mathbb{R}) < \infty$.

Before any density assertion, consider $s > 1/2$. Apply the analytic embedding of Corollary 7.3 and the invariant branch Proposition 9.2. They give

$$|p(z)| \le C(1 + |z|). \tag{13.5}$$

For the exact ordinary inverse normalization, canonicity gives $\widehat{\nu}_0 = e_s\nu$, and the sphere integral $S_{2,\eta} = 2\sqrt{\pi}\,\Gamma(1-s)/\Gamma(3/2-s)$ gives

$$\frac{2e_s}{(2s-1)S_{2,\eta}} = -\frac{\Gamma(1/2-s)}{4^s\sqrt{\pi}\Gamma(s)} = d_s.$$

Thus (9.3) is precisely $p = E_s * \nu + bz + c$, with the inverse kernel of Lemma 12.2. This representation and the linear bound precede density and allow atoms in the finite reduced measure. No finite planar pressure is assumed.

Bounded pressure. Convexity of u gives, for every z, r,

$$p(z+r) + p(z-r) - 2p(z) \ge -2ar^2. \tag{13.6}$$

Let $p_\varepsilon = p * \rho_\varepsilon$, with a nonnegative smooth mollifier supported in $(-\varepsilon, \varepsilon)$ and $0 < \varepsilon \le 1$. The same difference bound holds for p_ε. Its local amplitudes and ordinary-tail norms are uniformly bounded as $\varepsilon \downarrow 0$. On a fixed compact set Q, the smooth symmetric formula reads

$$\mathcal{A}_s^{(1)} p_\varepsilon(z) = \frac{c_{1,s}}{2} \int_{\mathbb{R}} \frac{2p_\varepsilon(z) - p_\varepsilon(z+r) - p_\varepsilon(z-r)}{|r|^{1+2s}}\,\mathrm{d}r.$$

On $|r| < 1$ the numerator is bounded above by $2ar^2$, with integrable majorant because $s < 1$. On $|r| \ge 1$ an upper bound follows from the local center amplitudes and the translated absolute tail norms. Those norms are uniformly comparable for $z \in Q$ and mollifier shifts of size at most one. Thus $\mathcal{A}_s^{(1)} p_\varepsilon \le C_Q$ on Q, with C_Q independent of ε. The integral for each fixed mollification is well defined; no uniform bound for its classical second derivative is asserted or needed. Since distributional convolution gives $\mathcal{A}_s^{(1)} p_\varepsilon = \nu * \rho_\varepsilon \ge 0$, passing against nonnegative tests gives

$$0 \le \nu \le C_Q\,\mathrm{d}z \quad \text{locally on } \mathbb{R}. \tag{13.7}$$

A compact neighborhood of J now supplies a global bounded density, which we continue to denote by ν. In particular empty or singleton contact carries zero pressure. If $\nu \ne 0$, write $J = [h - L, h + L]$ with $L > 0$. Its full pressure support has not yet been assumed.

The admissible harmonic ambiguity. For $s \le 1/2$, the density conclusion allows the compact inversion in Lemma 12.2. Its potential is continuous and has an ordinary tail. Applying Lemma 7.2 to $p - E_s * \nu$ leaves only a constant. Together with the already proved higher-order representation, this gives

$$p(z) = E_s * \nu(z) + bz + c, \qquad b = 0 \text{ if } 0 < s \le 1/2. \tag{13.8}$$

The restriction is strict at $s = 1/2$: a nonzero linear function has logarithmically divergent absolute tail. If $\nu = 0$, minimizing $az^2 + bz + c$ gives precisely the constant or affine alternatives in the statement, including their contact sets.

Positive pressure, centering, and the gauge. Suppose $\nu \ne 0$. By the whole-line composition identity (12.7), the affine term in (13.8) disappears on taking two derivatives, and

$$\mathcal{A}_t^{(1)} \nu = -p'' \quad \text{in } \mathcal{D}'(\mathbb{R}), \qquad t = 1 - s.$$

Because $p = -az^2$ on $I = (h - L, h + L)$, this is exactly

$$\mathcal{A}_t^{(1)}\nu = 2a \quad \text{in } \mathcal{D}'(I), \qquad \nu = 0 \text{ almost everywhere on } \mathbb{R} \setminus I, \qquad \nu \in L^\infty(\mathbb{R}). \tag{13.9}$$

Lemma 12.3, valid for every $0 < t < 1$, identifies

$$\nu(z) = \frac{a}{A_s^*}[L^2 - (z-h)^2]_+^{1-s}.$$

In particular its support is the whole contact interval, and its density is continuous and positive exactly in its interior.

Let $V_L = E_s * \nu_L$ be the centered potential used in Proposition 12.5. It is even and satisfies $V_L(z) = V_L(0) - az^2$ on $[-L, L]$. Equation (13.8) therefore reads $p(z) = V_L(z - h) + bz + c$. Comparing with $-az^2$ on the nondegenerate interval J gives the two coefficient identities

$$b = -2ah, \qquad c = ah^2 - V_L(0). \tag{13.10}$$

For $s \le 1/2$ the slope in (13.8) is zero, so $h = 0$ and $p = p_L$. This proves centering from the actual inverse representation. For $s > 1/2$ the permitted slope gives exactly $p(z) = p_L(z-h) - 2ahz + ah^2$.

Converse and degeneracies. Proposition 12.5 proves the exact pressure, C^1 endpoint gluing, strict exterior obstacle inequality, and ordinary tails for every centered model in all three regimes. For $s > 1/2$, affine functions have ordinary tails and their symmetric operator is zero. The transverse translation has the exact obstacle identity

$$p_L(Z - h) - 2ahZ + ah^2 + aZ^2 = p_L(Z - h) + a(Z - h)^2. \tag{13.11}$$

It therefore preserves the sign of the gap and translates its contact, while leaving the translated nonnegative pressure unchanged. Dimension reduction proves every planar converse, including local Radon finiteness. At $s \le 1/2$ a nonzero affine gauge is not an ordinary-tail symmetry, consistently with (13.10).

The endpoints determine h, L uniquely; below and at $1/2$ the center is already zero. The mass in (12.24) is positive for $L > 0$ and strictly increasing with L, since $3 - 2s > 0$. It is pressure per unit horizontal length; the planar product measure has infinite total mass whenever that one-dimensional mass is nonzero. Thus the positive and zero-pressure families are disjoint. By (13.5), every higher-order cylindrical profile has zero degree-λ blow-down, $\lambda = s + 1/2 > 1$, both locally uniformly and in the ordinary planar tail norm: its rescaling is bounded by $C(R^{-\lambda} + R^{1-\lambda}|x|)$, whose tail norm tends to zero because $2s > 1$.

As $L \downarrow 0$ the centered potentials tend locally uniformly to zero and their mass tends to zero. The higher-order gauged limit is $-2ahZ + ah^2$, exactly the tangent affine state, while the lower-order limit is zero. These limits carry no pressure. Strictly lifted affine states and positive constants have empty contact and are additional zero-pressure solutions. The full quadratic $p = -az^2$ is excluded by the ordinary tail. This accounts for every empty, singleton, and positive-length reduced contact possibility. □

14. Comparison with the full-line hitting kernel

Throughout this section $1/2 < s < 1$ and $0 < \delta < \kappa = 2s - 1$.

Lemma 14.1 (Full-line harmonic extension)**.** *For continuous $f : \mathbb{R} \to \mathbb{R}$ with $|f(q)| \le C(1 + |q|^\delta)$, define*

$$H_f(X, Z) = c_s|Z|^\kappa \int_{\mathbb{R}} \frac{f(q)}{((X - q)^2 + Z^2)^s}\,\mathrm{d}q, \qquad c_s = \frac{\Gamma(s)}{\sqrt{\pi}\Gamma(s - 1/2)}, \quad Z \ne 0, \tag{14.1}$$

and $H_f(X, 0) = f(X)$. Then H_f is continuous on $\mathbb{R}^2$, harmonic off $\{Z = 0\}$, and has growth $O(1 + |x|^\delta)$. The formula preserves positivity and constants.

Proof. The identity

$$\int_{\mathbb{R}} (1+q^2)^{-s}\,\mathrm{d}q = \frac{\sqrt{\pi}\Gamma(s-1/2)}{\Gamma(s)}$$

normalizes the kernel. Its δ moment is finite because $\delta < \kappa$. After the change of variables $q = X + |Z|v$, subadditivity of the δ power proves the growth bound. To verify continuity at $(X_0, 0)$, first restrict the integral to $|q - X_0| < \delta$ and use continuity of f; on the complement use the factor $|Z|^\kappa$, the integrable tail $q^{\delta-2s}$, and then let $\delta \downarrow 0$.

For completeness, harmonicity can be verified directly with the Fourier convention above. For f Schwartz, the Fourier transform in X of (14.1) is $\widehat{f}(\xi)h_\xi(Z)$, where a Gaussian integral gives

$$h_\xi(Z) = \frac{1}{\Gamma(\sigma)} \int_0^\infty v^{\sigma-1} \exp\left(-v - \frac{\xi^2 Z^2}{4v}\right) \mathrm{d}v.$$

For $\xi \neq 0$ let

$$Q_\xi(Z) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{i\zeta Z}(\xi^2 + \zeta^2)^{-s}\,\mathrm{d}\zeta.$$

The Laplace representation of the multiplier and the Gaussian Fourier integral yield

$$Q_\xi(Z) = \frac{1}{2\sqrt{\pi}\Gamma(s)} \int_0^\infty q^{\sigma-1} e^{-q\xi^2 - Z^2/(4q)}\,\mathrm{d}q, \qquad Q_\xi(0) = \frac{\Gamma(\sigma)}{2\sqrt{\pi}\Gamma(s)} |\xi|^{-\kappa}.$$

Thus $h_\xi(Z) = Q_\xi(Z)/Q_\xi(0)$. The full Fourier transform satisfies

$$\widehat{H_f}(\xi,\zeta) = \frac{2\sqrt{\pi}\Gamma(s)}{\Gamma(\sigma)} |\xi|^\kappa (\xi^2+\zeta^2)^{-s} \widehat{f}(\xi).$$

The multiplier is locally integrable at the origin, being homogeneous of degree -1 in two variables. Multiplication by $(\xi^2+\zeta^2)^s$ leaves an expression independent of ζ; hence $\mathcal{A}_s H_f$ is supported on $Z = 0$. General continuous f as stated follows by smooth truncation. The finite δ moment gives locally uniform convergence of the extensions off the line and a common ordinary-tail bound, so distributional harmonicity passes to the limit. The same bounded-approximation argument applies to indicator data off the line. Formula (14.1) is also the hyperplane hitting kernel of the isotropic stable process [17, formula (3)]. □

Lemma 14.2 (Comparison with a continuous slit trace)**.** *Let $\Omega = B_R \setminus \{Z = 0\}$, and let $v \in L^1_s(\mathbb{R}^2)$ be continuous in Ω and bounded above on $\mathbb{R}^2$. Suppose $\mathcal{A}_s v \leq 0$ in Ω, $v \leq 0$ almost everywhere outside B_R, and*

$$\limsup_{\Omega \ni x \to x_0} v(x) \leq 0 \quad \textit{for every } x_0 \in \partial\Omega.$$

Then $v \leq 0$ in Ω.

Proof. Change values on the slit, a set of planar measure zero, to zero. If $m = \sup_\Omega v > 0$, the upper boundary limits, compactness of $\partial\Omega$, and the exterior inequality place $\{v \geq m/2\}$ in a compact subset E of Ω. Choose a slightly larger compact neighborhood E' still inside Ω. Positive mollifications $v_\varepsilon = v * \rho_\varepsilon$ converge uniformly to v on E'. For all sufficiently small ε, every point where $v_\varepsilon \geq 3m/4$ lies in E': a finite boundary cover provides uniform upper control near the boundary, and the mollifications are nonpositive outside $B_{R+\varepsilon}$. The positive global maximum of v_ε is therefore attained at a point x_ε whose ε neighborhood lies in Ω. At that point,

$$\mathcal{A}_s v_\varepsilon = (\mathcal{A}_s v) * \rho_\varepsilon \leq 0.$$

The symmetric integral at a smooth global maximum is nonnegative. It is strictly positive here because the maximum is positive and $v_\varepsilon \leq 0$ on an open set outside a large ball. Local smoothness and the ordinary tail justify the integral, giving a contradiction. □

Proposition 14.3 (Full-line comparison and uniqueness)**.** *If $w \in C(\mathbb{R}^2)$, $w \geq 0$, $\mathcal{A}_s w \leq 0$ globally, and $w(x) \leq C(1+|x|^\delta)$, then*

$$w \leq H_f, \qquad f = w|_{\{Z=0\}}. \tag{14.2}$$

More generally, a signed continuous function D with zero line trace, $\mathcal{A}_s D \leq 0$ off the line, and $D(x) \leq C(1+|x|^\delta)$ satisfies $D \leq 0$, provided it has an ordinary tail. Consequently a continuous harmonic function off the line, of growth $O(1+|x|^\delta)$, is uniquely determined by its full-line trace.

Proof. For $R \geq 1$ put $F_R = H_{\mathbf{1}_{\{|q|>R\}}}$ off the line and use its indicator trace on the line, assigning zero at its two jump points. By Lemma 14.1, F_R is bounded and harmonic off the line. There is $c = c(s) > 0$ such that

$$F_R(x) \geq c \quad (|x| \geq R,\ Z \neq 0), \qquad F_R(x) \leq C|Z|^\kappa R^{-\kappa} \quad (|x| \leq R/2). \tag{14.3}$$

For the lower bound, if $|X| \geq R$, a tail starting at X contains half the kernel mass. Otherwise symmetry permits $0 \leq X < R$, and $X^2 + Z^2 \geq R^2$ implies $R - X \leq |Z|$. The $q \geq R$ integral is then at least $c_s \int_1^\infty (1+v^2)^{-s}\,\mathrm{d}v$. The upper bound follows by comparing $((X-q)^2+Z^2)^s$ with $|q|^{2s}$ when $|x| \leq R/2$ and $|q| > R$.

Define

$$G_R = A \sum_{j=0}^{\infty} (2^j R)^\delta F_{2^j R},$$

where A is chosen independently of R. The upper bound in (14.3) gives locally uniform convergence inside B_R because $\delta - \kappa < 0$. Globally, split the radii at $2|x|$: the low-radius terms sum to $C|x|^\delta$, while the high-radius terms are bounded by

$$C|Z|^\kappa \sum_{2^j R > 2|x|} (2^j R)^{\delta-\kappa} \leq C|x|^\delta$$

when $|x| \geq R/2$. For $|x| < R/2$ the direct geometric-series estimate applies. Thus G_R has an ordinary tail, and its partial sums have a common tail majorant. Harmonicity passes distributionally to the sum; interior regularity then gives smoothness off the line.

For an exterior off-line point, choose j with $2^j R \leq |x| < 2^{j+1} R$. One summand and the lower bound in (14.3) show that $G_R \geq D$ there if A is sufficiently large. On the line $D = 0$ and $G_R \geq 0$. Inside the line segment $|X| < R$, both have zero trace; at $(\pm R, 0)$, $D \to 0$ and $G_R \geq 0$. At every other point of the boundary circle the exterior comparison and continuity supply the required upper boundary limit. Therefore Lemma 14.2 applies to $D - G_R$ and gives $D \leq G_R$ in B_R. At each fixed off-line point, for all large R,

$$G_R(x) \leq C|Z|^\kappa R^{\delta-\kappa} \sum_{j=0}^{\infty} 2^{j(\delta-\kappa)} \longrightarrow 0.$$

This proves the signed comparison. Taking $D = w - H_f$ gives (14.2); applying signed comparison to both signs of the difference proves uniqueness. □

Lemma 14.4 (The principal line datum)**.** *With $H_0(X,Z) = (\sqrt{X^2+Z^2} - X)^\sigma$,*

$$H_{(-q)_+^\sigma}(X,Z) = 2^{-\sigma} H_0(X,Z).$$

Proof. Write $r = \sqrt{X^2+Z^2}$. In $\int_0^\infty u^\sigma ((X+u)^2+Z^2)^{-s}\,\mathrm{d}u$ put $u = re^v$, pair v and $-v$, and set $w = \sinh(v/2)$. The integral becomes

$$2^{1/2-\sigma} r^{1/2} \int_{\mathbb{R}} (r + X + 2rw^2)^{-s}\,\mathrm{d}w = 2^{-\sigma} c_s^{-1} (r+X)^{-\sigma}.$$

Multiplication by $c_s |Z|^{2\sigma}$ gives the identity off the line; continuity gives the line values. Here c_s is the normalization in (14.1). □

15. From strict power bounds to the critical endpoint

Throughout this section $1/2 < s < 1$, and we write

$$\sigma = s - \tfrac{1}{2}, \qquad \kappa = 2\sigma, \qquad \lambda = 1 + \sigma, \qquad \beta = 1 - \sigma,$$
$$\eta = 1 - 2s = -2\sigma, \quad b_G = 1 + \eta, \quad D = 3 + \eta.$$

Thus $2\sigma + b_G = 1$ and $1 < \lambda < 2s < 2$. The obstacle coefficient is $a > 0$; it is distinct from every extension exponent and from the principal amplitude $A(r)$ below.

Theorem 15.1 (Critical growth from the ordinary tail)**.** *Let $P \in C(\mathbb{R}^2) \cap L^1_s(\mathbb{R}^2)$ satisfy*

$$\begin{gathered} P(X,Z) \geq -aZ^2, \qquad \mu := \mathcal{A}_s P \geq 0 \text{ as a Radon measure}, \\ \operatorname{supp}\mu \subset K := \{P = -aZ^2\}, \qquad \partial_X P \not\equiv 0. \end{gathered} \tag{15.1}$$

Then

$$|P(x)| \leq C(1 + |x|^\lambda). \tag{15.2}$$

The same bound holds for its even Poisson extension on $\mathbb{R}^3$.

Embed P as $U = Q + \mathcal{E}_s P$ using Corollary 7.3. The preceding results then give convexity of $u = P + aZ^2$, the unique positive contact recession orientation, and a genuine obstacle-preserving normalization with an exact positive contact ray; see Theorem 8.2 and Lemma 9.3. They also give, separately for every $p \in (\lambda, 2s)$,

$$|P(x)| \leq C_p(1 + |x|^p), \qquad |\mathcal{E}_s P(q)| \leq C_p(1 + |q|^p), \tag{15.3}$$

by Corollary 7.3 followed by Proposition 11.3 and the canonical identity in Proposition 11.4. No bound uniform in $p \downarrow \lambda$ is asserted. We prove the endpoint by first obtaining a finite positive moment of the derivative pressure. Its principal projection controls the derivative on the negative axis; line comparison then controls the derivative everywhere. A final state record argument excludes the only mode that can remain at degree λ.

15.1. Regularity and monotonicity before a contact graph.

Fix for now any $\gamma \in (\lambda, 2s)$, and set $\delta = \gamma - 1 \in (\sigma, \kappa)$. All statements in this subsection use its bound in (15.3), not (15.2).

Lemma 15.2 (Local pressure and contact regularity)**.** *The pressure has a locally bounded Lebesgue density, and*

$$P \in C^{1,\kappa}_{\mathrm{loc}}(\mathbb{R}^2), \qquad \nabla(P + aZ^2) = 0 \quad \text{on } K.$$

The proper closed convex set K has nonempty interior.

Proof. Let P_ε be a positive smooth mollification. Convexity of $P + aZ^2$ gives

$$2P_\varepsilon(x) - P_\varepsilon(x + y) - P_\varepsilon(x - y) \leq 2ay_Z^2. \tag{15.4}$$

Its contribution to the symmetric integral for $\mathcal{A}_s P_\varepsilon$ on $|y| < 1$ is bounded above by $Ca \int_0^1 r^{1-2s}\,\mathrm{d}r < \infty$. For centers in a fixed compact set, the remaining adverse contribution is bounded by the local amplitude and the ordinary tail of P_ε^-. Both are bounded uniformly for small ε: translation through a bounded set preserves the tail weight up to a fixed factor. Thus $0 \leq \mathcal{A}_s P_\varepsilon \leq C$ on each smaller compact set. Pass to distributions to obtain $0 \leq \mu \leq C\,\mathrm{d}x$ locally. We henceforth also use μ for this density.

Let χ be compactly supported and equal to one near a fixed ball. The potential $v = K_s * (\chi\mu)$ has an ordinary tail and satisfies $\mathcal{A}_s v = \chi\mu$ by absolute Fubini against compact smooth tests. Thus $P - v$ is fractional harmonic in that ball. The positive ball Poisson formula gives its smoothness on smaller balls by differentiation: away from the smaller ball, the derivatives of the kernel retain its integrable boundary singularity and ordinary-tail decay. The Riesz kernel satisfies

$$|\nabla K_s(y)| \leq C|y|^{2s-3}, \qquad |D^2 K_s(y)| \leq C|y|^{2s-4}.$$

The first kernel is locally integrable in two dimensions because $s > 1/2$. For a gradient difference across distance $h < 1$, splitting into distances below $3h$ and above h gives

$$C \int_0^{3h} r^{2s-2} \, \mathrm{d}r + Ch \int_h^C r^{2s-3} \, \mathrm{d}r + Ch \le Ch^{2s-1}.$$

Approximation justifies differentiation of the potential and proves the claimed regularity. A differentiable nonnegative function attaining zero has zero gradient there, so $\nabla u = 0$ on K.

If $\mu = 0$, Lemma 7.2 makes P affine, and the obstacle for every X excludes its X-linear coefficient. This contradicts noncylindricity. A convex set with empty interior has planar measure zero and cannot support a nonzero locally bounded density. Consequently K has nonempty interior. Its closedness, convexity and properness also follow from the preceding geometry; properness follows directly from $\gamma < 2$. □

Work in the orientation of Lemma 9.3, so e_X is a recession direction of K and every escaping contact sequence points toward e_X. The supporting-plane inequality against $k + te_X \in K$, followed by $t \to \infty$, gives $P_X \le 0$. Define

$$w = -P_X. \tag{15.5}$$

Then w is continuous, nonnegative, nonzero, and zero on K. Horizontal convex secants with $h = 1 + |x|$ give

$$0 \le w(x) \le C(1 + |x|^{\delta}), \qquad \delta = \gamma - 1 < \kappa < 1 : \quad P_X(x) \ge \frac{P(x) - P(x - he_X)}{h}. \tag{15.6}$$

In particular w has an ordinary fractional tail.

Lemma 15.3 (Positive derivative pressure without a graph)**.** *There is a nonnegative Radon measure*

$$\nu := \partial_X \mu = -\mathcal{A}_s w, \qquad \operatorname{supp} \nu \subset K. \tag{15.7}$$

Proof. This is Corollary 11.5 for the analytically embedded solution. Its quotient-first argument was proved before local pressure density and supplies the positive measure without discarding any singular component. The intrinsic pressure is $e_s \mu$, so division by the positive constant e_s gives exactly (15.7). □

15.2. **Full transverse projection.**

Proposition 15.4 (Exclusion of a proper projection)**.** *The transverse projection $\pi_Z K$ is all of $\mathbb{R}$.*

Proof. Write $J_Z = \pi_Z K$. It is a nonempty interval. If it is proper, so is its closure. Choose a ball $B_{r_*}(c)$, $c = (0, Z_*)$, whose slightly enlarged closed ball has transverse coordinates outside $\overline{J_Z}$. For $T \ge 0$ put $P_T(X, Z) = P(X + T, Z)$. Every P_T is fractional harmonic near the closed ball, and

$$-aZ^2 \le P_T(X, Z) \le P(X, Z), \qquad P_T(X, Z) \downarrow \ell(Z). \tag{15.8}$$

The limit is finite; its independence of X follows by shifting the parameter tending to infinity. The ordinary ball formula at its center is

$$P_T(c) = \int_{B_{r_*}(c)^c} \mathcal{P}_c(y) P_T(y) \, \mathrm{d}y, \qquad \mathcal{P}_c(y) = \frac{\sin(\pi s)}{\pi^2} \frac{r_*^{2s}}{(|y - c|^2 - r_*^2)^s |y - c|^2}. \tag{15.9}$$

The positive kernel has an integrable boundary singularity and is bounded below on the exterior by a positive multiple of $(1 + |y|)^{-2-2s}$. Since $P_T^+ \le P^+$,

$$\int \mathcal{P}_c P_T^- \, \mathrm{d}y = \int \mathcal{P}_c P_T^+ \, \mathrm{d}y - P_T(c) \le \int \mathcal{P}_c P^+ \, \mathrm{d}y + aZ_*^2.$$

The right side is finite independently of T: local continuity handles the boundary singularity and the ordinary tail handles infinity. The sandwich controls the interior of the ball. Hence

$$\sup_{T\ge 0}\|P_T\|_{L^1_s(\mathbb{R}^2)}<\infty. \tag{15.10}$$

The quadratic lower bound was used locally and was never integrated against the global fractional-tail weight.

For $Z\in J_Z$, recession gives $\ell(Z)=-aZ^2$. Fatou in (15.10), followed by integration in X, yields

$$\int_{\mathbb{R}}|\ell(Z)|(1+|Z|)^{-1-2s}\,\mathrm{d}Z<\infty.$$

Indeed integration of the planar tail weight in X is comparable to $(1+|Z|)^{-1-2s}$. An unbounded proper interval contains a half-line, where the quadratic value would give the divergent integral $a\int^\infty Z^{1-2s}\,\mathrm{d}Z$. Thus a proper J_Z must be bounded, say $J_Z\subset[-H,H]$.

Finite derivative pressure in a strip. Under this temporary hypothesis, the sandwich gives common local amplitudes for P_T, and (15.10) gives common tails. The proof of Lemma 15.2, applied to their common semiconvexity on a neighborhood of $[0,1]\times[-H,H]$, bounds every translated pressure by one constant C. Countably many nonnegative integer translates give $0\le\mu\le C$ almost everywhere for $X\ge 0$.

The inequality $\partial_X\mu\ge 0$ gives monotone slices. To retain the measure content explicitly, take a countable dense family of nonnegative smooth tests ψ on bounded X intervals. Testing $\partial_X\mu$ by $\psi(X)\zeta(Z)$ with $\zeta\ge 0$ and using Fubini shows, outside one null set of Z, that $-\int\mu(X,Z)\psi'(X)\,\mathrm{d}X\ge 0$ for every such ψ. Local L^1 continuity in the test extends this to all nonnegative smooth ψ. The one-dimensional distributional monotonicity criterion supplies a non-decreasing representative $m_Z(X)$ of the slice. One-sided limits of local averages give a measurable choice. Its bound for $X\ge 0$ and monotonicity imply $0\le m_Z\le C$ on all of $\mathbb{R}$. Outside $[-H,H]$ it is zero. Let $\nu_Z=Dm_Z$ be its positive Stieltjes measure. Fubini and integration by parts identify

$$\int_{\mathbb{R}^2}\varphi\,\mathrm{d}\nu=\int_{-H}^{H}\int_{\mathbb{R}}\varphi(X,Z)\,\mathrm{d}\nu_Z(X)\,\mathrm{d}Z\qquad(\varphi\in C_c^\infty(\mathbb{R}^2)). \tag{15.11}$$

The right side is locally finite since each slice has total variation at most C. Increasing compact cutoffs then give

$$\nu(\mathbb{R}^2)=\int_{-H}^{H}\bigl[m_Z(+\infty)-m_Z(-\infty)\bigr]\,\mathrm{d}Z\le 2HC. \tag{15.12}$$

This is finiteness of the derivative pressure, not of μ.

A potential and its contact-ray average. Put $p_0=2-2s\in(0,1)$ and

$$Q(x)=k_s\int_{\mathbb{R}^2}|x-y|^{-p_0}\,\mathrm{d}\nu(y),\qquad K_s(x)=k_s|x|^{-p_0}.$$

Tonelli gives $Q\ge 0$ and $Q\in L^1_{\mathrm{loc}}$. Moreover

$$\sup_y\int_{\mathbb{R}^2}|x-y|^{-p_0}(1+|x|)^{-2-2s}\,\mathrm{d}x<\infty:$$

on $|x-y|<1$ use local kernel integrability and the bounded weight; on its complement use the integrable weight and a kernel at most one. Thus $Q\in L^1_s$. The same estimate and $|\mathcal{A}_s\varphi|\le C_\varphi(1+|x|)^{-2-2s}$ justify absolute Fubini against compact tests and give $\mathcal{A}_sQ=\nu$. Also, uniformly in ball centers,

$$\int_{B_R(x_0)}Q\,\mathrm{d}x\le C\nu(\mathbb{R}^2)R^{2-p_0}=C\nu(\mathbb{R}^2)R^{2s}. \tag{15.13}$$

This follows by integrating the radially decreasing kernel over a ball and comparing with its centered integral.

The function $w+Q$ is nonnegative, ordinary-tail, and entire fractional harmonic. Lemma 7.2 makes it a polynomial; alternatively its first conclusion already makes it affine. The estimate

$$\int_{B_R}(w+Q)\,\mathrm{d}x \le CR^{2+\delta}+C\nu(\mathbb{R}^2)R^{2s}, \qquad \delta<1,$$

excludes every positive degree by the absolute-integral growth of a nonzero polynomial. Consequently

$$w+Q=c \quad \text{almost everywhere}, \qquad c\ge 0. \tag{15.14}$$

This gives Q the continuous representative $c-w$; it does not claim pointwise finiteness of the raw measure potential everywhere.

Choose an actual contact ray $[X_0,\infty)\times\{Z_0\}$. For fixed $R>\max(1,X_0)$, continuity and $w=0$ on this ray give

$$\lim_{h\downarrow 0}\frac{1}{2hR}\int_{[R,2R]\times[Z_0-h,Z_0+h]} Q\,\mathrm{d}x = c.$$

For every source point y and height z,

$$\int_R^{2R}\left((X-y_X)^2+(z-y_Z)^2\right)^{-p_0/2}\mathrm{d}X \le \int_R^{2R}|X-y_X|^{-p_0}\,\mathrm{d}X \le C_{p_0}R^{1-p_0}.$$

The last bound includes y_X inside the interval because $p_0<1$. Tonelli bounds the tube average by $C\nu(\mathbb{R}^2)R^{-p_0}$. First take $h\downarrow 0$ at fixed R, and then $R\to\infty$. This gives $c=0$, forcing $w=0$ by positivity and continuity, a contradiction. Both proper-projection cases are impossible. □

Proposition 15.5 (Normalized convex epigraph)**.** *In the oriented, normalized coordinates of Lemma 9.3, $K=\{X\ge g(Z)\}$ for a finite continuous convex coercive function g, with*

$$g\ge 0, \qquad g(0)=0, \qquad K\subset\{X\ge 0\}, \qquad [0,\infty)\times\{0\}\subset K. \tag{15.15}$$

In these coordinates $P(X,0)=w(X,0)=0$ for $X\ge 0$.

Proof. Full projection, closedness and positive recession make each horizontal slice a terminal ray or a whole line. A whole line in K would, by the supporting-plane inequality against both its ends, force $P_X=0$ everywhere. Thus every slice is a finite terminal ray. Convexity of K makes its boundary function g convex and continuous. The minimum-X sublevel conclusion of Lemma 9.3 shows that g is bounded below and has compact sublevels; hence it is coercive and attains its minimum. The genuine obstacle gauge (9.4) has already placed a minimum-X contact point at the origin, with $K\subset\{X\ge 0\}$. Therefore $g(0)=\min g=0$ and (15.15) follows without any further change of coordinates. Regularity gives the assertion about w on the exact contact ray. □

At this stage contact already satisfies the sublinear horn bound

$$|Z|\le C(1+X^{\gamma/2}) \quad \text{on } K. \tag{15.16}$$

Indeed $aZ^2=-P(X,Z)\le C(1+X^\gamma+|Z|^\gamma)$, and $\gamma<2$ permits absorption of the last term. Consequently $R^{-1}(K\times\{0\})$ eventually avoids every compact subset of $\mathbb{R}^3\setminus\mathcal{L}$, where $\mathcal{L}=\{(X,0,0): X\ge 0\}$. Only this qualitative avoidance is needed for the endpoint state compactness below.

15.3. **A normal-scale pressure estimate.** The next two results are conditional on any actual state bound $|P(x)|\le C(1+|x|^\gamma)$ with

$$\lambda\le\gamma<2s, \qquad \delta=\gamma-1\in[\sigma,\kappa), \qquad \theta=\frac{\gamma-\kappa}{2-\kappa}\in[\tfrac12,1).$$

The horizontal secant proof of (15.6) then gives its bound with this same δ, including $\delta=\sigma$. The useful algebraic identity is

$$\kappa\theta+\gamma-\kappa=2\theta. \tag{15.17}$$

For the endpoint proof we will use a strictly supercritical γ; after the endpoint is proved, the same results apply at $\gamma = \lambda$ and $\theta = 1/2$.

Lemma 15.6 (Generalized line estimate and width)**.** *Suppose $\lambda \le \gamma < 2s$ and $|P(x)| \le C(1+|x|^\gamma)$. In the normalized coordinates, with δ, θ as just defined,*

$$w(X,Z) \le C|Z|^\kappa X^{\gamma-1-\kappa} \qquad (X \ge \max(1,|Z|)), \tag{15.18}$$

$$g(Z) \ge c|Z|^{1/\theta} - C, \tag{15.19}$$

$$|P(X,Z)| \le C(1+|Z|^\gamma + |Z|^\kappa X^{\gamma-\kappa}) \qquad (X \ge 0). \tag{15.20}$$

Consequently $|Z| \le C_K X^\theta$ on $K \cap \{X \ge 1\}$.

Proof. At the vertex, $w(0) = 0$ and Lemma 15.2 give $w(x) = O(|x|^\kappa)$ locally. Together with (15.6), this yields $w(x) \le C|x|^\delta$ globally. Its line trace $f(\xi) = w(\xi, 0)$ is zero for $\xi \ge 0$ and bounded by $C(-\xi)^\delta$ for $\xi < 0$. Proposition 14.3 applies with this original exponent $\delta < \kappa$: w is continuous, nonnegative, globally fractional subharmonic, and ordinary-tail. With the kernel of Lemma 14.1 it gives

$$w(X,Z) \le \mathcal{H}_f(X,Z) = \frac{\Gamma(s)}{\sqrt{\pi}\Gamma(\sigma)}|Z|^\kappa \int_{\mathbb{R}} \frac{f(\xi)}{((X-\xi)^2+Z^2)^s}\,\mathrm{d}\xi.$$

For $X \ge \max(1,|Z|)$, positivity and $u = Xv$ give

$$w(X,Z) \le C|Z|^\kappa \int_0^\infty u^\delta (X+u)^{-2s}\,\mathrm{d}u = C\,\mathrm{B}(\delta+1, \kappa-\delta)|Z|^\kappa X^{\delta-\kappa}.$$

Both beta parameters are positive. This proves (15.18); axis values follow by continuity.

Since $\gamma < 2$, for large $|Z|$ we have $u(0,Z) = aZ^2 + O(1+|Z|^\gamma) \ge (a/2)Z^2$. Horizontal absolute continuity and $u(g(Z),Z) = 0$ give

$$u(0,Z) = \int_0^{g(Z)} w(X,Z)\,\mathrm{d}X.$$

Split at $L = \max(1,|Z|)$. The first part is bounded by $C(1+|Z|^\gamma)$ from (15.6). The second is at most $C|Z|^\kappa g(Z)^{\gamma-\kappa}$ by (15.18), since $\gamma - \kappa > 0$. Absorbing the first term against aZ^2 proves (15.19), with an additive constant on bounded Z. Inverting this estimate gives the asserted contact width. Integrating the same derivative bounds from 0 to any $X \ge 0$, and using $|P(0,Z)| \le C(1+|Z|^\gamma)$, proves the absolute estimate (15.20). □

Proposition 15.7 (Pressure on the normal scale)**.** *Under the hypotheses and parameter range of Lemma 15.6, the original pressure density satisfies*

$$0 \le \mu(X,Z) \le C(1+X_+)^{2\theta(1-s)}\mathbf{1}_K(X,Z) \quad \textit{almost everywhere.} \tag{15.21}$$

Proof. For $R \ge 1$, define

$$p_R(\xi,z) = R^{-2\theta}P(R+R^\theta\xi, R^\theta z).$$

The gap $p_R + az^2$ is convex. We first prove on the entire rescaled plane the uniform majorant

$$|p_R(\xi,z)| \le C(1+|(\xi,z)|^\gamma). \tag{15.22}$$

If $X = R + R^\theta\xi \ge 0$, (15.20) gives

$$|p_R| \le C\left[R^{-2\theta} + R^{\theta(\gamma-2)}|z|^\gamma + R^{\kappa\theta-2\theta}|z|^\kappa(R+R^\theta\xi)^{\gamma-\kappa}\right].$$

The first two terms are uniformly bounded by $C(1+|z|^\gamma)$. By (15.17), the last is

$$C|z|^\kappa(1+R^{\theta-1}\xi)^{\gamma-\kappa} \le C|z|^\kappa(1+|\xi|)^{\gamma-\kappa} \le C(1+|(\xi,z)|^\gamma).$$

The parenthesis on the left is nonnegative in the present region. If instead $X<0$, then $\xi<-R^{1-\theta}$ and $|X|\le R^{\theta}|\xi|$. The original global γ bound gives

$$|p_R|\le C\left[R^{-2\theta}+R^{\theta(\gamma-2)}(|\xi|^{\gamma}+|z|^{\gamma})\right],$$

which has the same uniform bound. This proves (15.22) on both sides of the original axis. Since $\gamma<2s$, it supplies common ordinary tails as well as common local amplitudes.

Apply (15.4) to positive mollifications of p_R, with the same coefficient a. On a fixed compact set, the small increments contribute at most $Ca\int_0^1 r^{1-2s}\,\mathrm{d}r$; the large increments are controlled uniformly by (15.22). Positivity of $\mathcal{A}_s p_R$ and passage from mollifications give $0\le\mathcal{A}_s p_R\le C_Q$ on each fixed compact Q, uniformly in R. Scaling the original density gives

$$\mathcal{A}_s p_R(\xi,z)=R^{-2\theta(1-s)}\mu(R+R^{\theta}\xi,R^{\theta}z)\quad\text{almost everywhere.}$$

Choose the width constant C_K of Lemma 15.6 and fix $Q=[-2,2]\times[-2C_K-2,2C_K+2]$. For each sufficiently large integer n, the rectangle

$$[n-2n^{\theta},n+2n^{\theta}]\times[-(2C_K+2)n^{\theta},(2C_K+2)n^{\theta}]$$

contains $K\cap\{n\le X\le n+1\}$. The bound with $R=n$ therefore gives $\mu\le Cn^{2\theta(1-s)}\le C'X^{2\theta(1-s)}$ there, outside a null set. A countable union of these exceptional sets is still null. The remaining bounded-X contact is compact by coercivity and is covered by Lemma 15.2. Outside K the density is zero. This proves (15.21) without an uncountable union of exceptional sets, and without asserting that $\nu=\partial_X\mu$ has a density. □

15.4. **The derivative extension and its exact radial source.** For the remainder of the endpoint proof choose

$$0<\varepsilon<\sigma(1-\sigma),\qquad \gamma=\lambda+\varepsilon,\qquad \delta=\sigma+\varepsilon<\kappa,\qquad \theta=\frac12+\frac{\varepsilon}{2-2\sigma}.\tag{15.23}$$

Its actual state bound is supplied by (15.3). In addition to the preceding conclusions, this choice gives $2\theta<1+\sigma$, which is the strict tail-moment inequality used below.

For $q=(x,Y)=(X,Z,Y)$ put $\rho(q)=|Y|^{\eta}$ and $L_\eta=\operatorname{div}(\rho\nabla)$. The even Poisson extension, with its ordinary-tail interpretation, is

$$\mathcal{E}_s w(x,Y)=\frac{s}{\pi}|Y|^{2s}\int_{\mathbb{R}^2}\frac{w(y)}{(|x-y|^2+Y^2)^{1+s}}\,\mathrm{d}y\quad(Y\neq0),\qquad \mathcal{E}_s w(x,0)=w(x).\tag{15.24}$$

Its conormal constant is

$$e_s=2^{1-2s}\frac{\Gamma(1-s)}{\Gamma(s)}>0.$$

Set $W=\mathcal{E}_s w$. In particular W is the derivative extension, while the state extension below is denoted by E.

Lemma 15.8 (Derivative extension energy)**.** *The function W is continuous, nonnegative, even in Y, and belongs to $H^1_{\mathrm{loc}}(\mathbb{R}^3,\rho)$. It satisfies*

$$L_\eta W=2e_s\,\iota_{\#}\nu\ge0,\qquad \iota(X,Z)=(X,Z,0),\tag{15.25}$$

and, for every compact smooth cutoff ζ,

$$\int\zeta^2\rho|\nabla W|^2\le4\int\rho W^2|\nabla\zeta|^2.\tag{15.26}$$

Before any endpoint estimate, one has

$$0\le W(q)\le C(1+|q|^{\delta}),\qquad 0\le W(q)\le C|q|^{\kappa}.\tag{15.27}$$

Proof. Corollary 11.5 identifies this function with the intrinsic bulk derivative $W = -U_X$, proves its continuity, positivity, local weighted energy, and gives both (15.25) and (15.26). The exact conormal factor is e_s because $\widehat{\mu} = e_s\mu$. These conclusions were established by local-energy quotients, so no pressure-density or global squared-tail premise is used here.

The first bound in (15.27) follows from (15.6). At the normalized origin, $w(x) = O(|x|^\kappa)$ by contact regularity; since $\delta < \kappa$, the global bound also implies $w(x) \le C|x|^\kappa$ everywhere. Positivity, scaling and the finite κ moment of the Poisson kernel give its second bound. Here $\kappa < 2s$; no Poisson $2s$ moment is used. Local Hölder continuity alone was not treated as an H^s assertion. □

Proposition 15.9 (Exact principal source and positive amplitude)**.** *Put*

$$\phi(\omega) = (1-\omega_X)^\sigma, \qquad N = \int_{S^2} |\omega_Y|^\eta \phi(\omega)^2 \,\mathrm{d}S > 0,$$

and define

$$c(r) = \frac{1}{N}\int_{S^2} |\omega_Y|^\eta W(r\omega)\phi(\omega)\,\mathrm{d}S, \qquad A(r) = r^{-\sigma}c(r), \qquad J(r) = r^2A'(r). \tag{15.28}$$

Then A is locally absolutely continuous, J has a locally bounded-variation nondecreasing representative, and

$$DJ = G, \qquad G = \frac{2e_s}{N}\,\mathfrak{r}_\#\big[(\mathfrak{r}-X)^\sigma\nu\big] \quad \text{on } (0,\infty), \qquad \mathfrak{r}(X,Z) = \sqrt{X^2+Z^2}. \tag{15.29}$$

Moreover $A(0+) = J(0+) = 0$, A is nondecreasing, and

$$A_\infty := \lim_{r\to\infty} A(r) \in (0,\infty). \tag{15.30}$$

In fact its value has the exact moment representation

$$A_\infty = \frac{2e_s}{N}\int_{\mathbb{R}^2\setminus\{0\}} \frac{(\sqrt{X^2+Z^2}-X)^\sigma}{\sqrt{X^2+Z^2}}\,\mathrm{d}\nu(X,Z). \tag{15.31}$$

Proof. The angular form and both sides of the cutoff limit. Polar Fubini and Lemma 15.8 give weighted angular H^1 slices for almost every radius. Their positive-pole trace is zero. Indeed the bounded composed trace in Lemma 10.1,

$$H^1(\rho) \longrightarrow H^s(\mathbb{R}^2) \longrightarrow H^{s-1/2}(\mathbb{R}),$$

identifies the actual line trace of W with that of the continuous w, which is zero on the positive ray. On an annulus, smooth approximation converges in $L^2(\,\mathrm{d}r; H^1(S^2, |\omega_Y|^\eta))$. Bounded angular point trace and bounded ambient composed trace agree for the approximants and hence for their limits. Thus the slices belong to the exact zero-positive-pole form domain, with no condition at the negative pole.

The principal eigenfunction in Lemma 10.2 is ϕ, with eigenvalue $\Lambda = \sigma(\sigma + b_G) = \sigma(1-\sigma)$. Its form identity gives

$$\int_{S^2} |\omega_Y|^\eta \nabla_S W\cdot\nabla_S\phi\,\mathrm{d}S = \Lambda\int_{S^2}|\omega_Y|^\eta W\phi\,\mathrm{d}S \tag{15.32}$$

for almost every radius. This is an identity against zero-pole form-domain slices, not against arbitrary full-sphere tests.

Choose smooth χ_h, zero within angular distance h of e_X, one outside distance $2h$, with $|\nabla_S\chi_h| \le C/h$. Near that pole, writing v for angular distance, $\phi = O(v^{2\sigma})$ and $|\nabla_S\phi| = O(v^{2\sigma-1})$; weighted angular measure has radial order $v^{1+\eta}\,\mathrm{d}v$ with an integrable transverse angular factor. Thus

$$\int_{v<2h} |\omega_Y|^\eta\phi^2|\nabla_S\chi_h|^2\,\mathrm{d}S \le Ch^{\eta+4\sigma} = Ch^{2\sigma} \longrightarrow 0.$$

The lost gradient energy has the same vanishing order, and the lost mass vanishes as well. Hence $\chi_h\phi \to \phi$ in weighted H^1, and also uniformly, with error $O(h^{2\sigma})$. For any $\zeta \in C_c^\infty(0,\infty)$ the

annular tests $\zeta(r)\chi_h(\omega)\phi(\omega)$ are smooth. Their energy pairings converge by H^1 convergence. Their pairings with $\iota_\#\nu$ converge by uniform convergence and finite source mass on the fixed annulus. Passing both sides of (15.25) therefore gives equality for $\zeta(r)\phi(\omega)$, including signed ζ.

Define the positive radial Radon measure

$$m = \frac{2e_s}{N}\,\mathfrak{r}_\#\big[(1-X/\mathfrak{r})^\sigma \nu\big] \quad \text{on } (0,\infty).$$

Combining the resulting radial test identity with (15.32) gives, exactly in distributions,

$$(r^{b_G+1}c')' - \Lambda r^{b_G-1}c = m. \tag{15.33}$$

Since $c = r^\sigma A$ and $2\sigma + b_G = 1$, its left side is $r^{-\sigma}(r^2A')'$. Multiplication of the radial measure by r^σ proves (15.29). No restriction of ν to a circle and no absolute continuity of its radial pushforward has been used.

The origin flux. Radial energy puts c, A in ordinary H^1 on compact intervals inside $(0,\infty)$, so initially J is locally integrable. Equation (15.29) makes it locally BV and nondecreasing. The second bound in (15.27) gives $A(r) = O(r^{\kappa-\sigma}) = O(r^\sigma)$ at zero, hence $A(0+) = 0$. Let $J(0+)$ be its extended-real monotone limit. If it is negative, including $-\infty$, then $J \le -c_0$ on some interval $(0, r_0)$; integrating $A' = J/r^2$ from r to r_0 forces $A(r) \to +\infty$. If it is positive and finite, $J \ge c_0$ there instead forces $A(r) \to -\infty$. Both contradict the proved origin bound. Positive infinity is impossible for a nondecreasing function finite at interior radii. Thus $J(0+) = 0$, $J \ge 0$, and $A' \ge 0$. For its right-continuous representative, $J(r) = G((0,r])$.

The precise tail moment. For $T > r > 0$, Tonelli applied to $A' = J/r^2$ gives

$$A(T) - A(r) = J(r)(r^{-1} - T^{-1}) + \int_{(r,T]} (t^{-1} - T^{-1})\,\mathrm{d}G(t). \tag{15.34}$$

Thus boundedness of A at infinity follows from

$$\int_{\{\mathfrak{r} > r_0\}} F\,\mathrm{d}\nu < \infty, \qquad F(X,Z) = \frac{(\mathfrak{r}-X)^\sigma}{\mathfrak{r}}, \tag{15.35}$$

for one fixed $r_0 > 0$. For $X > 0$, write

$$F = \frac{|Z|^{2\sigma}}{(\mathfrak{r}+X)^\sigma \mathfrak{r}}, \qquad -\partial_X F = F\left(\frac{\sigma}{\mathfrak{r}} + \frac{X}{\mathfrak{r}^2}\right) \ge 0. \tag{15.36}$$

It follows that

$$F \le C|Z|^{2\sigma}X^{-\sigma-1}, \qquad 0 \le -\partial_X F \le C|Z|^{2\sigma}X^{-\sigma-2}. \tag{15.37}$$

Both F and $\partial_X F$ are continuous across $Z = 0$ on compact subsets of $X > 0$, with value zero. No Z derivative is claimed or needed.

Fix large L and a smooth nondecreasing cutoff $\chi_L(X)$, zero for $X \le L$ and one for $X \ge 2L$. Let $\zeta_T(X) = \zeta(X/T)$, where $0 \le \zeta \le 1$ is smooth and nonincreasing, one below 1, zero above 2, and $|\zeta_T'| \le C/T$; take $T > 4L$. Append a transverse compact cutoff equal to one on the entire contact portion $L \le X \le 2T$, which is bounded by coercivity. This produces a compactly supported test. Mollify F only in Z before applying $\nu = \partial_X \mu$. The function and its X derivative converge uniformly on this compact set, so locally finite ν and locally integrable μ permit passage to the limit. The exact identity is

$$\begin{aligned} I_{L,T} := \int \chi_L\zeta_T F\,\mathrm{d}\nu = &\int \mu\chi_L\zeta_T(-\partial_X F)\,\mathrm{d}X\,\mathrm{d}Z \\ &- \int \mu\chi_L'\zeta_T F\,\mathrm{d}X\,\mathrm{d}Z - \int \mu\chi_L\zeta_T' F\,\mathrm{d}X\,\mathrm{d}Z. \end{aligned} \tag{15.38}$$

There is no derivative of the transverse cutoff: integration by parts is only in X. Both measures vanish outside contact, where that cutoff might differ from one.

The pressure and width estimates give, for almost every large X,

$$\int_{\mathbb{R}} \mu(X,Z)|Z|^{2\sigma}\,\mathrm{d}Z \le CX^{\theta(1-2\sigma)+\theta(2\sigma+1)} = CX^{2\theta}. \tag{15.39}$$

Here $2(1-s) = 1-2\sigma$ was used. Hence the first integrand in (15.38), after transverse integration, is bounded by $CX^{2\theta-\sigma-2}$. Set $\tau = 1+\sigma-2\theta > 0$, as ensured by (15.23). The first term is at most $CL^{-\tau}$. The middle term is finite, supported on a fixed compact contact portion, and nonpositive. The upper cutoff term must be retained; it is nonnegative and at most

$$\frac{C}{T}\int_T^{2T} X^{2\theta-\sigma-1}\,\mathrm{d}X \le CT^{2\theta-\sigma-1} = CT^{-\tau} \longrightarrow 0. \tag{15.40}$$

Because $\zeta_T \uparrow 1$, monotone convergence for the nonnegative measure pairing gives

$$\int_{\{X\ge 2L\}} F\,\mathrm{d}\nu \le CL^{-\tau} < \infty.$$

The remaining source with $X \le 2L$ is compact by coercivity. On its part where $\mathfrak{r} > r_0$, the kernel F is bounded, so it contributes a finite amount. This proves (15.35). The strict inequality $2\theta < 1+\sigma$ controls both the tail integral and its outer cutoff error.

The limit and its positivity. Let $T \to \infty$ in (15.34). With the right-continuous value of $J(r_0)$ (including an atom at r_0),

$$A_\infty - A(r_0) = \frac{J(r_0)}{r_0} + \frac{2e_s}{N}\int_{\{\mathfrak{r}>r_0\}} F\,\mathrm{d}\nu < \infty. \tag{15.41}$$

The nonzero continuous nonnegative trace w has a strictly positive Poisson extension when $Y \neq 0$. Therefore $c(r) > 0$ for every $r > 0$, and the nondecreasing A has strictly positive limit. This proves (15.30). Finally $A(0+) = J(0+) = 0$ and Tonelli also give

$$A(r) = \int_{(0,r]} (t^{-1} - r^{-1})\,\mathrm{d}G(t).$$

Monotone convergence as $r \to \infty$ proves (15.31). Its near-origin integrability is a consequence of the radial identity and the proved origin behavior, not a hypothesis on the pressure near the vertex. □

15.5. **From the principal projection to the critical derivative.**

Proposition 15.10 (Global slit majorant)**.** *With*

$$H(q) = (|q| - X)^\sigma, \qquad H_0(X,Z) = H(X,Z,0),$$

one has

$$0 \le w(x) \le CH_0(x) \le C'|x|^\sigma \quad (x \in \mathbb{R}^2), \tag{15.42}$$

and

$$0 \le W(q) \le C|q|^\sigma \quad (q \in \mathbb{R}^3). \tag{15.43}$$

Proof. For $r > 0$ let $q_r = (-r, 0, 0)$. The ball $B_{r/2}(q_r)$ lies in $\{X < 0\}$ and is source-free by (15.15) and (15.25). Thus the even W is a nonnegative weak weighted harmonic function there, including across $Y = 0$; its local energy was proved in Lemma 15.8. The scale-invariant interior Harnack theorem for the A_2 weight $|Y|^\eta$ gives

$$W(r\omega) \ge C^{-1}W(q_r) = C^{-1}w(-r,0) \quad \text{if } |\omega + e_X| < \tfrac{1}{4};$$

see [13, Theorem 2.3.8 and Corollary 2.3.9]. The constants are independent of r by homogeneity of the weight. Continuity identifies its harmonic representative with the actual trace w. On this

fixed spherical cap, ϕ is bounded below and $\int |\omega_Y|^\eta \phi \, \mathrm{d}S$ is finite and positive. Integrating the comparison gives

$$w(-r,0) \le Cc(r) = Cr^\sigma A(r) \le Cr^\sigma A_\infty.$$

This is an interior source-free comparison on a fixed cap; no boundary comparison on a shrinking horn is involved.

The actual full-line trace consequently satisfies $0 \le f(\xi) = w(\xi,0) \le C(-\xi)_+^\sigma$. Apply Proposition 14.3 once more, using the already proved pre-endpoint growth exponent $\delta < \kappa$ in its infinity barrier. Its hypothesis is therefore available independently of the conclusion now sought. Positivity of the hitting kernel and Lemma 14.4 give

$$w \le \mathcal{H}_f \le C\mathcal{H}_{(-\xi)_+^\sigma} = C2^{-\sigma} H_0.$$

This proves (15.42). Positivity and the finite σ moment of the Poisson kernel then give $W = \mathcal{E}_s w \le C\mathcal{E}_s(|x|^\sigma) \le C'|q|^\sigma$, proving (15.43). □

15.6. **The endpoint state record contradiction.**

Proof of Theorem 15.1. Let $E = \mathcal{E}_s P$ and $M(R) = \sup_{B_R \subset \mathbb{R}^3} |E|$ for $R \ge 1$. The earlier state-extension conclusions used in Proposition 11.3 apply in these same normalized coordinates: E is continuous, even, locally in $H^1(\rho)$, has zero trace on $\mathcal{L}$, and satisfies $L_\eta E = -2e_s \iota_\# \mu$. Both E_+ and E_- are weak subsolutions. These signs come from the negative source for E_- and from positive-level truncations vanishing near compact source portions for E_+, since its source trace is $-aZ^2 \le 0$. Thus the phase compactness tool applies without assuming that E itself has a sign.

Suppose $M(R)/R^\lambda$ is unbounded. Choose $\lambda < p' < q < 2s$. The separate strict-power bound (15.3) with exponent p' gives $M(R)/R^q \to 0$. Lemma 10.3, with inner exponent λ and outer exponent q, supplies radii $R_j \to \infty$ and $M_j = M(R_j)$ such that $M_j/R_j^\lambda \to \infty$ and the normalized extensions $E_j(q') = E(R_j q')/M_j$ satisfy

$$\sup_{B_1} |E_j| = 1, \qquad \sup_{B_r} |E_j| \le r^\lambda \quad (R_j^{-1} \le r \le 1), \qquad \sup_{B_r} |E_j| \le r^q \quad (r \ge 1).$$

By (15.16) their source supports eventually avoid each compact set off $\mathcal{L}$. Lemma 10.4, in its qualitative source-avoidance form, gives a subsequence converging weakly in local $H^1(\rho)$ and strongly in local $L^2(\rho)$ to a nonzero limit E_∞. Here phase Caccioppoli bounds the energy, and phase Moser retains nonvanishing despite possible concentration near the collapsing source. The exact zero ray trace survives the bounded composed trace map. The Hardy-density assertion of Lemma 10.1 extends its equation off the ray to the zero-ray Friedrichs form domain. The inherited bounds are r^λ at the origin and r^q at infinity; the inner bound is needed only eventually for each fixed positive radius.

The complete spectrum in Lemma 10.2 has, below two, only degrees σ, 1, and λ. In the annular expansion, a coefficient of degree α is $a_\alpha r^\alpha + b_\alpha r^{-b_G-\alpha}$. The inner bound eliminates the negative branches and every $\alpha < \lambda$; the outer bound eliminates every $\alpha > q$. Consequently only the simple degree-λ mode can remain. Its generator and horizontal derivative, identified in Corollary 10.5, are

$$\Phi_\lambda(q) = (|q| - X)^\sigma (X + \sigma|q|), \qquad \partial_X \Phi_\lambda(q) = (1-\sigma^2)(|q| - X)^\sigma. \tag{15.44}$$

The spectral conclusion is $E_\infty = c_\lambda \Phi_\lambda$ with $c_\lambda \ne 0$.

On the other hand the planar traces $p_j(x) = P(R_j x)/M_j$ have, on every fixed planar ball,

$$|\partial_X p_j(x)| = \frac{R_j w(R_j x)}{M_j} \le C \frac{R_j^{1+\sigma}}{M_j} |x|^\sigma \longrightarrow 0$$

uniformly there, by (15.42) and the record normalization. Weak local extension $H^1(\rho)$ convergence gives weak local H^s convergence of the planar traces. For any $\psi \in C_c^\infty(\mathbb{R}^2)$, integration by parts

gives

$$\langle \partial_X \operatorname{Tr} E_\infty, \psi\rangle = \lim_j \langle \partial_X p_j, \psi\rangle = 0.$$

No local uniform convergence of state traces, or convergence of extension derivatives, was used. But the planar trace derivative in (15.44) is $c_\lambda(1-\sigma^2)H_0$, which is nonzero. This contradiction proves $M(R) = O(R^\lambda)$, hence (15.2). The fixed reflection, translation and affine obstacle correction preserve this bound because $\lambda > 1$, so it holds in the original coordinates as well. □

The principal coefficient (15.28) and its already proved limit (15.30) will now determine the canonical amplitude. Its positivity and finiteness have been obtained before any canonical blow-down is identified.

16. The explicit parabolic profile

Throughout this section assume $1/2 < s < 1$. Put

$$t = 1-s, \quad \sigma = s - \tfrac{1}{2}, \quad \beta = \tfrac{3}{2} - s, \quad \lambda = s + \tfrac{1}{2}, \qquad k_s = \frac{\Gamma(1-s)}{4^s \pi \Gamma(s)}.$$

Thus $0 < t, \sigma < 1/2$, $\beta + \lambda = 2$, and $\lambda < 2s$. The Fourier convention is $\widehat{f}(\xi) = \int e^{-ix\cdot\xi} f(x)\,\mathrm{d}x$, with inverse factor $(2\pi)^{-n}$. The operator $\mathcal{A}_s = (-\Delta)^s$ has symbol $|\xi|^{2s}$ and inverse kernel $K_s(x) = k_s|x|^{-2t}$ in two dimensions.

The lower-order canonical field, for $M > 0$, is

$$\Psi_M(x) = M \int_0^\infty [K_s(x - ue_X) - K_s(-ue_X) - \nabla K_s(-ue_X)\cdot x] u^\beta \,\mathrm{d}u. \tag{16.1}$$

This is a single combined integrand. The construction below proves its normalization and its role as the degree-λ limit.

Proposition 16.1 (Explicit model)**.** *For $b, C > 0$ set $q(X,Z) = X - bZ^2$ and $\mu_0 = Cq_+^t$. The combined integral*

$$P_0(x) = \int_{\mathbb{R}^2} [K_s(x-y) - K_s(-y) - \nabla K_s(-y)\cdot x]\mu_0(y)\,\mathrm{d}y \tag{16.2}$$

is absolutely convergent. With

$$A_s^* = \frac{2^{1-2s}\Gamma(3/2-s)\Gamma(2-s)}{\sqrt{\pi}}, \qquad a = Cb^t A_s^*, \tag{16.3}$$

it satisfies

$$P_0 \geq -aZ^2, \quad \mathcal{A}_s P_0 = \mu_0, \quad \{P_0 = -aZ^2\} = \{X \geq bZ^2\}.$$

Its gap $P_0 + aZ^2$ is convex, $P_0(0) = 0$, $\nabla P_0(0) = 0$, and $|P_0(x)| \leq C'(1+|x|^\lambda)$. Its canonical blow-down is Ψ_M from (16.1)*, with*

$$M = \frac{CB(1/2, 2-s)}{\sqrt{b}}, \quad D_s = \frac{B(1/2,2-s)}{A_s^*}, \quad b = (aD_s/M)^{1/\beta}, \quad C = \frac{a}{A_s^* b^t}. \tag{16.4}$$

16.1. **Interval identity and finite ellipses.** Put $B_1 = B(1/2, t+1)$, $B_2 = B(1/2 - t, t + 3/2)$, and

$$F(v) = \int_{-1}^1 |v-u|^{-2t}(1-u^2)^{t+1/2}\,\mathrm{d}u.$$

The shared angular lemma, Lemma 6.3, gives

$$F(v) = B_2(1 - 2tv^2) =: Q(v), \qquad |v| \leq 1, \tag{16.5}$$

together with endpoint C^1 matching and $F > Q$ outside the interval. The interval identity is due to Carrillo–Huang; the compact angular reduction was proved in Section 6 with its normalization and

strict global gap. Here the additional task is its noncompact parabolic degeneration. Furthermore, as $\delta \downarrow 0$,

$$F(1+\delta) - Q(1+\delta) = 2^{\beta} H_t \delta^{\lambda} + o(\delta^{\lambda}), \quad H_t = \frac{\Gamma(t+3/2)\Gamma(t+1/2)}{\Gamma(2t)\sigma(1+\sigma)}. \tag{16.6}$$

Indeed, rescaling $1-u=\delta r$ in $F''(1+\delta)$ gives

$$F''(1+\delta) \sim 2t(2t+1)2^{\beta} B(\beta+1, 2t+1-\beta)\delta^{\sigma-1}.$$

Two integrations and $\Gamma(2t+2) = 2t(2t+1)\Gamma(2t)$ give (16.6). The bounded quadratic contribution is $O(\delta^2)$, and the ratio in (16.6) is uniformly bounded for small δ.

For $L>0$, put $R_L^2 = L/(2b)$ and

$$q_L = X - \frac{X^2}{2L} - bZ^2 = \frac{L}{2}\left(1 - \frac{(X-L)^2}{L^2} - \frac{Z^2}{R_L^2}\right), \qquad \mu_L = C(q_L)_+^t.$$

The supporting ellipse has center $(L,0)$, semiaxes (L, R_L), and is tangent to zero. Let $V_L = K_s * \mu_L$ and subtract its affine jet at zero to obtain P_L. For this ellipse $\mathsf{H}_L = \operatorname{diag}(L^2, R_L^2)$ and $m_L = C(L/2)^t \pi L R_L/(t+1)$. The angular kernel identity (6.13) and the projection formula in Lemma 6.3 apply with this mass and center $(L,0)$. For $h^2 = L^2\theta_X^2 + R_L^2\theta_Z^2$, the projection of μ_L onto θ is

$$m_{L,\theta}(v) = C(L/2)^t \frac{LR_L}{h} B_1 \left[1 - \frac{(v - L\theta_X)^2}{h^2}\right]_+^{\beta}.$$

Fubini and (16.5) show that V_L equals a quadratic on the ellipse and lies above that same quadratic everywhere. Reflection removes the cross coefficient. Affine normalization at zero gives

$$P_L \geq -a_X(L)X^2 - a_Z(L)Z^2,$$

with equality on the ellipse and

$$a_i(L) = \frac{k_s C(L/2)^t L R_L B_1 B_2(2t)}{J} \int_{S^1} \theta_i^2 [L^2\theta_X^2 + R_L^2\theta_Z^2]^{-t-1} \, \mathrm{d}\theta. \tag{16.7}$$

The gradient of the global nonnegative C^1 gap is zero at its boundary point zero, so this normalization removes exactly the affine part of the interior quadratic. Near the two poles $\theta_X = 0$, substitute $\theta_X = (R_L/L)u$. Then

$$\int_{S^1} \theta_Z^2 h^{-2t-2} \, \mathrm{d}\theta \sim 2R_L^{-2t-1} L^{-1} \int_{\mathbb{R}} (1+u^2)^{-t-1} \, \mathrm{d}u.$$

Since the last integral is $\sqrt{\pi}\Gamma(t+1/2)/\Gamma(t+1)$, (16.7) gives $a_Z(L) \to Cb^t A_s^*$. Also $\theta_X^2 h^{-2t-2} \leq L^{-2t-2}|\theta_X|^{-2t}$, an integrable angular majorant; hence $a_X(L) = O(L^{-t-1/2}) \to 0$.

16.2. **Strict gap and convexity.** The limiting gap has the explicit expression

$$P_0(X,Z) + aZ^2 = E \int_{\mathbb{R}} [4b(bZ^2 - X) - (w + 2bZ)^2]_+^{\lambda} (1+w^2)^{-s} \, \mathrm{d}w, \tag{16.8}$$

where

$$E = \frac{2k_s C B_1 H_t}{J\sqrt{b}(4b)^{\lambda}} > 0.$$

Here is a uniform justification of the angular limit. For bounded x put $p = x\cdot\theta$ and $d_L = (|p - L\theta_X| - h)_+$. The finite-ellipse angular gap before multiplication by k_s/J is

$$C(L/2)^t L R_L B_1 h^{-2t}\big[F((p-L\theta_X)/h) - Q((p-L\theta_X)/h)\big].$$

On a fixed neighborhood $|\theta_X| < \varepsilon$, uniform for bounded x, it vanishes for all sufficiently large L, since

$$h^2 - (p - L\theta_X)^2 = L[\theta_Z^2/(2b) + 2\theta_X p] - p^2 > 0.$$

On $|\theta_X| \geq \varepsilon$, $h \sim L|\theta_X|$ uniformly, d_L remains bounded, and

$$d_L \longrightarrow [-\operatorname{sgn}(\theta_X)p - \theta_Z^2/(4b|\theta_X|)]_+$$

uniformly. Applying the bounded ratio form of (16.6) therefore gives the limiting angular gap

$$\frac{CB_1H_t}{\sqrt{b}|\theta_X|^{\beta+1}}[-\operatorname{sgn}(\theta_X)p - \theta_Z^2/(4b|\theta_X|)]_+^{\lambda}.$$

The substitution $w = \theta_Z/\theta_X$ on each hemisphere, together with $\beta + \lambda = 2$, gives (16.8). Its bracket is positive on a nonempty interval precisely when $X < bZ^2$, proving strict positivity outside the horn. Equivalently the bracket equals $-4bX - 4bZw - w^2$, an affine function of (X, Z) for fixed w. Since $\lambda > 1$, its positive part to the power λ is convex. Integration proves full convexity of the gap.

16.3. **Convergence, tails and the equation.** Near zero $\mu_0(y) \leq C|y|^t$, so $K_s\mu_0$ and $\nabla K_s\mu_0$ are locally integrable; the latter has radial bound $r^{-t}\,\mathrm{d}r$. The moving singularity at $y = x$ is integrable since $2t < 2$. On a longitudinal dyadic shell $X \asymp u$ at infinity the density is $O(u^t)$, width $O(\sqrt{u})$, length $O(u)$, and twice-subtracted kernel is $O_x(u^{-2t-2})$. Its contribution is $O_x(u^{-t-1/2})$, summable. This proves convergence of (16.2) as a single combined integrand; the subtraction terms at infinity are not separate integrals.

Because $0 \leq \mu_L \leq \mu_0$, the tail bound is uniform in L. On a bounded region $|(q_L)_+^t - q_+^t| \leq |X^2/(2L)|^t$, so the densities converge uniformly. The translated kernel has uniformly bounded local L^1 norm; cutting off the common tail thus proves $P_L \to P_0$ locally uniformly. Equality on the open horn passes from ellipses, and on its boundary follows by continuity; a nonzero boundary point need not lie in any finite ellipse.

For $r = |x| < 1$, splitting at $|y| = 2r$ and one gives $O(r^{2-t})$ for the three local terms, while Taylor's formula on $2r < |y| < 1$ gives $Cr^2 \int_{2r}^1 u^{-t-1}\,\mathrm{d}u = O(r^{2-t})$; the far part is $O(r^2)$. Thus $P_0(x) = O(r^{2-t}) = o(r)$ and its affine jet is zero. For $r = |x| \geq 2$, the outer Taylor estimate is $O(r^2 r^{-t-1/2}) = O(r^{\lambda})$. On $|y| < 2r$ the subtraction terms have the same bound by shell integration. The translated term is bounded using $\mu_0 \leq Cr^t$, the width $C\sqrt{r}$, and

$$\int_0^{Cr} |X - x_X|^{-2t}\,\mathrm{d}X \leq Cr^{1-2t},$$

uniformly in x_X. It is therefore $O(r^t\sqrt{r}\,r^{1-2t}) = O(r^{\lambda})$. These estimates are uniform for P_L and prove uniform ordinary-tail domination since $\lambda < 2s$. In particular

$$\sup_L \int_{|x|>R} |P_L(x)|(1+|x|)^{-2-2s}\,\mathrm{d}x \leq CR^{\lambda-2s}.$$

For compact smooth ϕ, $|\mathcal{A}_s\phi(x)| \leq C_\phi(1+|x|)^{-2-2s}$. Thus $\int P_L\mathcal{A}_s\phi \to \int P_0\mathcal{A}_s\phi$. Finite compact Riesz potentials solve $\mathcal{A}_sV_L = \mu_L$, and affine functions have zero fractional Laplacian and an ordinary tail because $2s > 1$. Passing the limit proves $\mathcal{A}_sP_0 = \mu_0$ in ordinary distributions.

16.4. **The canonical blow-down.** Direct transverse integration gives $\int \mu_0(X, Z)\,\mathrm{d}Z = MX_+^{\beta}$ with M in (16.4). The isotropically scaled pressure has density

$$\nu_r(X, Z) = Cr^{1/2}(X - brZ^2)_+^t,$$

and, on writing $v = \sqrt{r}Z$, converges vaguely to $MX_+^{\beta}\delta_0(Z)$. Its longitudinal marginal is exactly MX_+^{β} for every r. Kernel homogeneity expresses $r^{-\lambda}P_0(rx)$ as the combined potential of ν_r. To pass the singular kernel uniformly on compact x sets, first discard $X < \varepsilon$, whose derivative subtraction is $O(\varepsilon^{\sigma})$. On bounded X intervals the moving singularity is at most $C|X - x_X|^{-2t}$; its integral on $|X - x_X| < \delta$ is $O(\delta^{1-2t})$, uniformly in r, x. On $X > R$ Taylor's estimate gives $C_x \int_R^{\infty} X^{\beta-2t-2}\,\mathrm{d}X = O_x(R^{-\beta})$. The remainder is a bounded uniformly continuous kernel, where

vague convergence and a finite-net argument give compact-uniform convergence. Uniform global λ-growth supplies ordinary-tail convergence as well.

The resulting field is Ψ_M in (16.1). It is λ-homogeneous and has pressure $MX_+^\beta\delta_0$. It vanishes on the positive axis. On the negative axis, differentiating twice yields

$$\frac{\mathrm{d}^2}{\mathrm{d}v^2}\Psi_M(-v,0) = k_s M\frac{\Gamma(\beta+1)\Gamma(2t+1-\beta)}{\Gamma(2t)}v^{\sigma-1}, \quad v>0.$$

Homogeneity and the zero affine jet give

$$\Psi_M(X,0) = k_s M H_t(-X)_+^\lambda. \tag{16.9}$$

Combining $a = Cb^tA_s^*$ with the marginal coefficient proves (16.4) and completes Proposition 16.1.

17. Canonical asymptotics from the principal amplitude

We now use the critical bound and the finite positive principal limit already proved. No second argument for boundedness of that coefficient is needed. The canonical field is Ψ_M in (16.1).

Theorem 17.1 (Unique positive canonical limit)**.** *A noncylindrical $P \in C(\mathbb{R}^2)\cap\mathcal{T}_s$ satisfying* (2.7)*, with $1/2 < s < 1$, has a unique orientation $\varepsilon\in\{-1,1\}$ and a unique $M>0$ such that*

$$R^{-\lambda}P(\varepsilon RX, RZ) \longrightarrow \Psi_M(X,Z) \tag{17.1}$$

locally uniformly and in the ordinary fractional-tail norm. The orientation is the horizontal recession orientation of its contact. Fixed translations with their obstacle gauge leave M unchanged.

Corollary 17.2 (Critical layer estimates)**.** *In the normalized oriented coordinates of the preceding section,*

$$|P(x)| \le C(1+|x|^\lambda), \tag{17.2}$$

$$g(Z) \ge cZ^2 - C, \tag{17.3}$$

$$|P(X,Z)| \le C(1+|Z|^\lambda + |Z|^\kappa X^\beta) \quad (X\ge 0), \tag{17.4}$$

$$w(X,Z) \le C|Z|^\kappa X^{-\sigma} \quad (X \ge \max(1,|Z|)), \tag{17.5}$$

$$0 \le \mu(X,Z) \le C(1+X)^{1-s}\mathbf{1}_K \quad \textit{almost everywhere.} \tag{17.6}$$

In particular,

$$K \subset \{X\ge 0,\ |Z| \le C\sqrt{1+X}\}. \tag{17.7}$$

Proof. Theorem 15.1 gives the first estimate. Apply the generalized width and normal-pressure calculations of Lemma 15.6 and Proposition 15.7 with this actual exponent $\gamma=\lambda$. Then $\delta=\sigma$, $\theta=1/2$, and $\gamma-\kappa=\beta$. Its state, derivative, width, and pressure conclusions are precisely the remaining estimates. This use is after the endpoint has been established. □

Lemma 17.3 (Canonical derivative normalization)**.** *Let $H(q) = (|q|-X)^\sigma$ and $H_0(X,Z) = H(X,Z,0)$. Then*

$$-\partial_X\Psi_1 = d_1H_0, \qquad d_1 = \lambda D_1/2^\sigma > 0, \tag{17.8}$$

where $\Psi_1(X,0) = D_1(-X)_+^\lambda$ is given by (16.9)*.*

Proof. Differentiate the combined integral (16.1) and integrate by parts on truncated u intervals. This gives

$$-\partial_X\Psi_1(x) = \beta\int_0^\infty [K_s(-ue_X) - K_s(x-ue_X)]u^{-\sigma}\,\mathrm{d}u.$$

The boundary terms are $O(u^\sigma)$ at zero and $O_x(u^{-\beta})$ at infinity. At zero the integrable exponent is $2-2s+\sigma=\beta<1$; at infinity the difference gains an inverse power. At an interior positive-ray point the singular exponent in u is $2-2s<1$. Splitting about that point proves continuity

uniformly on compact sets. Scaling gives degree σ and growth $C|x|^\sigma$. The fractional Laplacian is supported on the horizontal line, and the actual line trace is $\lambda D_1(-X)_+^\sigma$. The signed uniqueness in Proposition 14.3, followed by Lemma 14.4, proves the claimed normalization. □

17.1. **Unique derivative limits and recovery of the state.**

Proposition 17.4 (Full derivative convergence)**.** *In the normalized coordinates,*

$$w_R(x) := R^{-\sigma} w(Rx) \longrightarrow A_\infty H_0(x) \tag{17.9}$$

locally uniformly on the whole plane and in $L^1_s(\mathbb{R}^2)$. The extensions $W_R(q) = R^{-\sigma}W(Rq)$ converge strongly in local $L^2(\rho)$ and weakly in local $H^1(\rho)$ to $A_\infty H(q)$.

Proof. Take any sequence $R_j \to \infty$. The homogeneous bound (15.43) and the scaled Caccioppoli inequality give uniform local weighted H^1 bounds for W_{R_j}. The weight $|Y|^\eta$ is an A_2 weight, so the local weighted Rellich argument used in Lemma 10.4 applies. More explicitly, weighted Sobolev gain bounds the contribution of a thin layer $|Y| < \delta$ by its vanishing weighted measure; ordinary Rellich on its complement gives compactness there. A diagonal extraction yields a limit W_∞ strongly in local $L^2(\rho)$ and weakly in local $H^1(\rho)$. It is even and satisfies $0 \le W_\infty(q) \le C|q|^\sigma$ almost everywhere.

The support in (15.25) and the horn collapse (17.7) show that W_{R_j} is weakly harmonic on every compact set away from $\mathcal{L}$ for all sufficiently large j. Passing the weak energy pairing gives harmonicity of W_∞ there. The bounded composed trace of Lemma 10.1 preserves its zero ray trace under weak convergence. The annular smooth approximation used in Proposition 15.9 identifies this trace with the zero positive-pole angular trace. Finally the Hardy and density assertions of Lemma 10.1 extend the equation from tests supported away from $\mathcal{L}$ to the zero-ray Friedrichs form domain. These arguments require only the present energy, trace, and support properties; no bound on a density of the rescaled measure is needed.

Choose any Q with $\sigma < Q < 1$. The second Liouville window of Lemma 10.2 applies: the limit is locally bounded, locally weighted H^1, even, slit-harmonic, and has zero ray trace, with growth at most $C(1 + |q|^Q)$. Thus $W_\infty = dH$. The spectrum has principal degree σ and next degree one; boundedness at the origin excludes the negative radial branches. No symmetry in Z has been imposed.

On any fixed annulus $r_0 < r < r_1$, the principal coefficient of W_{R_j} is

$$\frac{1}{N}\int_{S^2} |\theta_Y|^\eta W_{R_j}(r\theta)\phi(\theta)\,\mathrm{d}S = r^\sigma A(R_j r).$$

Monotonicity and convergence of A imply uniform convergence of this expression on the annulus to $A_\infty r^\sigma$. Cauchy–Schwarz on the sphere shows that strong annular weighted L^2 convergence passes these projections in radial $L^2(r^{D-1}\,\mathrm{d}r)$. The limiting coefficient of dH is dr^σ, so $d = A_\infty$. In particular it is nonzero. Every subsequential limit is the same, proving full strong local weighted L^2 convergence and full weak local energy convergence.

It remains to identify and improve the planar convergence. After localization, the bounded trace map $H^1(\rho) \to H^s(\mathbb{R}^2)$ identifies the weak planar trace limit with $A_\infty H_0$. On a compact planar set disjoint from L, the functions w_R are fractional harmonic for sufficiently large R. Their common bound $C|x|^\sigma$ gives common ordinary tails. Interior Poisson estimates therefore make them locally uniformly precompact on that set, and the weak trace identification forces each limit to be $A_\infty H_0$.

Near a compact positive-ray segment, homogeneity in (15.42) gives $0 \le w_R \le CH_0$ uniformly in R. The continuous right side vanishes on the ray, so the whole boundary layer is uniformly small. At the tip the bound $w_R(x) \le C|x|^\sigma$ supplies the same conclusion in a small ball. A finite covering of any planar compact set, using these layers and the preceding off-ray convergence, proves (17.9) locally uniformly everywhere. Its common growth bound is integrable against the ordinary-tail weight, so dominated convergence also proves L^1_s convergence. □

Completion of Theorem 17.1. Set

$$M = A_\infty/d_1 > 0, \tag{17.10}$$

with d_1 from (17.8), and put $P_R(x) = R^{-\lambda}P(Rx)$. These functions satisfy

$$|P_R(x)| \le C(R^{-\lambda} + |x|^\lambda), \qquad R \ge 1. \tag{17.11}$$

On compact sets disjoint from L, horn collapse removes their pressure for large R; interior fractional-harmonic estimates and the common ordinary tail make them locally uniformly precompact there.

We check compactness across the entire slit as well. Scaling (17.4) and using $\kappa + \beta = \lambda$ gives

$$|P_R(X, Z)| \le C\big(R^{-\lambda} + |Z|^\lambda + |Z|^\kappa X^\beta\big), \qquad X \ge 0. \tag{17.12}$$

For X in a bounded positive interval the right side becomes arbitrarily small by first choosing $|Z|$ small and then R large. On the ray itself $P_R(X, 0) = 0$. At the tip, (17.11) yields

$$\limsup_{R\to\infty} \sup_{|x|\le\delta} |P_R(x)| \le C\delta^\lambda.$$

Starting with any sequence $R_j \to \infty$, diagonal off-ray compactness, followed by these two layer bounds, therefore gives a subsequence converging locally uniformly on the whole plane to a continuous Q. The common majorant in (17.11) and $\lambda < 2s$ give convergence in L^1_s as well. In particular,

$$|Q(x)| \le C|x|^\lambda \qquad (x \in \mathbb{R}^2). \tag{17.13}$$

Differentiation in distributions and Proposition 17.4 imply

$$\partial_X Q = \lim_j \partial_X P_{R_j} = -A_\infty H_0 = \partial_X \Psi_M.$$

Hence $F = Q - \Psi_M$ is independent of X. For every compact smooth test supported off L, the pressure of P_{R_j} eventually vanishes on its support. Local uniform and ordinary-tail convergence let us pass the fractional pairing, so $\operatorname{supp} \mathcal{A}_s Q \subset L$. The explicit canonical pressure has the same support, and consequently

$$\operatorname{supp} \mathcal{A}_s F \subset L.$$

Since F is invariant under all horizontal translations, so is $\mathcal{A}_s F$. If its support were nonempty, translating any support point in both horizontal directions would put an entire horizontal line in that support. No such line is contained in L. Therefore $\mathcal{A}_s F = 0$ on the whole plane.

Lemma 7.2 now gives $F = \ell Z + c$. The homogeneous growth of Ψ_M and (17.13) imply $|F(x)| \le C|x|^\lambda$ also arbitrarily close to zero. First $x \to 0$ gives $c = 0$; then $x = (0, Z)$ and $Z \to 0$ give $\ell = 0$, since $\lambda > 1$. Thus $Q = \Psi_M$ with exactly the affine normalization in (16.1). Every subsequence has this same limit, which proves the full locally uniform and ordinary-tail convergence.

Undoing the fixed translation (9.4) does not alter the limit: its affine correction is $o(R^\lambda)$, and a fixed displacement of the original variables becomes a displacement of size $O(R^{-1})$ in a locally uniformly convergent rescaled family. Ordinary-tail convergence in the original coordinates follows again from their common critical-growth majorant. Undoing the possible reflection gives (17.1). The selected orientation is unique by Lemma 9.3; equivalently, the nonzero ray pressure of a canonical limit singles out that orientation. The coefficient A_∞ fixes M by (17.10), proving its uniqueness and its invariance under fixed obstacle translations. □

18. The pressure marginal and asymptotic curvature

Work in the normalized oriented noncylindrical class, with the canonical amplitude M just established. The ordinary-tail convergence in Theorem 17.1 permits fractional pairing against compact tests. Since the canonical pressure is $MX_+^\beta \delta_0(Z)$, it yields

$$R^{2s-\lambda}\mu(R\,\cdot) \rightharpoonup MX_+^\beta \delta_0(Z). \tag{18.1}$$

Here the left side denotes the density when one is used, or equivalently the measure $R^{2s-\lambda-2}\mu(R\,\cdot)$ on Borel sets. Positivity upgrades distributional convergence to vague Radon convergence.

Proposition 18.1 (Marginal regular variation)**.** *The marginal*

$$m(X) = \int_{\mathbb{R}} \mu(X, Z)\,\mathrm{d}Z$$

has a locally finite nondecreasing representative, and

$$\frac{m(R)}{R^{\beta}} \longrightarrow M. \tag{18.2}$$

For every $B < \infty$, also

$$\sup_{|\xi|\le B} \left| R^{-\beta} m(R + \sqrt{R}\,\xi) - M \right| \longrightarrow 0. \tag{18.3}$$

Proof. On each bounded X interval, (17.3) confines contact to a bounded Z interval. Lemma 15.2 therefore makes m locally integrable. Integrating (15.7) against transverse cutoffs shows that m is distributionally nondecreasing, and hence has a nondecreasing representative. For $0 < A < B < \infty$, the pressure support in the isotropically scaled slab $A \le X/R \le B$ satisfies $|Z|/R \le CR^{-1/2}$. A fixed transverse cutoff equal to one near zero thus captures all the pressure in this slab when R is large. The weak limit (18.1), with continuous approximation to the slab indicator, gives

$$R^{-\beta-1} \int_{RA}^{RB} m(X)\,\mathrm{d}X \longrightarrow M \int_A^B x^{\beta}\,\mathrm{d}x. \tag{18.4}$$

The two slab endpoints have zero limiting mass, justifying the indicator approximation. For $0 < \varepsilon < 1$, monotonicity bounds $m(R)$ below by its average on $[(1-\varepsilon)R, R]$ and above by its average on $[R, (1+\varepsilon)R]$. Divide by R^{β}, apply (18.4), and let $\varepsilon \downarrow 0$ to obtain (18.2). Finally squeeze $m(R+\sqrt{R}\,\xi)$ between the values at $R - B\sqrt{R}$ and $R + B\sqrt{R}$. The two normalized endpoint values tend to M, which proves (18.3). □

19. Mesoscopic selection and asymptotic curvature

For $R > 0$ define

$$p_R(\xi, z) = R^{-1} P(R + \sqrt{R}\,\xi, \sqrt{R}\,z), \qquad u_R(\xi, z) = p_R(\xi, z) + az^2. \tag{19.1}$$

Proposition 19.1 (Compactness, dimension reduction, and mass)**.** *Every sequence $R \to \infty$ has a subsequence for which p_R converges locally uniformly and in $L^1_s(\mathbb{R}^2)$ to a function $p(z)$ independent of ξ. The limit solves the one-dimensional obstacle problem with obstacle $-az^2$ and satisfies*

$$-az^2 \le p(z) \le 0, \qquad p(0) = 0, \qquad |p(z)| \le C|z|^{\kappa}. \tag{19.2}$$

Its pressure $\nu = \mathcal{A}_s^{(1)} p$ is compactly supported and has mass $\nu(\mathbb{R}) = M$.

Proof. The rescaled gap u_R is convex. On every fixed compact set, for all large R, monotonicity gives

$$-az^2 \le p_R(\xi, z) \le R^{-1} P(0, \sqrt{R}\,z).$$

The upper bound tends uniformly to zero on compact sets by (17.2) and $\lambda/2 < 1$. Convex functions bounded uniformly on an enlarged compact set are equi-Lipschitz on the smaller set: each directional subgradient is bounded by the oscillation divided by the distance to the larger boundary, using its two secants. Therefore u_R and p_R have subsequential locally uniform limits. On a fixed compact set $X = R + \sqrt{R}\,\xi$ is comparable to R, and (17.5) gives

$$|\partial_{\xi} p_R(\xi, z)| = R^{-1/2} w(R + \sqrt{R}\,\xi, \sqrt{R}\,z) \le CR^{-1/2}|z|^{\kappa}.$$

Integration along horizontal lines shows that every limit is independent of ξ.

We next verify the global tail convergence. In the region $X = R + \sqrt{R}\,\xi \geq 0$, (17.4) yields

$$|p_R(\xi, z)| \leq C\left[R^{-1} + R^{\lambda/2-1}|z|^\lambda + |z|^\kappa(1 + \xi/\sqrt{R})_+^{1-\sigma}\right] \leq C(1 + |(\xi, z)|^\lambda). \tag{19.3}$$

For the final inequality use $R \geq 1$ and $\kappa + 1 - \sigma = \lambda$. In the region $X < 0$, $\xi < -\sqrt{R}$ and $|X| \leq \sqrt{R}\,|\xi|$. The original growth bound gives the same majorant, since $R^{\lambda/2-1} \leq 1$. The majorant in (19.3) is integrable against the two-dimensional fractional weight because $\lambda < 2s$. Thus local convergence is also ordinary-tail convergence. The obstacle sandwich gives $-az^2 \leq p \leq 0$; positive-ray contact gives $p_R(0, 0) = 0$. At a fixed (ξ, z), the sharper first bound in (19.3) gives $|p(z)| \leq C|z|^\kappa$.

The operator passes to distributions by the tail convergence. The limit pressure is nonnegative. On any compact subset of $\{p(z) + az^2 > 0\}$, local uniform gap convergence eventually makes all approximating gaps positive, so their pressures vanish there. Lemma 12.1 now gives $\mathcal{A}_s p = \mathrm{d}\xi \otimes \nu$, with $\nu = \mathcal{A}_s^{(1)} p$, proving the one-dimensional obstacle problem. At any contact point,

$$az^2 \leq C|z|^\kappa.$$

Since $\kappa < 2$, contact is compact. It is a convex zero set of $p + az^2$ and contains zero, so it is a compact interval before any use of the explicit profile.

In density notation the rescaled pressure is

$$\mu_R(\xi, z) = R^{s-1}\mu(R + \sqrt{R}\,\xi, \sqrt{R}\,z). \tag{19.4}$$

For Borel sets its affine pushforward factor is R^{s-2}, accounting also for the two-dimensional Jacobian R. Its transverse marginal is

$$\int_{\mathbb{R}} \mu_R(\xi, z)\,\mathrm{d}z = R^{-\beta} m(R + \sqrt{R}\,\xi) \longrightarrow M,$$

uniformly for bounded ξ by (18.3). For ξ in a fixed compact interval, the width bound (17.3) confines $\operatorname{supp}\mu_R$ to a fixed bounded z interval. Choose a smooth compact transverse cutoff χ equal to one there and also on the compact contact set of p. For any $\varphi \in C_c^\infty(\mathbb{R})$, the weak pressure limit and the marginal convergence give respectively

$$\lim_{R\to\infty} \int \varphi(\xi)\chi(z)\,\mathrm{d}\mu_R = \left(\int \varphi\,\mathrm{d}\xi\right)\nu(\mathbb{R}), \qquad \lim_{R\to\infty} \int \varphi(\xi)\chi(z)\,\mathrm{d}\mu_R = M\int \varphi\,\mathrm{d}\xi.$$

Hence $\nu(\mathbb{R}) = M$. □

Theorem 19.2 (Centered mesoscopic selection and asymptotic parabola)**.** *For the noncylindrical solution in the normalized coordinates above, with p_L from* (12.23) *and L from* (12.25)*,*

$$R^{-1}P(R + \sqrt{R}\,\xi, \sqrt{R}\,z) \longrightarrow p_L(z) \tag{19.5}$$

locally uniformly and in $L^1_s(\mathbb{R}^2)$. If

$$\{Z : g(Z) \leq R\} = [z_-(R), z_+(R)],$$

then

$$\frac{z_-(R)}{\sqrt{R}} \longrightarrow -L, \qquad \frac{z_+(R)}{\sqrt{R}} \longrightarrow L. \tag{19.6}$$

Consequently the two ends have the same asymptotic curvature:

$$\lim_{Z\to+\infty} \frac{g(Z)}{Z^2} = \lim_{Z\to-\infty} \frac{g(Z)}{Z^2} = b, \qquad b = L^{-2} = \left(\frac{aD_s}{M}\right)^{1/\beta}. \tag{19.7}$$

Proof. By Proposition 19.1, every subsequential limit is a one-dimensional obstacle solution with compact pressure of mass M, growth $O(1+|z|^{\kappa})$, and value zero at zero. Proposition 12.5 constructs such a solution, and Proposition 12.7 makes it unique. Thus every subsequential limit is p_L, proving full locally uniform convergence. The common majorant (19.3) also gives full ordinary-tail convergence. In particular the centering in the limit follows from the mass, growth and normalization conditions; symmetry of the original solution has not been imposed.

Coercivity and convexity make each section a compact interval. The width estimate bounds $z_{\pm}(R)/\sqrt{R}$. Along any subsequence on which an endpoint converges, its gap value is zero. Local uniform convergence of $u_R(0,z)$ therefore puts the limiting endpoint in the zero set of $p_L(z)+az^2$, namely $[-L,L]$. This gives the exterior bounds on the two endpoints.

For the reverse positive-endpoint bound, fix any small interval $I \subset (-L,L)$ arbitrarily close to L. The pressure ν_L is strictly positive on I. Choose a nonnegative test $\varphi(\xi)\psi(z)$ with nonzero factors supported in $(-2,-1)$ and I, respectively. Its limiting pressure pairing is

$$\left(\int \varphi \,\mathrm{d}\xi\right)\left(\int \psi \,\mathrm{d}\nu_L\right) > 0.$$

Hence for every sufficiently large R there is an actual contact point with $X = R+\sqrt{R}\,\xi < R$ and $Z/\sqrt{R} \in I$. Recession propagates that point horizontally to $X = R$. Therefore $z_+(R)/\sqrt{R} \geq \inf I$ for all large R. Letting I approach L gives $\liminf z_+(R)/\sqrt{R} \geq L$. Tests near $-L$ similarly give $\limsup z_-(R)/\sqrt{R} \leq -L$. Together with the exterior bounds this proves (19.6).

A finite convex coercive g is eventually strictly increasing on the positive side and strictly decreasing on the negative side. To see this, choose a fixed secant of positive slope to the right of a minimum and use monotonicity of convex secants; choose a negative-slope secant on the left for the other side. Thus, for sufficiently large $|Z|$, at $R = g(Z)$ the point Z is the corresponding endpoint of the sublevel interval. Since $R \to \infty$, (19.6) gives $Z^2/g(Z) \to L^2$ at both ends. Formula (12.25) yields (19.7). □

20. Affine matching for a half-line potential

The passage from a homogeneous leading term to an affine-order remainder requires information on the contact trace. Regular variation of the projected density alone supplies no rate at this finer scale. We first prove an analytic statement which uses that trace explicitly.

Theorem 20.1 (Half-line affine matching)**.** *Let $0 < \alpha < \frac{1}{2}$, $p = 2\alpha - 1$, and $c > 0$. Suppose that a measurable, possibly signed density m on $[0,\infty)$ satisfies*

$$|m(u)| \leq C(1+u)^{1-\alpha} \quad \text{a.e.}, \qquad \frac{m(u)}{u^{1-\alpha}} \longrightarrow M. \tag{20.1}$$

Put $\chi(u) = \mathbf{1}_{[1,\infty)}(u)$, and, for real A_0, B_0, define

$$\begin{aligned} V(x,z) &= A_0 + B_0 x \\ &\quad + c\int_0^{\infty} \left\{ \left((x-u)^2+z^2\right)^{p/2} - \chi(u)\left(u^p - pxu^{p-1}\right) \right\} m(u)\,\mathrm{d}u. \end{aligned} \tag{20.2}$$

The braces are integrated together. Write $q(x) = V(x,0)$, and assume

$$q(x) = \ell x + o(x) \qquad (x \longrightarrow +\infty). \tag{20.3}$$

Define the homogeneous potential

$$\Phi_M(x,z) = cM\int_0^{\infty} \left\{ \left((x-u)^2+z^2\right)^{p/2} - u^p + pxu^{p-1} \right\} u^{1-\alpha}\,\mathrm{d}u. \tag{20.4}$$

Then Φ_M is homogeneous of degree $1+\alpha$, its positive-axis trace vanishes, and

$$q(-r) = MC_\alpha r^{1+\alpha} - \ell r + o(r), \tag{20.5}$$

$$C_\alpha = \frac{c\,\Gamma(2-\alpha)\Gamma(1-\alpha)}{\alpha(1+\alpha)\Gamma(1-2\alpha)} > 0. \tag{20.6}$$

Moreover,

$$\lim_{R\to\infty} \frac{1}{R} \sup_{x^2+z^2\le R^2} |V(x,z) - \Phi_M(x,z) - \ell x| = 0. \tag{20.7}$$

In particular the remainder has a global bound $C(1+|(x,z)|)$.

Proof. Truncation and convergence. Replace m by $m_N = m\mathbf{1}_{[0,N]}$, keeping the same cutoff and affine terms, and denote the resulting objects by V_N, q_N. For $R = \max(2, \sqrt{x^2+z^2})$, the part $u \le 2R$ of the constant subtraction is $O(R^{1+\alpha})$, and its linear subtraction is bounded by

$$CR \int_1^{2R} u^{\alpha-1}\,\mathrm{d}u \le CR^{1+\alpha}.$$

Since $p < 0$, the unsubtracted planar kernel is bounded by $|x-u|^p$. Its integral is at most $CR^{1-\alpha}\int_0^{2R}|x-u|^p\,\mathrm{d}u \le CR^{1+\alpha}$. For $u > 2R$, the Taylor remainder in the braces is at most CR^2u^{p-2}, and hence its integral is bounded by

$$CR^2 \int_{2R}^{\infty} u^{\alpha-2}\,\mathrm{d}u \le CR^{1+\alpha}.$$

All these bounds hold uniformly in N, also for signed m. Thus

$$|V_N(x,z)| \le C(1+|(x,z)|)^{1+\alpha}. \tag{20.8}$$

The combined tails converge locally uniformly. On compact sets disjoint from the source ray, derivatives of every fixed order also converge locally uniformly. The trace is continuous: on compact intervals the singular kernel belongs to L^1, its translations are continuous in L^1, and the density is bounded; the remaining tails converge uniformly. In particular $r^{-\alpha}q_N(-r) \to r^{-\alpha}q(-r)$ in C^2 on compact subintervals of $(0,\infty)$.

A twice-differentiated half-line identity. Set $k_\alpha = \sin(\pi\alpha)/\pi$. The ordinary half-line Poisson kernel for $(-\Delta)^\alpha$, on the negative half-line, is

$$k_\alpha \frac{r^\alpha}{x^\alpha(r+x)}, \qquad r, x > 0.$$

For clarity, its precise application here is the elementary identity

$$(r+u)^p = k_\alpha r^\alpha \int_0^\infty \frac{|x-u|^p x^{-\alpha}}{r+x}\,\mathrm{d}x, \qquad r, u > 0. \tag{20.9}$$

Here is a direct verification, so the identity requires no uniqueness principle for growing half-line data. Substitute $x = rv/(1-v)$ and put $\theta = u/(r+u)$. It remains to evaluate

$$J(\theta) = \int_0^1 |v-\theta|^{2\alpha-1}[v(1-v)]^{-\alpha}\,\mathrm{d}v.$$

For the part $0 < v < \theta$, the substitution $v = \theta w$ gives

$$\theta^\alpha \int_0^1 w^{-\alpha}(1-w)^{2\alpha-1}(1-\theta w)^{-\alpha}\,\mathrm{d}w = \alpha B(1-\alpha, 2\alpha) B_\theta(\alpha,\alpha),$$

where $B_\theta(\alpha,\alpha) = \int_0^\theta q^{\alpha-1}(1-q)^{\alpha-1}\,\mathrm{d}q$. To check the equality, expand $(1-\theta w)^{-\alpha}$ in its positive binomial series and integrate termwise. The coefficient on the left is $B(1-\alpha,2\alpha)(\alpha)_n(1-\alpha)_n/((1+\alpha)_n n!)$, and $(\alpha)_n/(1+\alpha)_n = \alpha/(\alpha+n)$ is precisely the coefficient obtained by integrating the

binomial series for $(1-q)^{\alpha-1}$ on the right; $(z)_n$ denotes the rising factorial. Reflection gives the second part with θ replaced by $1-\theta$. Since the beta integrand is symmetric,

$$J(\theta) = \alpha B(1-\alpha, 2\alpha)B(\alpha,\alpha) = \Gamma(\alpha)\Gamma(1-\alpha) = \frac{\pi}{\sin(\pi\alpha)}.$$

This proves (20.9), with all integrals convergent because $0 < \alpha < \frac{1}{2}$.

Apply (20.9) to the uncompensated potential of a compact density,

$$v_N(x) = c\int_0^N |x-u|^p m(u)\,\mathrm{d}u.$$

Fubini's theorem is applicable to its absolute value, and gives

$$v_N(-r) = k_\alpha r^\alpha \int_0^\infty \frac{v_N(x)x^{-\alpha}}{r+x}\,\mathrm{d}x.$$

Divide by r^α before differentiating twice. The result is

$$[r^{-\alpha}v_N(-r)]'' = 2k_\alpha \int_0^\infty \frac{v_N(x)x^{-\alpha}}{(r+x)^3}\,\mathrm{d}x.$$

The same differentiated identity holds for the affine functions 1 and x separately, because the beta integral gives

$$2k_\alpha \int_0^\infty \frac{x^{-\alpha}}{(r+x)^3}\,\mathrm{d}x = \alpha(1+\alpha)r^{-\alpha-2},$$
$$2k_\alpha \int_0^\infty \frac{x^{1-\alpha}}{(r+x)^3}\,\mathrm{d}x = \alpha(1-\alpha)r^{-\alpha-1}.$$

These are the second derivatives of $r^{-\alpha}$ and $-r^{1-\alpha}$, respectively. Notice that the undifferentiated Poisson integral of x need not converge; it has not been used. Subtracting the finite affine terms in q_N therefore preserves the differentiated identity. In its right side, the bound (20.8) gives an integrable majorant $O(x^{-2})$ at infinity and $O(x^{-\alpha})$ at zero. Passing to the limit yields

$$H''(r) = 2k_\alpha \int_0^\infty \frac{q(x)x^{-\alpha}}{(r+x)^3}\,\mathrm{d}x, \qquad H(r) = r^{-\alpha}q(-r). \tag{20.10}$$

Recovering the negative trace. Write $q(x) = \ell x + e(x)$ for $x > 0$. On a fixed compact interval the contribution of e to (20.10) is $O(r^{-3})$. On its complement choose the interval sufficiently large that $|e(x)| \le \varepsilon x$, and use the second beta integral above. It follows that

$$H''(r) = \ell\alpha(1-\alpha)r^{-\alpha-1} + o(r^{-\alpha-1}).$$

Thus H'' is integrable at infinity and H' has a finite limit B. Integrating first from infinity and then once more gives

$$H'(r) = B - \ell(1-\alpha)r^{-\alpha} + o(r^{-\alpha}), \qquad H(r) = Br - \ell r^{1-\alpha} + o(r^{1-\alpha}).$$

The finite constant of the second integration is absorbed because $1-\alpha > 0$. Therefore

$$q(-r) = Br^{1+\alpha} - \ell r + o(r). \tag{20.11}$$

This integration uses no rate in (20.1). For example, if the positive trace is identically zero, (20.10) shows exactly that the negative trace has the form $Ar^\alpha + Br^{1+\alpha}$.

Identification of B without a subtraction at the origin. Split the original definition exactly at one:

$$\begin{aligned} q(-r) = A_0 - B_0 r &+ c\int_0^1 (r+u)^p m(u)\,\mathrm{d}u \\ &+ c\int_1^\infty \left[(r+u)^p - u^p - pru^{p-1}\right] m(u)\,\mathrm{d}u. \end{aligned} \tag{20.12}$$

For $r \geq 1$, the compact integral is $O(r^p)$. In the last integral put $u = rv$, and define

$$B_\alpha(v) = (1+v)^p - v^p - pv^{p-1}, \qquad f_r(v) = \mathbf{1}_{\{v \geq 1/r\}} r^{\alpha-1} m(rv).$$

After division by $r^{1+\alpha}$ that integral is $c\int_0^\infty B_\alpha(v) f_r(v)\,\mathrm{d}v$. For every $v > 0$, $f_r(v) \to Mv^{1-\alpha}$, and on its support

$$|f_r(v)| \leq C 2^{1-\alpha} v^{1-\alpha}.$$

The product $|B_\alpha(v)| v^{1-\alpha}$ is $O(v^{\alpha-1})$ near zero and $O(v^{\alpha-2})$ at infinity. Dominated convergence applies. Taylor's integral remainder also supplies its sign and exact integral:

$$\begin{aligned} B_\alpha(v) &= p(p-1)\int_0^1 (1-\rho)(v+\rho)^{p-2}\,\mathrm{d}\rho > 0, \\ \int_0^\infty B_\alpha(v) v^{1-\alpha}\,\mathrm{d}v &= p(p-1)B(2-\alpha, 1-\alpha)\int_0^1 (1-\rho)\rho^{\alpha-1}\,\mathrm{d}\rho \\ &= \frac{\Gamma(2-\alpha)\Gamma(1-\alpha)}{\alpha(1+\alpha)\Gamma(1-2\alpha)}. \end{aligned}$$

Tonelli's theorem is legitimate because the remainder is nonnegative. Comparison with (20.11) gives $B = MC_\alpha$.

For the power density $Mu^{1-\alpha}$, the origin subtractions in (20.4) converge. If $Q_{Mu^{1-\alpha}}$ denotes (20.2) with this density and zero affine part, then

$$\Phi_M = Q_{Mu^{1-\alpha}} - \frac{cM}{1+\alpha} + \frac{cMp}{\alpha} x. \tag{20.13}$$

A change of variables proves homogeneity. Its negative trace is exactly $MC_\alpha r^{1+\alpha}$. On the positive axis homogeneity gives $\Phi_M(x,0) = Dx^{1+\alpha}$. Apply (20.10) to Φ_M; the left side vanishes, whereas the right side is a strictly positive finite factor times D. Hence $D = 0$. In contrast, origin subtraction is not legitimate for an arbitrary bounded density: if $m = \mathbf{1}_{[0,1]}$, then $\int_0^1 u^{p-1} m(u)\,\mathrm{d}u = \infty$. Equation (20.12) avoids this difficulty entirely.

Reconstruction in the physical plane. Set

$$F(x,z) = V(x,z) - \Phi_M(x,z) - \ell x, \qquad e(x) = F(x,0).$$

The preceding argument shows $e(x) = o(|x|)$ at both ends of the line and $|e(x)| \leq C(1+|x|)$. By (20.13), F is in the same cutoff potential class, with density $m(u) - Mu^{1-\alpha}$ and affine coefficients

$$A_0 + \frac{cM}{1+\alpha}, \qquad B_0 - \ell - \frac{cMp}{\alpha}. \tag{20.14}$$

For $z > 0$ put

$$J_{x,z}(y) = \left((x-y)^2 + z^2\right)^{-1/2-\alpha}, \qquad d_\alpha = \frac{\Gamma(1/2+\alpha)}{\sqrt{\pi}\,\Gamma(\alpha)}.$$

For completeness, the fundamental-solution convolution needed here follows by a direct calculation valid for every $0 < \alpha < 1$ and $n > 2\alpha$. Put $A = n/2 + \alpha$, $B = n/2 - \alpha > 0$, and

$$\Pi_{n,\alpha}(x,z) = \frac{\Gamma(A)}{\pi^{n/2}\Gamma(\alpha)} \frac{z^{2\alpha}}{(|x|^2+z^2)^A}, \qquad z > 0.$$

We claim

$$\int_{\mathbb{R}^n} \Pi_{n,\alpha}(x-u,z)|u|^{-2B}\,du = (|x|^2+z^2)^{-B}.$$

Indeed, Schwinger's gamma-integral formula and the Gaussian integral give

$$\begin{aligned} I &:= \int_{\mathbb{R}^n} (|x-u|^2+z^2)^{-A}|u|^{-2B}\,du \\ &= \frac{\pi^{n/2}}{\Gamma(A)\Gamma(B)} \int_0^\infty\!\!\int_0^\infty t^{A-1}r^{B-1}(t+r)^{-n/2} \exp\!\left(-tz^2 - \frac{tr}{t+r}|x|^2\right) dr\,dt \\ &= \frac{\pi^{n/2}\Gamma(n/2)}{\Gamma(A)\Gamma(B)} \int_0^1 v^{B-1}(1-v)^{\alpha-1}(z^2+v|x|^2)^{-n/2}\,dv \\ &= \frac{\pi^{n/2}\Gamma(\alpha)}{\Gamma(A)} z^{-2\alpha}(|x|^2+z^2)^{-B}. \end{aligned}$$

In the second equality use $h = t+r$, $v = r/(t+r)$ and integrate h. For the last equality use Euler's beta integral with $B+\alpha = n/2$; equivalently substitute $w = v(1+q)/(1+qv)$, $q = |x|^2/z^2$. All parameters in the beta integrals are positive, and all integrands before evaluation are nonnegative, so Tonelli's theorem justifies the interchanges. Multiplication by the displayed Poisson prefactor proves the claim. Taking $n = 1$, $0 < \alpha < 1/2$ gives exactly the fundamental solution extension used in the half-line reconstruction. The case $n = 2$, $0 < \alpha < 1$ is covered as well. This calculation uses no uniqueness theorem for growing data. Compact signed source densities are then handled by absolute-value Fubini before the compensated and differentiated limits already described in the proof.

Consequently, for each uncompensated compact-density potential,

$$v_N(x,z) = d_\alpha z^{2\alpha} \int_{\mathbb{R}} v_N(y,0) J_{x,z}(y)\,\mathrm{d}y.$$

Divide by $z^{2\alpha}$ before differentiating. For $j = x, z$ this gives

$$\partial_j[z^{-2\alpha} v_N(x,z)] = d_\alpha \int_{\mathbb{R}} v_N(y,0)\partial_j J_{x,z}(y)\,\mathrm{d}y.$$

Both differentiated identities hold for affine functions as well. Indeed, direct integration gives

$$\begin{aligned} \int_{\mathbb{R}} J_{x,z}\,\mathrm{d}y &= z^{-2\alpha}/d_\alpha, & \int_{\mathbb{R}} \partial_x J_{x,z}\,\mathrm{d}y &= 0, \\ \int_{\mathbb{R}} y\partial_x J_{x,z}\,\mathrm{d}y &= z^{-2\alpha}/d_\alpha, & \int_{\mathbb{R}} \partial_z J_{x,z}\,\mathrm{d}y &= -2\alpha z^{-2\alpha-1}/d_\alpha, \\ \int_{\mathbb{R}} y\partial_z J_{x,z}\,\mathrm{d}y &= -2\alpha x z^{-2\alpha-1}/d_\alpha. \end{aligned}$$

For the x identity integrate by parts using $\partial_x J = -\partial_y J$; the endpoint term is $yJ = O(|y|^{-2\alpha})$. For the z identities use the mass formula and oddness after translating by x. Every differentiated integral is absolutely convergent. The bound (20.8) is integrable against both differentiated kernels, with decay $O(|y|^{-1-\alpha})$ and $O(|y|^{-2-\alpha})$, respectively. The truncation limit therefore gives

$$\nabla_{x,z}[z^{-2\alpha}F(x,z)] = d_\alpha \int_{\mathbb{R}} e(y)\nabla_{x,z} J_{x,z}(y)\,\mathrm{d}y. \tag{20.15}$$

Define the absolutely convergent compensated extension

$$E_e(x,z) = d_\alpha z^{2\alpha} \int_{\mathbb{R}} e(y)\left[J_{x,z}(y) - \mathbf{1}_{\{|y|\ge 1\}}|y|^{-1-2\alpha}\right]\mathrm{d}y. \tag{20.16}$$

Its gradient after division by $z^{2\alpha}$ equals the right side of (20.15). Connectedness of $\{z > 0\}$ yields

$$F(x,z) = E_e(x,z) + Az^{2\alpha}, \tag{20.17}$$

where the finite constant is, for example,

$$A = F(0,1) - d_\alpha \int_{\mathbb{R}} e(y) \left[(1+y^2)^{-1/2-\alpha} - \mathbf{1}_{\{|y|\geq 1\}} |y|^{-1-2\alpha} \right] \mathrm{d}y.$$

Evenness in z gives the same formula in the lower half-plane with $|z|$. No uniqueness assertion for an uncompensated, growing Poisson integral is needed.

Uniformity up to the source ray. Let $R = \sqrt{x^2+z^2} \geq 2$ and $0 < z \leq R$. Given $\varepsilon > 0$, choose $T \geq 1$ with $|e(y)| \leq \varepsilon |y|$ for $|y| > T$. For $|y| > 2R$, Taylor expansion gives

$$|J_{x,z}(y) - |y|^{-1-2\alpha}| \leq CR|y|^{-2-2\alpha}.$$

The contribution of that region to (20.16) is at most $C\varepsilon z^{2\alpha} R^{1-2\alpha} \leq C\varepsilon R$, provided $2R > T$. On $T < |y| \leq 2R$, use $|y| \leq |x| + |y-x|$, the mass-one property, and

$$\int_{|y|\leq 2R} d_\alpha z^{2\alpha} J_{x,z}(y) |y-x| \, \mathrm{d}y \leq C z^{2\alpha} R^{1-2\alpha}.$$

This bounds the raw Poisson contribution by $C\varepsilon R$. The subtraction on the same region is at most

$$C\varepsilon z^{2\alpha} \int_1^{2R} v^{-2\alpha} \, \mathrm{d}v \leq C\varepsilon R.$$

On $|y| \leq T$ the raw Poisson part is bounded by the compact supremum of $|e|$, while the subtraction is $O_T(z^{2\alpha})$. Combining these estimates with (20.17) gives

$$\sup_{x^2+z^2=R^2,\ z>0} |F(x,z)| \leq C\varepsilon R + C_T(1+R^{2\alpha}) + |A| R^{2\alpha}.$$

Because $2\alpha < 1$, division by R, followed by $R \to \infty$ and $\varepsilon \to 0$, proves uniform sublinearity on the upper semicircle. The trace estimate and reflection cover the entire circle. Finally, uniform sublinearity on circles and local boundedness imply the same assertion on balls: for every $\varepsilon > 0$, $|F(x,z)| \leq \varepsilon |(x,z)| + C_\varepsilon$. □

21. The pressure potential and its transverse spreading

We return to physical coordinates $x = (X, Z) \in \mathbb{R}^2$, and fix

$$\frac{1}{2} < s < 1, \qquad t = 1-s, \quad \sigma = s - \frac{1}{2}, \quad \beta = \frac{3}{2} - s, \quad \lambda = s + \frac{1}{2}, \quad \kappa = 2s - 1 = 2\sigma.$$

The obstacle is $\psi(X,Z) = -aZ^2$, $a > 0$, and the Fourier convention is $\widehat{f}(\xi) = \int_{\mathbb{R}^2} e^{-ix\cdot\xi} f(x) \, \mathrm{d}x$, with inverse factor $(2\pi)^{-2}$. Thus $\mathcal{A}_s = (-\Delta)^s$ has multiplier $|\xi|^{2s}$ and inverse kernel

$$K_s(x) = k_s |x|^{-2t}, \qquad k_s = \frac{\Gamma(1-s)}{4^s \pi \Gamma(s)}. \tag{21.1}$$

The ordinary fractional tail norm used below is

$$\|f\|_{L^1_s(\mathbb{R}^2)} = \int_{\mathbb{R}^2} \frac{|f(x)|}{(1+|x|)^{2+2s}} \, \mathrm{d}x. \tag{21.2}$$

Let P be a continuous ordinary-tail obstacle solution, with $P \geq \psi$, $\mu = \mathcal{A}_s P \geq 0$ supported on $K = \{P = \psi\}$, and with $P + aZ^2$ convex. Assume the prescribed nonzero thin-front blow-down $r^{-\lambda} P(r\cdot) \to \Psi_M$ locally uniformly, with $M > 0$. The geometric and mesoscopic conclusions proved

above, after the genuine obstacle translation placing a vertex at the origin, are

$$K = \{X \geq g(Z)\}, \quad g \text{ finite and convex}, \quad g \geq 0, \quad g(0) = 0, \tag{21.3}$$

$$K \subset \{X \geq 0,\ |Z| \leq C\sqrt{1+X}\}, \tag{21.4}$$

$$m(X) = \int_{\mathbb{R}} \mu(X, Z)\,\mathrm{d}Z, \qquad m(X)/X^{\beta} \longrightarrow M, \tag{21.5}$$

$$p_R(\xi, z) = R^{-1}P(R + \sqrt{R}\,\xi, \sqrt{R}\,z) \longrightarrow p_L(z), \tag{21.6}$$

$$R^{s-1}\mu(R + \sqrt{R}\,\xi, \sqrt{R}\,z)\,\mathrm{d}\xi\,\mathrm{d}z \rightharpoonup \mathrm{d}\xi\,\nu_L(z)\,\mathrm{d}z. \tag{21.7}$$

Here μ is a locally bounded two-dimensional density, $P \in C^{1,\kappa}_{\mathrm{loc}}$, convergence in (21.6) is local uniform and in L^1_s, and

$$\begin{aligned} \nu_L(z) &= \frac{a}{A^*_s}(L^2 - z^2)^t_+, \qquad A^*_s = \frac{2^{1-2s}\Gamma(3/2 - s)\Gamma(2 - s)}{\sqrt{\pi}}, \\ D_s &= \frac{B(1/2, 2 - s)}{A^*_s}, \qquad b = \left(\frac{aD_s}{M}\right)^{1/\beta}, \qquad L^2 = b^{-1}. \end{aligned} \tag{21.8}$$

We shall use the common tail majorant supplied by mesoscopic compactness,

$$|p_R(v)| \leq C(1 + |v|^{\lambda}), \qquad R \geq 1, \qquad \lambda < 2s. \tag{21.9}$$

In particular $P(X, 0) = 0$ for every $X \geq 0$.

Lemma 21.1 (Bounds for the density and its moments)**.** *The pressure and its longitudinal marginal obey*

$$0 \leq \mu(X, Z) \leq C(1 + X)^t, \qquad 0 \leq m(X) \leq C(1 + X)^{\beta} \quad \text{a.e. on } X \geq 0. \tag{21.10}$$

If

$$n_1(X) = \int_{\mathbb{R}} |Z|\mu(X, Z)\,\mathrm{d}Z, \qquad n_2(X) = \int_{\mathbb{R}} Z^2\mu(X, Z)\,\mathrm{d}Z,$$

then

$$n_1(X) \leq C(1 + X)^{\beta+1/2}, \qquad n_2(X) \leq C(1 + X)^{\beta+1}. \tag{21.11}$$

Proof. On $0 \leq X \leq B$, (21.4) confines the pressure to a fixed compact rectangle. Its local L^{∞} norm therefore bounds m on that interval. Combining this fact with (21.5) proves the second estimate in (21.10); no choice of values on a null set affects any potential below.

The sharper two-dimensional estimate is already (17.6), obtained from the common normal-scale pressure lemma at the proved critical exponent. Finally, (21.11) follows from (21.4) and the bound for m. □

Keep the single cutoff $\chi(u) = \mathbf{1}_{[1,\infty)}(u)$, and define

$$Q_m(x) = \int_0^{\infty} \left[K_s(x - (u, 0)) - \chi(u)\{K_s(-u, 0) + \nabla K_s(-u, 0) \cdot x\}\right] m(u)\,\mathrm{d}u, \tag{21.12}$$

$$\begin{aligned} S_{\mu}(x) = \int_0^{\infty} \int_{\mathbb{R}} \big[&K_s(x - (u, z)) - K_s(x - (u, 0)) \\ &-\chi(u)\{K_s(-u, -z) - K_s(-u, 0)\}\big]\,\mu(u, z)\,\mathrm{d}z\,\mathrm{d}u. \end{aligned} \tag{21.13}$$

The potential S_{μ} measures the effect of spreading each longitudinal section away from the axis. Only a constant is subtracted from it; subtraction of an additional, possibly divergent derivative at the vertex is unnecessary.

Proposition 21.2 (Growth and sources of the cutoff potentials)**.** *The integrals* (21.12)–(21.13) *converge with their indicated grouping and define continuous functions. They satisfy*

$$|Q_m(x)| \le C(1+|x|^{\lambda}), \qquad |S_\mu(x)| \le C(1+|x|), \tag{21.14}$$

$$\mathcal{A}_s Q_m = m(X)\mathbf{1}_{\{X\ge 0\}}\delta_0(Z), \qquad \mathcal{A}_s S_\mu = \mu - m(X)\mathbf{1}_{\{X\ge 0\}}\delta_0(Z). \tag{21.15}$$

On every closed cone whose directions avoid the positive X-axis,

$$|S_\mu(x)| \le C_{\mathrm{cone}}(1+|x|^{s}+|x|^{\sigma}). \tag{21.16}$$

Consequently there are real constants A_X, A_Z, A_0 such that

$$P(x) = Q_m(x) + S_\mu(x) + A_X X + A_Z Z + A_0. \tag{21.17}$$

The growth estimates hold for the longitudinally truncated potentials with constants independent of the truncation N. Their sources are the corresponding truncated densities, and the potentials converge locally uniformly and in $L^1_s(\mathbb{R}^2)$ as $N\to\infty$.

Proof. For Q_m, the proof of (20.8) applies with $\alpha=\sigma$, $p=-2t$, and $c=k_s$. In particular its far Taylor remainder is bounded by $C|x|^2u^{\beta-2t-2}$.

For S_μ, on a fixed compact set of x and for large u, expand in x after retaining the constant subtraction. The first normal moment is bounded by

$$C_x n_1(u) u^{-2t-2} = O_x(u^{-t-1}),$$

and the second-moment term by

$$C_x n_2(u) u^{-2t-3} = O_x(u^{-t-3/2}).$$

Both tails are integrable. Local singularities of the two-dimensional kernel are integrable against bounded μ; those of the collapsed kernel are integrable against bounded m, since $2t<1$. The same translation-continuity argument used for the half-line trace, now also for the locally integrable two-dimensional kernel, proves continuity.

We give the growth estimates in some detail because the narrow region near the positive ray has a different size. Suppose first that $x/|x|$ lies in a closed cone separated from that ray, and write $r=|x|$. Then $|x-(u,0)|\ge c(r+u)$. For sufficiently large u, the transverse width $C\sqrt{u}$ is a small perturbation of this distance, so Taylor expansion in z is uniform. On $1\le u\le Ar$, where A is a fixed sufficiently large constant, its first-moment contribution is at most

$$C\int_1^{Ar} n_1(u)(r+u)^{-2t-1}\,\mathrm{d}u \le Cr^s.$$

The second-moment term is at most $C\int_1^{Ar} n_2(u)(r+u)^{-2t-2}\,\mathrm{d}u \le Cr^\sigma$. The constant subtraction in this region is bounded by

$$C\int_1^{Ar} n_2(u)u^{-2t-2}\,\mathrm{d}u \le Cr^\sigma,$$

because the first normal derivative of K_s vanishes on the axis. For $u>Ar$, use the difference between the Taylor coefficients at x and at zero before estimating. The first and second normal terms are bounded respectively by

$$Cr\int_{Ar}^{\infty} n_1(u)u^{-2t-2}\,\mathrm{d}u \le Cr^s, \qquad Cr\int_{Ar}^{\infty} n_2(u)u^{-2t-3}\,\mathrm{d}u \le Cr^\sigma.$$

Higher remainders are absorbed in these bounds using $|z|/u\to 0$; the bounded-u part has a uniform bound. This proves (21.16), without a symmetry assumption on μ.

To obtain the global bound, the only additional region is $X > 0$, X comparable to r, x near the positive ray, and u comparable to r. On this region (21.10) gives $\mu(u, z) \le Cr^t$ and $|z| \le C\sqrt{r}$. The exact one-dimensional translation identity is

$$\int_{\mathbb{R}} \left|(v^2 + A^2)^{-t} - (v^2 + B^2)^{-t}\right| \, \mathrm{d}v = c_t \big||A|^\kappa - |B|^\kappa\big| \le c_t |A - B|^\kappa, \tag{21.18}$$

where

$$c_t = \frac{2t}{\kappa} B\left(\frac{1}{2}, t + \frac{1}{2}\right) > 0. \tag{21.19}$$

For its proof, the integrand before taking the absolute value has one sign independent of v. Differentiate its convergent difference with respect to a positive parameter A, integrate $-2tA(v^2 + A^2)^{-t-1}$, and then integrate the resulting $-2tB(1/2, t+1/2)A^{-2t}$ from 0 to A. Subtraction for two values and continuity at zero give the equality; κ-Hölder continuity of $|A|^\kappa$ gives the inequality.

At fixed z, apply (21.18) with $A = Z - z$, $B = Z$, and $v = X - u$. Extending the nonnegative majorant from the comparable-u interval to the whole line gives

$$Cr^t \int_{|z| \le C\sqrt{r}} |z|^\kappa \, \mathrm{d}z \le Cr^{t+(\kappa+1)/2} = Cr.$$

Its constant subtraction is $O(r^\sigma)$. For u well below r or well above r the Taylor estimates already used apply. This proves the second estimate in (21.14). Each estimate integrates a nonnegative majorant over a subset when the density is truncated, so its constant is independent of N.

At finite N, the fundamental-kernel identity gives exactly the sources $m\mathbf{1}_{[0,N]}\delta_0$ and $\mu\mathbf{1}_{[0,N]} - m\mathbf{1}_{[0,N]}\delta_0$. The affine and constant subtraction terms are annihilated by $\mathcal{A}_s$; their ordinary tails converge because $2s > 1$. The locally uniform convergence of the combined tails and the uniform growth bounds imply convergence in (21.2), since $\lambda < 2s$ and $1 < 2s$. For $\varphi \in C_c^\infty(\mathbb{R}^2)$, $|\mathcal{A}_s\varphi(x)| \le C_\varphi(1 + |x|)^{-2-2s}$, so passage in $\int Q_{m,N}\mathcal{A}_s\varphi$ and $\int S_{\mu,N}\mathcal{A}_s\varphi$ proves (21.15).

The difference $P - Q_m - S_\mu$ is therefore entire s-harmonic and has growth $O(1 + |x|^\lambda)$. The distributional Fourier equation, tested away from frequency zero where $|\xi|^{2s}$ is smooth and invertible, shows that its Fourier support is contained in $\{0\}$. It is a polynomial; growth with $\lambda < 2$ restricts its degree to at most one. This proves (21.17) without requiring a product of a singular Fourier distribution with $|\xi|^{2s}$ at zero. □

Proposition 21.3 (The linear defect on the contact ray)**.** *With the normalization above, let*

$$A_* = \frac{a}{b(2s-1)} = \frac{aL^2}{\kappa}. \tag{21.20}$$

Then

$$S_\mu(R, 0) = -A_* R + o(R), \qquad Q_m(R, 0) = (A_* - A_X)R + o(R). \tag{21.21}$$

Proof. The longitudinally integrated kernel is

$$k_s \int_{\mathbb{R}} \left[(\xi^2 + z^2)^{-t} - |\xi|^{-2t}\right] \, \mathrm{d}\xi = -d_s |z|^\kappa, \qquad d_s = -\frac{\Gamma(1/2 - s)}{4^s \sqrt{\pi}\, \Gamma(s)} > 0. \tag{21.22}$$

This follows from the parameter differentiation used in (21.18); indeed $d_s = k_s c_t$.

Fix $H > 0$ and restrict first to $|u - R| \le H\sqrt{R}$. Under $u = R + \sqrt{R}\,\xi$, $z = \sqrt{R}\,\zeta$, the pressure, area, and kernel factors have sizes R^t, R, and R^{-t}, respectively. Thus the raw kernel difference, divided by R, converges by (21.7) to

$$k_s \int_{-H}^{H} \int_{\mathbb{R}} \left[(\xi^2 + \zeta^2)^{-t} - |\xi|^{-2t}\right] \nu_L(\zeta) \, \mathrm{d}\zeta \, \mathrm{d}\xi.$$

Here is a justification despite the singular kernel. The scaled pressures are bounded uniformly by (21.10), and have uniformly bounded normal support. The two-dimensional singularity is uniformly integrable because $2t < 2$. The collapsed term is uniformly integrable in ξ because its normal

marginal is bounded and $2t < 1$. One may consequently truncate both singularities, use weak measure convergence for the bounded kernels, and then remove the truncations. No pointwise convergence of the pressures is required.

The remaining local tail $H\sqrt{R} < |u - R| \le R/2$ has no first normal term on $Z = 0$. Its second-moment bound, after division by R, is

$$\frac{C}{R}\int_{H\sqrt{R}<|u-R|\le R/2} n_2(u)|u-R|^{-2t-2}\,\mathrm{d}u \le CH^{-2t-1}.$$

For $u \notin [R/2, 3R/2]$, the raw difference and the cutoff subtraction together are $O(1 + R^\sigma)$. To see this, on $u < R/2$ estimate the raw difference by $Cn_2(u)(R-u)^{-2t-2}$ and integrate the cutoff subtraction by $Cn_2(u)u^{-2t-2}$ for $u \ge 1$. On $u > 3R/2$ retain their difference and use $CRn_2(u)u^{-2t-3}$. Each integral is $O(R^\sigma)$, up to a bounded compact contribution. The cutoff subtraction in the central region is also $O(R^\sigma)$. Its individual unbounded pieces have therefore never been separated at infinity.

Let $R \to \infty$ and then $H \to \infty$. Formula (21.22) gives

$$\lim_{R\to\infty}\frac{S_\mu(R,0)}{R} = -d_s\int_{\mathbb{R}} |z|^\kappa \nu_L(z)\,\mathrm{d}z.$$

The last integral and the constants can be evaluated explicitly:

$$\int_{\mathbb{R}} |z|^\kappa \nu_L(z)\,\mathrm{d}z = \frac{a}{A_s^*}B(s,2-s)L^2, \qquad d_s\kappa B(s,2-s) = A_s^*. \tag{21.23}$$

For the first identity substitute $z^2 = L^2 v$ on $z > 0$; the exponent of L is $\kappa + 2t + 1 = 2$. For the second, use $-\Gamma(-\sigma) = \Gamma(1-\sigma)/\sigma$ and the gamma definitions. This proves the first limit in (21.21). Since $P(R,0) = 0$, the decomposition (21.17) gives the second limit, with $A_0/R \to 0$. □

22. Relative matching of the actual profiles

The coordinate z in the auxiliary extension in the proof of Theorem 20.1 was a physical coordinate. From now on the extension coordinate is a new variable $Y > 0$, with exponent

$$\eta = 1 - 2s \in (-1, 0), \qquad D = 3 + \eta = 4 - 2s.$$

For a function on $\mathbb{R}^2$ its actual order-s extension is

$$\widetilde{f}(x,Y) = \int_{\mathbb{R}^2} \Pi_s(x-y,Y)f(y)\,\mathrm{d}y, \qquad \Pi_s(x,Y) = \frac{s}{\pi}\frac{Y^{2s}}{(|x|^2+Y^2)^{1+s}}. \tag{22.1}$$

It solves $\operatorname{div}(Y^\eta \nabla \widetilde{f}) = 0$ for $Y > 0$. The kernel has mass one and first moment

$$\int_{\mathbb{R}^2} |x|\Pi_s(x,Y)\,\mathrm{d}x = \frac{\Gamma(3/2)\Gamma(s-1/2)}{\Gamma(s)}Y < \infty. \tag{22.2}$$

These are the usual extension normalizations of [4]. Let $B_R^+ = \{(x,Y) : |(x,Y)| < R,\ Y > 0\}$ and $\partial B_R^+ = \{(x,Y) : |(x,Y)| = R,\ Y > 0\}$.

Lemma 22.1 (Poisson transfer of the transverse remainder)**.** *For every fixed upper ball, bounded upper annulus, and upper hemisphere,*

$$\frac{\widetilde{S}_\mu(Rx,RY)}{R} \longrightarrow 0 \quad \text{in } L^2(Y^\eta), \qquad R \longrightarrow \infty. \tag{22.3}$$

For a hemisphere $L^2(Y^\eta)$ uses surface measure, and for balls and annuli it uses volume measure. If f has a global linear bound and $f(x) = o(|x|)$ uniformly as $|x| \to \infty$, then

$$\sup_{|(x,Y)|\le R,\ Y\ge 0} |\widetilde{f}(x,Y)| = o(R). \tag{22.4}$$

Proof. By (21.16), $R^{-1}S_\mu(Rx) \to 0$ for every x outside the positive ray, while (21.14) gives the common majorant $C(R^{-1}+|x|)$. At a fixed (x,Y) with $Y>0$, scale the Poisson integral and use (22.2). Dominated convergence applies, since the exceptional ray has planar measure zero. The extension has the common bound

$$R^{-1}|\widetilde{S}_\mu(Rx,RY)| \le C(R^{-1}+|x|+Y).$$

The weight Y^η is locally integrable in volume and on the upper hemisphere because $\eta>-1$. A second application of dominated convergence proves (22.3), including balls that meet the origin. This argument asserts no convergence of gradients.

For the second statement, write $|f(x)| \le \varepsilon|x| + C_\varepsilon$ and apply the mass and first-moment identities. Then $|\widetilde{f}(x,Y)| \le \varepsilon(|x|+C_sY)+C_\varepsilon$. Take the supremum on a ball, divide by R, and let first $R\to\infty$ and then $\varepsilon\to 0$. □

Proposition 22.2 (The common longitudinal coefficient)**.** *For the canonical homogeneous field*

$$\Psi_M(X,Z) = k_sM\int_0^\infty \left[|(X,Z)-(u,0)|^{-2t} - u^{-2t} - 2tXu^{-2t-1}\right]u^\beta\,\mathrm{d}u, \tag{22.5}$$

the profile has the expansion

$$P(x) = \Psi_M(x) + A_*X + A_ZZ + S_\mu(x) + o(|x|), \tag{22.6}$$

where the final remainder is uniform in direction. If P_0 is the centered parabolic profile with contact set $\{X \ge bZ^2\}$ and parameters a, M, there is a real ℓ such that

$$\frac{\widetilde{P}(Rx,RY) - \widetilde{P}_0(Rx,RY) - \ell RZ}{R} \longrightarrow 0 \quad \text{in } L^2(Y^\eta) \tag{22.7}$$

on every fixed upper ball, bounded upper annulus, and upper hemisphere. The trace difference $P-P_0-\ell Z$ has a global linear bound.

Proof. The power of u in the worst origin term of (22.5) is $\sigma-1>-1$, and that in its far Taylor remainder is $\sigma-2<-1$. Thus this combined integral fixes its affine gauge without ambiguity. A direct computation of the terms omitted on $0<u<1$ gives

$$Q_{Mu^\beta} = \Psi_M + \frac{k_sM}{1+\sigma} + \frac{2tk_sM}{\sigma}X. \tag{22.8}$$

This is precisely (20.13) with $\alpha=\sigma$, $p=-2t$, $1-\alpha=\beta$, and $c=k_s$. There is no dimensional conversion factor: the half-line theorem uses the same two-dimensional kernel as (21.1).

Lemma 21.1 and (21.5) verify both density hypotheses of Theorem 20.1. The required trace is the second conclusion of Proposition 21.3. Apply that theorem to the literal Q_m, with no externally added affine term. It gives

$$Q_m(x) = \Psi_M(x) + (A_* - A_X)X + o(|x|)$$

uniformly in the physical plane. Substitution in (21.17) cancels A_X and proves (22.6).

The model has pressure

$$\mu_0(X,Z) = \frac{a}{A_s^*b^t}(X-bZ^2)_+^t, \qquad m_0(X) = MX_+^\beta.$$

It has the same L and A_*. Reflection in Z makes the normal coefficient in its decomposition vanish, since P_0, Q_{m_0}, S_{μ_0} are all even in Z. Hence, with $\ell = A_Z$,

$$P - P_0 - \ell Z = S_\mu - S_{\mu_0} + o(|x|).$$

The global linear bound follows from (21.14) and Theorem 20.1. Lemma 22.1 proves (22.7), and the Poisson extension preserves the affine function Z by (22.2) and symmetry. □

Lemma 22.3 (Fixed translations at affine order)**.** *For every fixed $v \in \mathbb{R}^2$,*

$$\sup_{|x|\le R} |P_0(x-v) - P_0(x)| = o(R). \tag{22.9}$$

The same estimate holds for the order-s extensions on balls in the closed upper half-space.

Proof. Put $U_0 = P_0 + aZ^2$. It is convex and C^1, and the growth of P_0, with $\lambda < 2$, gives

$$R^{-2}U_0(Rx) \longrightarrow aZ^2$$

locally uniformly. We recall the elementary gradient consequence of convexity. For a differentiable convex function F_R and a coordinate vector e_j,

$$\frac{F_R(x) - F_R(x-\delta e_j)}{\delta} \le \partial_j F_R(x) \le \frac{F_R(x+\delta e_j) - F_R(x)}{\delta}.$$

Apply this with $F_R(x) = R^{-2}U_0(Rx)$ on a slightly enlarged compact set. Let $R \to \infty$ and then $\delta \downarrow 0$; the smooth quadratic limit makes both differences converge uniformly to its partial derivative. Consequently

$$\sup_{|x|\le CR} |\nabla P_0(x)| = o(R) \qquad \text{for every fixed } C.$$

Integrate the gradient on the segment of displacement v to obtain (22.9). Its global linear bound and the second assertion of Lemma 22.1 prove the extension statement. □

Theorem 22.4 (Uniform relative affine matching)**.** *For a profile in the class and normalization of Section 21, there is a unique $\ell \in \mathbb{R}$ such that*

$$\sup_{|x|\le R} |P(x) - P_0(x) - \ell Z| = o(R). \tag{22.10}$$

Define genuine translated obstacle profiles by

$$P_{d,h}(X,Z) = P_0(X-d, Z-h) - 2ahZ + ah^2, \qquad K_{d,h} = \{X \ge d + b(Z-h)^2\}. \tag{22.11}$$

Take $h = -\ell/(2a)$ and $d = -bh^2$, and let W be the even reflection of the order-s extension of $w = P - P_{d,h}$. Then $0 \in \partial K \cap \partial K_{d,h}$, and

$$\sup_{B_R} |W| = o(R). \tag{22.12}$$

Writing $L_\eta = \operatorname{div}(|Y|^\eta \nabla)$ and $W_\pm = \max(\pm W, 0)$, one has

$$W_\pm \in H^1_{\mathrm{loc}}(\mathbb{R}^3, |Y|^\eta), \qquad L_\eta W_\pm \ge 0 \quad \text{in distributions.} \tag{22.13}$$

The matching also holds in weighted ball, annular, and sphere L^2. For balls centered at the origin, or at any fixed point of $\{Y = 0\}$,

$$\int_{B_R} |Y|^\eta |\nabla W|^2 \, \mathrm{d}q = o(R^D), \tag{22.14}$$

$$I_\pm(R) := \int_{B_R} |Y|^\eta |\nabla W_\pm|^2 |q|^{-(D-2)} \, \mathrm{d}q = o(R^2), \tag{22.15}$$

$$I_+(R) I_-(R) = o(R^4). \tag{22.16}$$

In the expressions for a different center, $|q|$ denotes distance to that center. Each Green integral is finite on bounded balls; no finite energy at infinity is asserted.

Proof. Alignment within the obstacle problem. Formula (22.11) preserves the fixed obstacle, because on its contact set

$$-a(Z-h)^2 - 2ahZ + ah^2 = -aZ^2.$$

Translation invariance and the annihilation of affine functions show that it is an actual obstacle solution, with pressure $\mu_{d,h}(X,Z) = \mu_0(X-d, Z-h)$. With $h = -\ell/(2a)$ its normal affine

term is ℓZ. Lemma 22.3 shows that any fixed d preserves (22.7). The choice $d = -bh^2$ ensures $d + b(0-h)^2 = 0$, and hence common boundary contact at the origin. It does not require the vertex of the translated parabola to remain at the origin.

It follows from Proposition 22.2 and Lemma 22.3 that the final actual difference has a global linear trace bound and

$$\int_{B_{2R}} |Y|^\eta W^2 \,\mathrm{d}q = o(R^{D+2}). \tag{22.17}$$

Indeed this is exactly the rescaling of the weighted unit-ball convergence; the even reflection multiplies its integral by two. The sphere and bounded-annulus conclusions follow at the same time.

Local extension regularity and the exact source. The traces are locally C^1 and have ordinary fractional tails. We record why their extensions have locally bounded gradients up to $Y = 0$. Multiply the trace by a smooth cutoff equal to one near a chosen compact set. Its compactly supported part has bounded Lipschitz norm. Tangential differentiation of the Poisson convolution is then bounded by the norm of its tangential derivative. For the normal derivative, subtract the trace value at the center of the kernel and use

$$\int_{\mathbb{R}^2} |h|\,|\partial_Y \Pi_s(h, Y)| \,\mathrm{d}h \le C_s, \qquad Y > 0. \tag{22.18}$$

Scaling reduces this bound to $Y = 1$; its integral at infinity converges precisely because $2s > 1$. The complementary far-tail piece is smooth in tangential variables on the compact set, and its normal derivative is $O(Y^{2s-1})$. Its estimates follow by differentiating the kernel away from its singularity and using (21.2). Thus ∇W is locally bounded on either side of $Y = 0$, and $W \in H^1_{\mathrm{loc}}(\mathbb{R}^3, |Y|^\eta)$.

Lemma 3.6 supplies the exact distributional conormal identity (3.9), with $c_{s,\mathrm{ext}} = e_s = 2^{1-2s}\Gamma(1-s)/\Gamma(s)$. The local energy just proved makes it a weak source identity here.

For smooth compactly supported upper-half-space tests up to the boundary, the weak identity is therefore

$$\int_{Y>0} Y^\eta \nabla W \cdot \nabla \varphi \,\mathrm{d}q = c_{s,\mathrm{ext}} \int_{\mathbb{R}^2} \varphi(x, 0)(\mu - \mu_{d,h})(x) \,\mathrm{d}x. \tag{22.19}$$

Even reflection gives the exact distributional identity

$$L_\eta W = -2c_{s,\mathrm{ext}}(\mu - \mu_{d,h})\delta_{\{Y=0\}}. \tag{22.20}$$

Complementarity and the two phase inequalities. On $\operatorname{supp}\mu$, $w = \psi - P_{d,h} \le 0$, while on $\operatorname{supp}\mu_{d,h}$, $w = P - \psi \ge 0$. Hence

$$w(\mu - \mu_{d,h}) \le 0. \tag{22.21}$$

Choose smooth convex functions f_ε increasing to q_+, with $0 \le f_\varepsilon' \le 1$ and $f_\varepsilon'(q) = 0$ for $q \le 0$. By (22.20) and the weak chain rule,

$$L_\eta f_\varepsilon(W) = -2c_{s,\mathrm{ext}} f_\varepsilon'(w)(\mu - \mu_{d,h})\delta_{\{Y=0\}} + |Y|^\eta f_\varepsilon''(W)|\nabla W|^2 \ge 0.$$

The source term is nonnegative because $\mu = 0$ on $\{w > 0\}$ and the derivative vanishes on $\{w \le 0\}$. In particular no sign has been assigned to the source on $\{w = 0\}$. Weighted local H^1 approximation and the locally bounded trace-source densities justify the chain rule and passage to the limit. This proves $L_\eta W_+ \ge 0$; apply the same argument to $-W$ to obtain $L_\eta W_- \ge 0$.

Caccioppoli and Green energies. For a compactly supported cutoff ζ, use $\varphi = \zeta^2 W$ in (22.19). Local weighted H^1 approximation and local boundedness of the densities justify this test. The right side is nonpositive by (22.21). Expanding the left side and using Cauchy–Schwarz and Young's inequality gives, after reflection,

$$\int \zeta^2 |Y|^\eta |\nabla W|^2 \,\mathrm{d}q \le 4 \int |Y|^\eta W^2 |\nabla \zeta|^2 \,\mathrm{d}q. \tag{22.22}$$

Choose $\zeta = 1$ on B_R, supported in B_{2R}, with $|\nabla\zeta| \le C/R$. Formula (22.17) proves (22.14).

The local Green integrals are finite: near their center the gradient is bounded, and the polar radial exponent of the integrand is $D-1-(D-2)=1$. To estimate their growth, fix $\varepsilon>0$ and then choose R_ε so that $\int_{B_\rho}|Y|^\eta|\nabla W|^2\le\varepsilon\rho^D$ for $\rho\ge R_\varepsilon$. On each dyadic shell of outer radius ρ above that scale, $|q|^{-(D-2)}\le C\rho^{-(D-2)}$, so its Green energy is at most $C\varepsilon\rho^2$. Summing the geometric series up to R gives at most $C\varepsilon R^2$. The fixed inner Green energy divided by R^2 tends to zero. Since $|\nabla W_\pm|\le|\nabla W|$ a.e., this proves (22.15) and (22.16). Fixed translations in X,Z preserve the weight, local regularity, and the large-ball estimates, so the same conclusions hold at every fixed center on $Y=0$.

Uniform sublinearity from the actual phase inequalities. The weight $|Y|^\eta$ belongs to A_2, since $-1<\eta<1$. For a ball separated from $Y=0$ the weight is comparable to a constant; for balls meeting a fixed multiple of that plane, rescaling bounds the averages of $|Y|^\eta$ and $|Y|^{-\eta}$ by constants times R^η and $R^{-\eta}$, respectively. Their product is uniformly bounded. At a center on $Y=0$ its exact ball mass is

$$\int_{B_R}|Y|^\eta\,\mathrm{d}q=\frac{4\pi}{(1+\eta)(3+\eta)}R^D.$$

The weighted local subsolution estimate of Fabes–Kenig–Serapioni [13, Theorem 2.3.1, p. 96], applied to the verified nonnegative subsolutions $W_\pm$, gives

$$\operatorname*{ess\,sup}_{B_R}W_\pm\le C_s\left(R^{-D}\int_{B_{2R}}|Y|^\eta W_\pm^2\,\mathrm{d}q\right)^{1/2}=o(R). \tag{22.23}$$

Continuity changes essential supremum into supremum. This proves (22.12). It has been applied only to the difference of two actual obstacle solutions, whose phases satisfy (22.13); the separate transverse potential does not have this property, and in fact (21.21) precludes its uniform sublinearity.

Finally, on $Y=0$,

$$P-P_0-\ell Z=w+\big[P_0(\,\cdot-(d,h))-P_0\big]+ah^2.$$

The first term is uniformly $o(R)$ by (22.12), the second by Lemma 22.3, and the constant is harmless. This proves (22.10). If ℓ_1 and ℓ_2 both satisfy it, evaluate their difference at $(X,Z)=(0,R)$ to obtain $|\ell_1-\ell_2|R=o(R)$, hence $\ell_1=\ell_2$. □

23. Ordering for sources in a thin horn

Use the notation and slit tools of Section 10.

Theorem 23.1 (Thin-horn ordering)**.** *Let W be continuous, even in Y, and locally in $H^1(\rho)$ on $\mathbb{R}^3$. Suppose W_+ and W_- are weak L_η-subsolutions,*

$$L_\eta W=f(X,Z)\delta_{\{Y=0\}},\qquad f\in L^\infty_{\mathrm{loc}}(\mathbb{R}^2).$$

Outside a fixed compact set suppose

$$\operatorname{supp}f\subset\{X\ge-C_0,\ |Z|\le C_1\sqrt{1+X_+}\},\qquad |f(X,Z)|\le C_2(1+X_+)^{1-s}.$$

Suppose further that W has zero trace on $\mathcal{L}$ and $M(R):=\sup_{B_R}|W|=o(R)$. Then, for every $\gamma>\sigma$, $M(R)=O(R^\gamma)$. Moreover, W has one sign. The constants may depend on s,γ,W, and the fixed horn constants.

The source bound controls one angular coefficient. Compactness then improves the growth twice; an elementary two-phase energy inequality applies at the improved exponent. The spectral and compactness facts, including their boundary conditions, were proved in Section 10.

23.1. **The degree-one projection.** Let

$$K_Z = \int_{S^2} |\theta_Y|^\eta \theta_Z^2 \, \mathrm{d}S = \frac{4\pi}{(\eta+1)(\eta+3)}, \qquad c_Z(r) = K_Z^{-1} \int_{S^2} |\theta_Y|^\eta W(r\theta)\theta_Z \, \mathrm{d}S.$$

Lemma 23.2. *Under the hypotheses of Theorem 23.1,* $|c_Z(r)| \le C(1+r^s)$.

Proof. Testing the source identity against a radial test times the smooth eigenfunction θ_Z yields

$$(r^{b_G+1}c_Z')' - (b_G+1)r^{b_G-1}c_Z = \frac{r}{K_Z}\int_{S^1} f(r\omega)\omega_Z \, \mathrm{d}\omega.$$

The planar source has polar measure $r \, \mathrm{d}r \, \mathrm{d}\omega$, without an angular weight. Equivalently,

$$c_Z'' + \frac{b_G+1}{r}c_Z' - \frac{b_G+1}{r^2}c_Z = \frac{g_Z(r)}{r^2}, \qquad g_Z(r) = \frac{r^{1-\eta}}{K_Z}\int_{S^1} f(r\omega)\omega_Z \, \mathrm{d}\omega. \tag{23.1}$$

Polar Fubini defines the source integral for almost every radius. This is a distributional ODE and provides the absolutely continuous derivative representative of the continuous coefficient. The large circle meets the horn in an arc of length $O(r^{-1/2})$ with $|\omega_Z| = O(r^{-1/2})$. Consequently

$$|g_Z(r)| \le Cr^{2s}r^{1-s}r^{-1/2}r^{-1/2} = Cr^s.$$

Local boundedness of f gives $g_Z(r) = O(r^{2s})$ near zero. Variation of parameters therefore gives the convergent expression

$$c_Z(r) = -\frac{1}{b_G+2}\left[r\int_r^\infty g_Z(u)u^{-2} \, \mathrm{d}u + r^{-b_G-1}\int_0^r g_Z(u)u^{b_G} \, \mathrm{d}u\right]. \tag{23.2}$$

Indeed the displayed particular solution is $O(r)$ at zero and $O(r^s)$ at infinity. Subtracting it from c_Z gives $Ar + Br^{-b_G-1}$; boundedness at zero removes B, and $c_Z = o(r)$ removes A. The integrals use $s < 1$ at infinity and $2s > 1$ at zero. This proves the bound for the entire coefficient, not merely an estimate on its forcing. □

23.2. **Two growth improvements.**

Growth conclusion of Theorem 23.1. Suppose $M(R)/R^s$ is unbounded. Choose $p = s$ and $1 < Q < 1+\sigma$. Since $M = o(R)$, Lemmas 10.3 and 10.4 apply. The nonzero limit has the form $U = AH + BZ$ by Lemma 10.2. On any fixed radial interval $\varepsilon < r < T$, its approximating Z projection satisfies

$$\left|\frac{c_Z(R_j r)}{A_j}\right| \le \frac{C}{A_j} + C\frac{R_j^s}{A_j}r^s \longrightarrow 0.$$

Strong weighted L^2 convergence on annuli passes angular projections in radial $L^2(r^{D-1}\,\mathrm{d}r)$, by Cauchy–Schwarz on the sphere. As H is even in Z, the remaining projection is Br; thus $B = 0$. The inner estimate gives $|A|r^\sigma\|H\|_{L^2(S^2,\rho)} \le Cr^s$ for almost every small r, forcing $A = 0$. This contradicts nontriviality. Hence $M(R) = O(R^s)$.

Now fix $\sigma < \gamma < s$. If $M(R)/R^\gamma$ were unbounded, choose $p = \gamma$ and $s < Q < 1$. The first improvement gives $M(R)/R^Q \to 0$. The resulting nonzero limit has outer growth below one, so it equals AH. Its inner order $\gamma > \sigma$ forces $A = 0$, again a contradiction. This proves every $\gamma \in (\sigma, s)$; larger exponents follow from $O(R^s)$. No endpoint estimate at $\gamma = \sigma$ is needed. □

23.3. **An elementary two-phase threshold.**

Lemma 23.3. *Let V_1, V_2 be nonnegative, continuous, even weighted H^1_{loc} subsolutions on $\mathbb{R}^3$ with $V_1V_2 = 0$ and $V_1(0) = V_2(0) = 0$. If $\sup_{B_R}(V_1+V_2) = o(R^{1/2})$, then one phase vanishes identically.*

Proof. On the full even weighted sphere the first nonconstant eigenvalue is $b_G + 1$. To verify it, put $u = |\theta_Y| \in [0, 1]$ and use azimuthal Fourier modes of frequency $m \geq 0$. Their profiles are $(1-u^2)^{m/2}$ times Jacobi polynomials of degree j in u^2, with eigenvalues $(m+2j)(m+2j+b_G)$. The Jacobi weight is $z^{(\eta-1)/2}(1-z)^m$, whose endpoint parameters exceed -1. Fourier and polynomial completeness show that these are all modes. The eigenvalue $b_G + 1$ is realized by θ_X, θ_Z.

For nonzero disjoint nonnegative angular functions f, g, choose $c > 0$ so that $h = f - cg$ has zero weighted mean. Disjointness gives zero cross mass and cross energy almost everywhere. Thus its Rayleigh quotient is a weighted average of those of f, g, and Poincaré gives $\max\{Q(f), Q(g)\} \geq b_G + 1$. With $\gamma_0(q) = \sqrt{b_G^2/4 + q} - b_G/2$ we obtain

$$\gamma_0(Q(f)) + \gamma_0(Q(g)) \geq 1. \tag{23.3}$$

Define $\Gamma(q) = |q|^{-b_G}$ and $I_i(r) = \int_{B_r} \rho\Gamma|\nabla V_i|^2$. Regularize by $\Gamma_\varepsilon = \min\{|q|^{-b_G}, \varepsilon^{-b_G}\}$; its distributional L_η is nonpositive on the corner sphere. Testing the subsolution inequality with $V_i\Gamma_\varepsilon$ and radial cutoffs, then integrating by parts, gives for almost every r

$$I_i(r) \leq r^{-b_G} \int_{\partial B_r} \rho V_i \partial_r V_i + \frac{b_G}{2} r^{-b_G-1} \int_{\partial B_r} \rho V_i^2. \tag{23.4}$$

The kernel Laplacian term has favorable sign and is discarded before monotone convergence as $\varepsilon \downarrow 0$. This also proves local finiteness of I_i. If retained, the central term tends to zero by continuity and $V_i(0) = 0$; no prescribed Hölder modulus is used. Put $H_i = \int_{\partial B_r} \rho V_i^2$, $T_i = \int_{\partial B_r} \rho|\nabla_\tau V_i|^2$, $A_i = \int_{\partial B_r} \rho|\partial_r V_i|^2$. Where $I_i, H_i > 0$, Cauchy–Schwarz in (23.4) yields

$$\frac{rI_i'}{I_i} \geq \frac{u_i^2 + q_i}{u_i + b_G/2} \geq 2\gamma_0(q_i), \quad u_i = r\sqrt{A_i/H_i}, \quad q_i = r^2 T_i/H_i.$$

The last inequality is the square identity $u_i^2 + q_i - 2\gamma_0(q_i)(u_i + b_G/2) = (u_i - \gamma_0(q_i))^2$. Equation (23.3) proves that $r^{-2}I_1(r)I_2(r)$ is nondecreasing. If $H_i = 0$, (23.4) forces $I_i = 0$; continuity extends monotonicity across such radii.

For completeness, the outer cutoff test with $V_i\Gamma_\varepsilon\chi_R^2$ gives

$$I_i(R) \leq C \sup_{B_{2R}} V_i^2.$$

In this test integrate the kernel cross term: the part $\chi_R^2 L_\eta \Gamma_\varepsilon$ is nonpositive; all remaining terms of $L_\eta(\Gamma_\varepsilon\chi_R^2)$ lie in the outer annulus and are bounded by $C\rho R^{-b_G-2}$. Their integral scales as $R^{D-b_G-2} = 1$. Passing $\varepsilon \downarrow 0$ proves the bound without estimating a singular kernel gradient by its absolute value near zero. The assumed growth gives $I_i(R) = o(R)$; hence the monotone nonnegative product $R^{-2}I_1I_2$ tends to zero. It vanishes identically. If both phases were nonzero, a sufficiently large ball would have positive energy for each, since a constant zero-energy phase on that ball must equal its value zero at the center. This is a contradiction. □

Completion of Theorem 23.1. Choose $\sigma < \gamma < 1/2$. The growth conclusion gives $M(R) = O(R^\gamma) = o(R^{1/2})$. Apply Lemma 23.3 to W_+, W_-; their common zero is the origin on the ray. One phase vanishes. □

24. Proof of parabolic rigidity

Theorem 24.1 (Rigidity with the canonical field)**.** *Suppose $1/2 < s < 1$, $P \in C(\mathbb{R}^2) \cap \mathcal{T}_s$ satisfies* (2.7)*, its gap is convex, and $r^{-\lambda}P(r\,\cdot) \to \Psi_M$ locally uniformly for some $M > 0$. With $b = (aD_s/M)^{1/\beta}$ and $P_0 = P_b$, there are unique $d, h \in \mathbb{R}$ with*

$$P(X, Z) = P_0(X - d, Z - h) - 2ahZ + ah^2. \tag{24.1}$$

Conversely these profiles satisfy the stated hypotheses.

Proof. We now combine the asymptotic geometry, affine matching, and ordering theorem. Translate a minimizing contact point (d_*, h_*) to zero with

$$P_{\mathrm{new}}(X, Z) = P(X + d_*, Z + h_*) + 2ah_*Z + ah_*^2.$$

Then the gap is just the translated original gap. The ordinary tail and the prescribed blow-down remain unchanged, since $2s > 1$ and $\lambda > 1$. The geometry already proved gives $K = \{X \geq g(Z)\}$ with $g \geq 0$, $g(0) = 0$, and $g(Z) \geq cZ^2 - C$. We work in this normalization.

The matching theorem gives a unique $\ell \in \mathbb{R}$ such that

$$\sup_{|x| \leq R} |P(x) - P_0(x) - \ell Z| = o(R).$$

Set $h = -\ell/(2a)$, $d = -bh^2$, and

$$P_{d,h}(X, Z) = P_0(X - d, Z - h) - 2ahZ + ah^2.$$

The actual extension W of $w = P - P_{d,h}$ is continuous, even, locally weighted H^1, and satisfies $\sup_{B_R} |W| = o(R)$. The phase signs established in the matching argument give $L_\eta W_\pm \geq 0$. Its reflected source is

$$L_\eta W = -2e_s(\mu - \mu_{d,h})\delta_{\{Y=0\}}, \qquad e_s = 2^{1-2s}\Gamma(1-s)/\Gamma(s) > 0.$$

The actual contact and the fixed translated model lie in one horn $X \geq -C_0$, $|Z| \leq C_1\sqrt{1 + X_+}$. Each pressure is locally bounded and bounded by $C(1 + X_+)^{1-s}$. For the actual pressure this is (17.6); its normal-scale proof already includes the countable covering that makes the density bound an almost-everywhere statement on the whole horn. The translated model satisfies the same bound directly from its density.

Both contacts contain the whole positive axis: the actual one has $g(0) = 0$, and the model threshold at $Z = 0$ is $d + bh^2 = 0$. Thus $W(X, 0, 0) = 0$ for all $X \geq 0$. Every hypothesis of Theorem 23.1 holds, so W and its trace have one sign.

We finish with the ordered-contact argument. Reverse the labels if necessary so that $v = P_1 - P_2 \geq 0$. Then $K_1 \subset K_2$. Both contacts have zero on their boundaries. A supporting line to the larger closed convex set K_2 at zero supplies a ball in the opposite half-plane, tangent at zero and disjoint from both contact sets. Hence v is fractional harmonic in that ball. If $v \not\equiv 0$, choose a compact patch E away from zero where $v \geq \varepsilon > 0$. Shrink the tangent ball to make E disjoint from its closure. The positive ball Poisson formula, integrating only over E, gives

$$v(\delta n) \geq c\delta^s$$

for small $\delta > 0$ on the inward normal into the ball: its numerator is comparable to δ^s, and the denominator is uniformly bounded above on E. Local C^1 regularity and contact with the same smooth obstacle instead give $v(0) = 0$, $\nabla v(0) = 0$, and $v(\delta n) = o(\delta) = o(\delta^s)$, a contradiction. Thus $P = P_{d,h}$. Since the normalized contact is contained in $X \geq 0$, equality with the model forces its minimum X coordinate $d = -bh^2$ to be nonnegative. Therefore $h = d = 0$ and $P = P_0$. Undoing the initial translation proves (24.1). The contact parabola has a unique vertex, giving uniqueness of d, h. Conversely Proposition 16.1 and translation with the stated affine gauge verify every hypothesis for this family. This proves Theorem 24.1.

□

24.1. **Completion of the intrinsic rank-one branch.**

Proposition 24.2 (Rank-one exhaustion)**.** *Suppose* (1.1) *holds and the automatic matrix has rank one. After an orthogonal thin change of coordinates, write* $\mathsf{B} = ae_Z \otimes e_Z$, $a > 0$. *Then* $P = u - aZ^2 \in \mathcal{T}_s$, $U - Q = \mathcal{E}_s P$, *and* $\widehat{\mu} = e_s \mathcal{A}_s P$. *The solution is exactly a rank-one polynomial from Section 2.1, a strip from Section 2.4, or a parabola from Section 2.5. Their stated parameters are unique, and every such model is an intrinsic solution.*

Proof. Proposition 4.1 supplies the full quadratic limit in the exact topology of (2.6). The orthogonal coordinate change preserves the weight, source sign, energy, continuity, and growth. Proposition 8.4 gives trace convexity. Proposition 11.4, with its compact, invariant, and noninvariant branches, proves the ordinary tail, canonical extension, and pressure conversion before any classification theorem is used. Thus P satisfies (2.7).

If $u_X = 0$, Theorem 13.1 gives precisely the affine and positive-strip alternatives. In the remaining case, Lemma 9.3 fixes the unique horizontal orientation and a legitimate obstacle gauge. Proposition 11.3 provides every strict exponent between $s + 1/2$ and $2s$. Theorem 15.1 gives the critical bound and Theorem 17.1 the unique positive canonical amplitude M. Every hypothesis of Theorem 24.1 now holds; it identifies the profile with a translated parabolic model, with $b = (aD_s/M)^{1/\beta}$. Undoing the horizontal reflection gives (2.4). The exact gauge identity is

$$P_b(\varepsilon X - d, Z - h) - 2ahZ + ah^2 + aZ^2 = P_b(\varepsilon X - d, Z - h) + a(Z - h)^2.$$

Its physical vertex is $(\varepsilon d, h)$, and the coefficient b is unrestricted and positive for fixed a. The coordinate-to-geometric map in Section 2.5 gives the stated catalogue.

The affine coefficients are unique. Positive-strip endpoints recover h, L, with no horizontal center. A parabolic contact has a unique recession orientation and boundary $X = d + b(Z - h)^2$ in that orientation; its minimum and curvature recover d, h, b. Its range direction is fixed by B, and reversing the transverse normal changes no state. Pressure sign and invariance distinguish the three alternatives.

For the intrinsic converses, affine and translated-strip remainders have growth at most one; the centered parabolic remainder has growth $\lambda = s + 1/2$, and its affine translation gauge has lower growth. Thus in each case $|P(x)| \le C(1 + |x|^p)$ for a $p < 2s$. The positive Poisson kernel has a finite pth moment, giving $|\mathcal{E}_s P(q)| \le C(1 + |q|^p)$. The exact nonnegative ordinary pressure, contact set, and converse were proved in Theorem 13.1 and Proposition 16.1. Lemma 3.6 gives local weighted energy and the source $-2\iota_*(e_s\mu)$. Adding Q produces nonnegative trace, evenness, continuity and quadratic growth. The strict remainder power gives full uniform blowdown; phase Caccioppoli gives strong local energy convergence after scaling. Translation and rotation preserve all these properties, with the displayed gauge preserving the fixed obstacle. Finally Proposition 11.4 has eliminated an extra zero-trace vertical mode. These are full bulk converses, not merely trace identities. No all-order classification theorem is used in this proof. □

25. Completion of the global classification

Proof of Theorem 1.1. Proposition 4.1 gives the unique PSD matrix B, the full convergence, and its power remainder. If $\widehat{\mu} = 0$, Lemma 4.3 gives precisely the polynomial family, including every rank and every nonnegative constant. Assume henceforth that the pressure is nonzero.

The ranks $0, 1, 2$ exhaust the PSD 2×2 matrices. At rank zero, Proposition 5.3 gives the positive full-contact or regular halfspace family. At rank one, Proposition 24.2 gives a strip or a parabola. At rank two, Proposition 6.5 gives an ellipse. Each branch begins with the same intrinsic energy class and the full limit supplied by Proposition 4.1; no pressure density, ordinary tail, or contact geometry has entered as an extra hypothesis.

The polynomial converse is Lemma 4.3. The full-contact and regular converses, including their explicit normal fluxes and local energy, are proved in Proposition 5.3. The strip and parabolic bulk converses are in Proposition 24.2, and the ellipse converse is in Proposition 6.5. Each has the displayed source, complementarity, growth, and quadratic limit. Thus every listed parameter choice belongs to the intrinsic class.

For completeness, the alternatives and their intrinsic data can be recovered without a coordinate choice. The unique blowdown recovers B in every family. Zero pressure identifies the polynomial family, whose literal coefficients avoid any center ambiguity in a null direction. Positive pressure

and full contact distinguish the negative vertical state at rank zero; its density determines c_0. For a regular state, the oriented positivity halfspace determines e, h_{reg}, and a positive trace value determines its amplitude.

A strip contact has a one-dimensional lineality space and two parallel boundary lines; these determine its centerline and width. A parabolic contact has exactly one recession ray and a unique vertex, which recover the oriented opening, vertex, and curvature. In both cases range B recovers the transverse line, but its unit normal is determined only up to sign. Replacing (n, h) by $(-n, -h)$ leaves a strip unchanged, and $n \mapsto -n$ leaves a parabola unchanged because its centered potential is even in the transverse coordinate. No axis translation is assigned to a strip.

For an ellipse, the unique affine coefficient in the decaying-potential representation fixes its center, since B is invertible. The source fixes $m_{\mathrm{ell}} = \widehat{\mu}(\mathbb{R}^2)/e_s$, and the unique matrix map fixes H and ℓ_{ell}. Shape matrices, rather than eigenvectors, handle a circular contact without redundant parameters. Under a thin orthogonal map R, both matrices transform by conjugation, centers and oriented directions by R, and scalar data remain fixed. The isotropic kernels and arc-length matrix formula have the same covariance. Fixed translations change only the center/gauge data and leave the leading matrix unchanged.

All positive-pressure amplitudes, widths, and masses in the catalogue are strictly positive. The zero state, zero-width strip limits, and pressure-free contact lines or points belong only to the polynomial family. Pressure, rank, and these contact distinctions make the six families pairwise disjoint. This proves exhaustion, converses, and parameter uniqueness. □

Corollary 25.1 (Sharp remainder powers)**.** *For every solution in Theorem 1.1,*

$$\sup_{B_R} |U - Q_{\mathsf{B}}| \le C_U(1 + R^{1+s}).$$

If $\mathsf{B} \neq 0$*, then the stronger estimate*

$$\sup_{B_R} |U - Q_{\mathsf{B}}| \le C_U(1 + R^{s+1/2})$$

holds. The powers $1 + s$ *in the full class and* $s + 1/2$ *in the class with nonzero quadratic matrix are optimal.*

Proof. This is a consequence of the completed catalogue. Polynomial and strip remainders grow at most linearly, and an ellipse's remainder relative to its unshifted quadratic is affine plus a bounded decaying potential. A parabolic remainder has growth $s + 1/2$ by Proposition 16.1 and its finite-moment Poisson extension. Fixed translations of a quadratic add only an affine term, and rotations preserve these bounds. At rank zero, full-contact states have degree $2s < 1 + s$, constants have degree zero, and regular states have degree $1 + s$. This proves both estimates. A nonzero regular state attains growth R^{1+s} on a ray in its positivity halfspace. A parabolic state has a nonzero canonical trace limit of degree $s + 1/2$, by (16.9); its affine translation terms have smaller degree. It therefore cannot have a smaller remainder power. This proves optimality in the two stated classes. Neither estimate is used in the preceding classification. □

26. The all-order finite-tail companion

The intrinsic theorem has now been proved. We record the related ordinary fractional problem, whose operator domain is different and whose range includes all $0 < s < 1$.

Theorem 26.1 (All-order finite-tail classification)**.** *Fix* $a > 0$ *and* $0 < s < 1$*. Under exactly* (2.7)*, the following alternatives are exhaustive, disjoint, and have unique parameters.*

(1) *If* $0 < s \le 1/2$*, either* $P = c \ge 0$ *with zero pressure, or* $P(X, Z) = p_L(Z)$ *for a unique* $L > 0$*, where* p_L *is the centered interval profile of Proposition 12.5. Its contact is* $\mathbb{R} \times [-L, L]$ *and its pressure is* $\mathrm{d}X \otimes (a/A_s^*)(L^2 - Z^2)_+^{1-s}\,\mathrm{d}Z$*.*

(2) *If $1/2 < s < 1$, the remainders are precisely the rank-one polynomial, strip, and parabolic instances of Sections 2.1, 2.4, and 2.5, expressed relative to the fixed quadratic aZ^2. Explicitly these are the affine states $P = \ell Z + c$ with $c \geq \ell^2/(4a)$; the strips $p_L(Z-h) - 2ahZ + ah^2$ with $h \in \mathbb{R}$, $L > 0$; and the parabolic coordinate formula (2.4) with unique $\varepsilon \in \{-1, 1\}$, $b > 0$, and $d, h \in \mathbb{R}$. The fractional pressures are the catalogue's intrinsic pressures divided by e_s.*

Every listed profile satisfies (2.7). In every case the gap $P + aZ^2$ is jointly convex and $M_P(R) = o(R^{2s})$. The parabolic family is exactly the noncylindrical class.

Proof. Proposition 7.1 and Theorem 8.2 prove the little-o bound and convexity under the original ordinary-tail hypotheses. For $s \leq 1/2$, Theorem 13.1 gives horizontal invariance and the constant or centered-strip alternatives, including all converses and endpoint cases.

For $s > 1/2$, the direct construction in Corollary 7.3 gives $U = Q + \mathcal{E}_s P$ with the intrinsic hypotheses and full quadratic matrix $ae_Z \otimes e_Z$. Apply the already proved Theorem 1.1. Its rank and pressure distinctions leave exactly the three stated fixed-coordinate families. The catalogue and the interval/parabolic model proofs give their ordinary tails and exact pressures, so every listed remainder satisfies the original hypotheses. Parameter uniqueness follows from the same contact and coefficient recovery. This implication uses the complete theorem only after its proof; the rank-one proof used the analytic embedding and interval tools, not the present classification. □

At $s \leq 1/2$ no affine translation gauge has an absolute ordinary tail, so contact is forced to be centered. The half-order logarithmic profile and its normalization are retained in Proposition 12.5; strict little-o sublinearity, rather than an $O(R)$ bound, is what gives invariance there. The tail condition is an operator-domain condition and does not by itself fix uniqueness data at infinity.

AI assistance and verification status

Fu Xiaonan is the sole human author. OpenAI tools contributed mathematical ideas and proof development to the intrinsic local-energy and automatic quadratic-blowdown arguments, the zero-rank operator interface, and the compact-potential and ellipse-normalization arguments (Sections 3–6). They contributed to the ordinary-tail preparation, intrinsic convexity and contact branches, positive-derivative slit analysis, transverse-channel recovery, and canonical-extension arguments (Sections 7–11); the interval, cylindrical, critical-moment, and canonical-asymptotic proofs; and the model, mesoscopic, matching, signed-ordering, and classification arguments (Sections 12–26). They also assisted with literature searches, drafting, calculations, notation integration, and internal checks throughout. These contributions include mathematical ideas and proof development, not only language editing. The human author remains responsible for the mathematics, citations, and final manuscript. Independent human verification of every argument and citation has not yet been completed, and this draft is not represented as submission-ready. AI checks are not formal proof certification or a substitute for that verification.

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